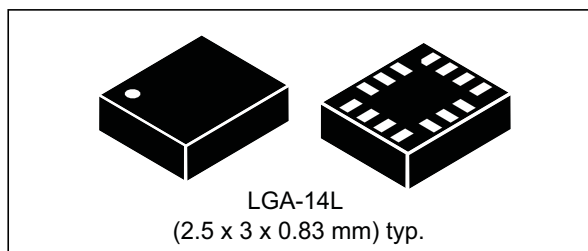


## iNEMO inertial module: always-on 3D accelerometer and 3D gyroscope

Datasheet - production data



### Features

- Power consumption: 0.4 mA in combo normal mode and 0.65 mA in combo high-performance mode
- “Always-on” experience with low power consumption for both accelerometer and gyroscope
- Smart FIFO up to 4 kbyte based on features set
- Android M compliant
- Hard, soft ironing for external magnetic sensor corrections
- $\pm 2/\pm 4/\pm 8/\pm 16$  g full scale
- $\pm 125/\pm 250/\pm 500/\pm 1000/\pm 2000$  dps full scale
- Analog supply voltage: 1.71 V to 3.6 V
- Independent IOs supply (1.62 V)
- Compact footprint, 2.5 mm x 3 mm x 0.83 mm
- SPI & I<sup>2</sup>C serial interface with main processor data synchronization feature
- Pedometer, step detector and step counter
- Significant motion and tilt function
- Standard interrupts: free-fall, wakeup, 6D/4D orientation, click and double-click
- Embedded temperature sensor
- ECOPACK<sup>®</sup>, RoHS and “Green” compliant

### Applications

- Motion tracking and gesture detection
- Collecting sensor data
- Indoor navigation
- IoT and connected devices
- Intelligent power saving for handheld devices
- Vibration monitoring and compensation

### Description

The LSM6DSL is a system-in-package featuring a 3D digital accelerometer and a 3D digital gyroscope performing at 0.65 mA in high-performance mode and enabling always-on low-power features for an optimal motion experience for the consumer.

The LSM6DSL supports main OS requirements, offering real, virtual and batch sensors with 4 kbyte for dynamic data batching.

ST’s family of MEMS sensor modules leverages the robust and mature manufacturing processes already used for the production of micromachined accelerometers and gyroscopes.

The various sensing elements are manufactured using specialized micromachining processes, while the IC interfaces are developed using CMOS technology that allows the design of a dedicated circuit which is trimmed to better match the characteristics of the sensing element.

The LSM6DSL has a full-scale acceleration range of  $\pm 2/\pm 4/\pm 8/\pm 16$  g and an angular rate range of  $\pm 125/\pm 250/\pm 500/\pm 1000/\pm 2000$  dps.

High robustness to mechanical shock makes the LSM6DSL the preferred choice of system designers for the creation and manufacturing of reliable products.

The LSM6DSL is available in a plastic land grid array (LGA) package.

Table 1. Device summary

| Part number | Temp. range [°C] | Package                   | Packing     |
|-------------|------------------|---------------------------|-------------|
| LSM6DSL     | -40 to +85       | LGA-14L<br>(2.5x3x0.83mm) | Tray        |
| LSM6DSLTR   | -40 to +85       |                           | Tape & Reel |

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# 1 Overview

The LSM6DSL is a system-in-package featuring a high-performance 3-axis digital accelerometer and 3-axis digital gyroscope.

The integrated power-efficient modes are able to reduce the power consumption down to 0.65 mA in high-performance mode, combining always-on low-power features with superior sensing precision for an optimal motion experience for the consumer thanks to ultra-low noise performance for both the gyroscope and accelerometer.

The LSM6DSL delivers best-in-class motion sensing that can detect orientation and gestures in order to empower application developers and consumers with features and capabilities that are more sophisticated than simply orienting their devices to portrait and landscape mode.

The event-detection interrupts enable efficient and reliable motion tracking and contextual awareness, implementing hardware recognition of free-fall events, 6D orientation, click and double-click sensing, activity or inactivity, and wakeup events.

The LSM6DSL supports main OS requirements, offering real, virtual and batch mode sensors. In addition, the LSM6DSL can efficiently run the sensor-related features specified in Android, saving power and enabling faster reaction time. In particular, the LSM6DSL has been designed to implement hardware features such as significant motion detection, tilt, pedometer functions, timestamping and to support the data acquisition of an external magnetometer with ironing correction (hard, soft).

The LSM6DSL offers hardware flexibility to connect the pins with different mode connections to external sensors to expand functionalities such as adding a sensor hub, etc.

Up to 4 kbyte of FIFO with dynamic allocation of significant data (i.e. external sensors, timestamp, etc.) allows overall power saving of the system.

Like the entire portfolio of MEMS sensor modules, the LSM6DSL leverages the robust and mature in-house manufacturing processes already used for the production of micromachined accelerometers and gyroscopes. The various sensing elements are manufactured using specialized micromachining processes, while the IC interfaces are developed using CMOS technology that allows the design of a dedicated circuit which is trimmed to better match the characteristics of the sensing element.

The LSM6DSL is available in a small plastic land grid array (LGA) package of 2.5 x 3.0 x 0.83 mm to address ultra-compact solutions.

## 2 Embedded low-power features

The LSM6DSL has been designed to be fully compliant with Android, featuring the following on-chip functions:

- 4 kbyte data buffering
  - 100% efficiency with flexible configurations and partitioning
  - Possibility to store timestamp
- Event-detection interrupts (fully configurable):
  - Free-fall
  - Wakeup
  - 6D orientation
  - Click and double-click sensing
  - Activity / inactivity recognition
- Specific IP blocks with negligible power consumption and high-performance:
  - Pedometer functions: step detector and step counters
  - Tilt (refer to [Section 2.1: Tilt detection](#) for additional information)
  - Absolute Wrist Tilt (refer to [Section 2.2: Absolute wrist tilt](#) for additional information)
  - Significant Motion Detection
- Sensor hub
  - Up to 6 total sensors: 2 internal (accelerometer and gyroscope) and 4 external sensors
- Data rate synchronization with external trigger for reduced sensor access and enhanced fusion

### 2.1 Tilt detection

The tilt function helps to detect activity change and has been implemented in hardware using only the accelerometer to achieve both the targets of ultra-low power consumption and robustness during the short duration of dynamic accelerations.

It is based on a trigger of an event each time the device's tilt changes. For a more customized user experience, in the LSM6DSL the tilt function is configurable through:

- a programmable average window
- a programmable average threshold

The tilt function can be used with different scenarios, for example:

- a) Triggers when phone is in a front pants pocket and the user goes from sitting to standing or standing to sitting;
- b) Doesn't trigger when phone is in a front pants pocket and the user is walking, running or going upstairs.

## 2.2 Absolute wrist tilt

The LSM6DSL implements in hardware the Absolute Wrist Tilt (AWT) function which allows detecting when the angle between a selectable accelerometer semi-axis and the horizontal plane becomes higher than a specific user-selectable value.

Configurable threshold and latency parameters are associated with the AWT function: the threshold parameter defines the amplitude of the tilt angle; the latency parameter defines the minimum duration of the AWT event to be recognized. The AWT interrupt signal is generated if the tilt angle is higher than the threshold angle for a period of time equal to or greater than the latency period.

The AWT function is based on the accelerometer sensor only and works at 26 Hz, so the accelerometer ODR must be set at a value of 26 Hz or higher.

By default, the AWT algorithm is applied to the positive X-axis.

In order to enable the AWT function it is necessary to set to 1 both the FUNC\_EN bit and the WRIST\_TILT\_EN bit of [CTRL10\\_C \(19h\)](#).

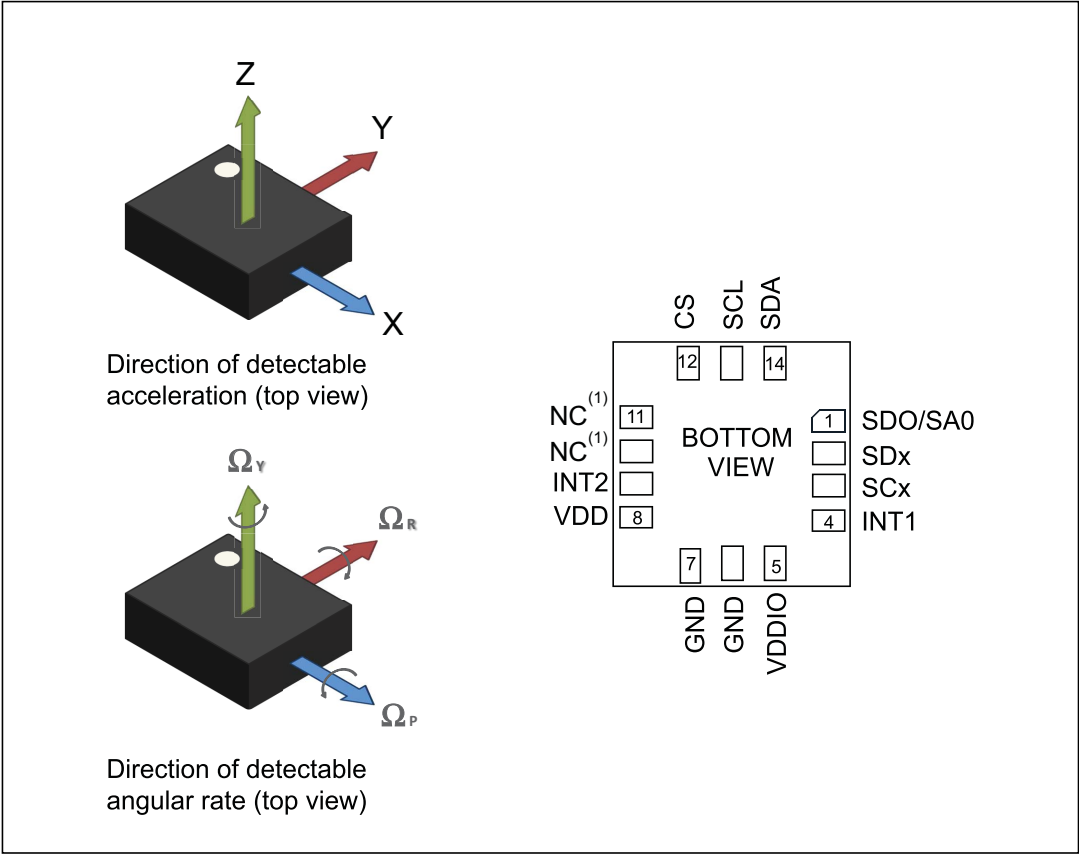
The AWT interrupt signal can be driven to the INT2 interrupt pin by setting to 1 the INT2\_WRIST\_TILT bit of the [DRDY\\_PULSE\\_CFG\\_G \(0Bh\)](#) register; it can also be checked by reading the WRIST\_TILT\_IA bit of the [FUNC\\_SRC2 \(54h\)](#) register (it will also clear the interrupt signal if latched).

[WRIST\\_TILT\\_IA \(55h\)](#) is the status register to be used to detect which axis has triggered the AWT event (not applicable when using one axis side only).

The full description and an example is given in the dedicated application note.

### 3 Pin description

Figure 1. Pin connections



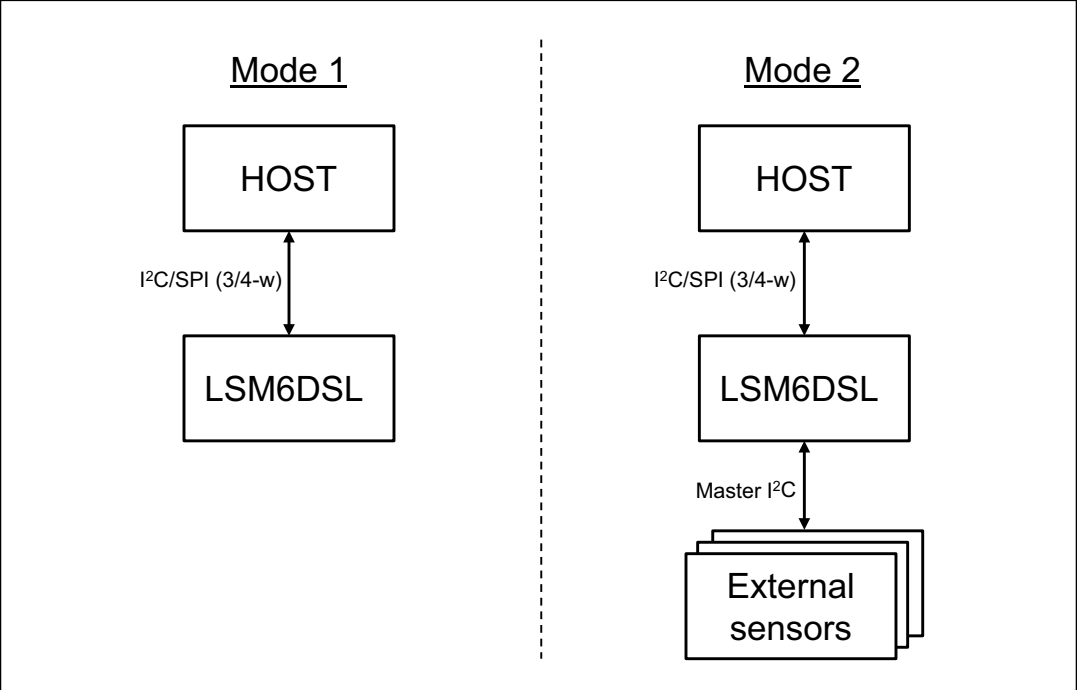
1. Leave pin electrically unconnected and soldered to PCB.

3.1 Pin connections

The LSM6DSL offers flexibility to connect the pins in order to have two different mode connections and functionalities. In detail:

- **Mode 1:** I<sup>2</sup>C slave interface or SPI (3- and 4-wire) serial interface is available;
- **Mode 2:** I<sup>2</sup>C slave interface or SPI (3- and 4-wire) serial interface and I<sup>2</sup>C interface master for external sensor connections are available;

Figure 2. LSM6DSL connection modes



In the following table each mode is described for the pin connections and function.

Table 2. Pin description

| Pin# | Name                 | Mode 1 function                                                                                                                                           | Mode 2 function                                                                                                                                              |
|------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | SDO/SA0              | SPI 4-wire interface serial data output (SDO)<br>I <sup>2</sup> C least significant bit of the device address (SA0)                                       | SPI 4-wire interface serial data output (SDO)<br>I <sup>2</sup> C least significant bit of the device address (SA0)                                          |
| 2    | SDx                  | Connect to VDDIO or GND                                                                                                                                   | I <sup>2</sup> C serial data master (MSDA)                                                                                                                   |
| 3    | SCx                  | Connect to VDDIO or GND                                                                                                                                   | I <sup>2</sup> C serial clock master (MSCL)                                                                                                                  |
| 4    | INT1                 | Programmable interrupt 1                                                                                                                                  |                                                                                                                                                              |
| 5    | VDDIO <sup>(1)</sup> | Power supply for I/O pins                                                                                                                                 |                                                                                                                                                              |
| 6    | GND                  | 0 V supply                                                                                                                                                |                                                                                                                                                              |
| 7    | GND                  | 0 V supply                                                                                                                                                |                                                                                                                                                              |
| 8    | VDD <sup>(1)</sup>   | Power supply                                                                                                                                              |                                                                                                                                                              |
| 9    | INT2                 | Programmable interrupt 2 (INT2) / Data enable (DEN)                                                                                                       | Programmable interrupt 2 (INT2)/<br>Data enable (DEN)/<br>I <sup>2</sup> C master external synchronization signal (MDRDY)                                    |
| 10   | NC <sup>(2)</sup>    | Leave unconnected                                                                                                                                         |                                                                                                                                                              |
| 11   | NC <sup>(2)</sup>    | Leave unconnected                                                                                                                                         |                                                                                                                                                              |
| 12   | CS                   | I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled; 0: SPI communication mode / I <sup>2</sup> C disabled) | I <sup>2</sup> C/SPI mode selection<br>(1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) |
| 13   | SCL                  | I <sup>2</sup> C serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                        | I <sup>2</sup> C serial clock (SCL)<br>SPI serial port clock (SPC)                                                                                           |
| 14   | SDA                  | I <sup>2</sup> C serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                            | I <sup>2</sup> C serial data (SDA)<br>SPI serial data input (SDI)<br>3-wire interface serial data output (SDO)                                               |

1. Recommended 100 nF filter capacitor.

2. Leave pin electrically unconnected and soldered to PCB.

## 4 Module specifications

### 4.1 Mechanical characteristics

@ Vdd = 1.8 V, T = 25 °C unless otherwise noted.

**Table 3. Mechanical characteristics**

| Symbol   | Parameter                                                                  | Test conditions    | Min. | Typ. <sup>(1)</sup> | Max. | Unit     |
|----------|----------------------------------------------------------------------------|--------------------|------|---------------------|------|----------|
| LA_FS    | Linear acceleration measurement range                                      |                    |      | ±2                  |      | g        |
|          |                                                                            |                    |      | ±4                  |      |          |
|          |                                                                            |                    |      | ±8                  |      |          |
|          |                                                                            |                    |      | ±16                 |      |          |
| G_FS     | Angular rate measurement range                                             |                    |      | ±125                |      | dps      |
|          |                                                                            |                    |      | ±250                |      |          |
|          |                                                                            |                    |      | ±500                |      |          |
|          |                                                                            |                    |      | ±1000               |      |          |
|          |                                                                            |                    |      | ±2000               |      |          |
| LA_So    | Linear acceleration sensitivity <sup>(2)</sup>                             | FS = ±2            |      | 0.061               |      | mg/LSB   |
|          |                                                                            | FS = ±4            |      | 0.122               |      |          |
|          |                                                                            | FS = ±8            |      | 0.244               |      |          |
|          |                                                                            | FS = ±16           |      | 0.488               |      |          |
| G_So     | Angular rate sensitivity <sup>(2)</sup>                                    | FS = ±125          |      | 4.375               |      | mdps/LSB |
|          |                                                                            | FS = ±250          |      | 8.75                |      |          |
|          |                                                                            | FS = ±500          |      | 17.50               |      |          |
|          |                                                                            | FS = ±1000         |      | 35                  |      |          |
|          |                                                                            | FS = ±2000         |      | 70                  |      |          |
| G_So%    | Sensitivity tolerance <sup>(3)</sup>                                       | at component level |      | ±1                  |      | %        |
| LA_SoDr  | Linear acceleration sensitivity change vs. temperature <sup>(4)</sup>      | from -40° to +85°  |      | ±0.01               |      | %/°C     |
| G_SoDr   | Angular rate sensitivity change vs. temperature <sup>(4)</sup>             | from -40° to +85°  |      | ±0.007              |      | %/°C     |
| LA_TyOff | Linear acceleration zero-g level offset accuracy <sup>(5)</sup>            |                    |      | ±40                 |      | mg       |
| G_TyOff  | Angular rate zero-rate level <sup>(4)</sup>                                |                    |      | ±3                  |      | dps      |
| LA_OffDr | Linear acceleration zero-g level change vs. temperature <sup>(4)</sup>     |                    |      | ±0.1                |      | mg/°C    |
| G_OffDr  | Angular rate typical zero-rate level change vs. temperature <sup>(4)</sup> |                    |      | ±0.015              |      | dps/°C   |
| Rn       | Rate noise density in high-performance mode <sup>(6)</sup>                 |                    |      | 4                   |      | mdps/√Hz |

Table 3. Mechanical characteristics (continued)

| Symbol | Parameter                                                           | Test conditions        | Min. | Typ. <sup>(1)</sup>                                                                         | Max. | Unit                           |
|--------|---------------------------------------------------------------------|------------------------|------|---------------------------------------------------------------------------------------------|------|--------------------------------|
| RnRMS  | Gyroscope RMS noise in normal/low-power mode <sup>(7)</sup>         |                        |      | 75                                                                                          |      | mdps                           |
| An     | Acceleration noise density in high-performance mode <sup>(8)</sup>  | FS = $\pm 2\text{ g}$  |      | 80                                                                                          |      | $\mu\text{g}/\sqrt{\text{Hz}}$ |
|        |                                                                     | FS = $\pm 4\text{ g}$  |      | 80                                                                                          |      |                                |
|        |                                                                     | FS = $\pm 8\text{ g}$  |      | 90                                                                                          |      |                                |
|        |                                                                     | FS = $\pm 16\text{ g}$ |      | 130                                                                                         |      |                                |
| RMS    | Acceleration RMS noise in normal/low-power mode <sup>(9)(10)</sup>  | FS = $\pm 2\text{ g}$  |      | 1.8                                                                                         |      | mg(RMS)                        |
|        |                                                                     | FS = $\pm 4\text{ g}$  |      | 2.0                                                                                         |      |                                |
|        |                                                                     | FS = $\pm 8\text{ g}$  |      | 2.4                                                                                         |      |                                |
|        |                                                                     | FS = $\pm 16\text{ g}$ |      | 3.0                                                                                         |      |                                |
| LA_ODR | Linear acceleration output data rate                                |                        |      | 1.6 <sup>(11)</sup><br>12.5<br>26<br>52<br>104<br>208<br>416<br>833<br>1666<br>3332<br>6664 |      | Hz                             |
| G_ODR  | Angular rate output data rate                                       |                        |      | 12.5<br>26<br>52<br>104<br>208<br>416<br>833<br>1666<br>3332<br>6664                        |      |                                |
| Vst    | Linear acceleration self-test output change <sup>(12)(13)(14)</sup> |                        | 90   |                                                                                             | 1700 | mg                             |
|        | Angular rate self-test output change <sup>(15)(16)</sup>            | FS = 250 dps           | 20   |                                                                                             | 80   | dps                            |
|        |                                                                     | FS = 2000 dps          | 150  |                                                                                             | 700  | dps                            |
| Top    | Operating temperature range                                         |                        | -40  |                                                                                             | +85  | °C                             |

1. Typical specifications are not guaranteed.
2. Sensitivity values after factory calibration test and trimming
3. Subject to change.
4. Measurements are performed in a uniform temperature setup and they are based on characterization data in a limited number of samples. Not measured during final test for production.
5. Values after factory calibration test and trimming.

6. Gyroscope rate noise density in high-performance mode is independent of the ODR and FS setting.
7. Gyroscope RMS noise in normal/low-power mode is independent of the ODR and FS setting.
8. Accelerometer noise density in high-performance mode is independent of the ODR.
9. Accelerometer RMS noise in normal/low-power mode is independent of the ODR.
10. Noise RMS related to  $BW = ODR / 2$  (for  $ODR / 9$ , typ value can be calculated by  $Typ * 0.6$ ).
11. This ODR is available when accelerometer is in low-power mode.
12. The sign of the linear acceleration self-test output change is defined by the STx\_XL bits in [CTRL5\\_C \(14h\)](#), [Table 64](#) for all axes.
13. The linear acceleration self-test output change is defined with the device in stationary condition as the absolute value of:  
 $OUTPUT[LSb] \text{ (self-test enabled)} - OUTPUT[LSb] \text{ (self-test disabled)}$ . 1LSb = 0.061 mg at  $\pm 2 g$  full scale.
14. Accelerometer self-test limits are full-scale independent.
15. The sign of the angular rate self-test output change is defined by the STx\_G bits in [CTRL5\\_C \(14h\)](#), [Table 63](#) for all axes.
16. The angular rate self-test output change is defined with the device in stationary condition as the absolute value of:  
 $OUTPUT[LSb] \text{ (self-test enabled)} - OUTPUT[LSb] \text{ (self-test disabled)}$ . 1LSb = 70 mdps at  $\pm 2000$  dps full scale.

## 4.2 Electrical characteristics

@ V<sub>dd</sub> = 1.8 V, T = 25 °C unless otherwise noted.

**Table 4. Electrical characteristics**

| Symbol               | Parameter                                                                | Test conditions                              | Min.                     | Typ. <sup>(1)</sup> | Max.                     | Unit |
|----------------------|--------------------------------------------------------------------------|----------------------------------------------|--------------------------|---------------------|--------------------------|------|
| V <sub>dd</sub>      | Supply voltage                                                           |                                              | 1.71                     | 1.8                 | 3.6                      | V    |
| V <sub>dd_IO</sub>   | Power supply for I/O                                                     |                                              | 1.62                     |                     | 3.6                      | V    |
| I <sub>ddHP</sub>    | Gyroscope and accelerometer current consumption in high-performance mode | ODR = 1.6 kHz                                |                          | 0.65                |                          | mA   |
| I <sub>ddNM</sub>    | Gyroscope and accelerometer current consumption in normal mode           | ODR = 208 Hz                                 |                          | 0.45                |                          | mA   |
| I <sub>ddLP</sub>    | Gyroscope and accelerometer current consumption in low-power mode        | ODR = 52 Hz                                  |                          | 0.29                |                          | mA   |
| LA_I <sub>ddHP</sub> | Accelerometer current consumption in high-performance mode               | ODR < 1.6 kHz<br>ODR ≥ 1.6 kHz               |                          | 150<br>160          |                          | μA   |
| LA_I <sub>ddNM</sub> | Accelerometer current consumption in normal mode                         | ODR = 208 Hz                                 |                          | 85                  |                          | μA   |
| LA_I <sub>ddLM</sub> | Accelerometer current consumption in low-power mode                      | ODR = 52 Hz<br>ODR = 12.5 Hz<br>ODR = 1.6 Hz |                          | 25<br>9<br>4.5      |                          | μA   |
| I <sub>ddPD</sub>    | Gyroscope and accelerometer current consumption during power-down        |                                              |                          | 3                   |                          | μA   |
| T <sub>on</sub>      | Turn-on time                                                             |                                              |                          | 35                  |                          | ms   |
| V <sub>IH</sub>      | Digital high-level input voltage                                         |                                              | 0.7 * V <sub>DD_IO</sub> |                     |                          | V    |
| V <sub>IL</sub>      | Digital low-level input voltage                                          |                                              |                          |                     | 0.3 * V <sub>DD_IO</sub> | V    |
| V <sub>OH</sub>      | High-level output voltage                                                | I <sub>OH</sub> = 4 mA <sup>(2)</sup>        | V <sub>DD_IO</sub> - 0.2 |                     |                          | V    |
| V <sub>OL</sub>      | Low-level output voltage                                                 | I <sub>OL</sub> = 4 mA <sup>(2)</sup>        |                          |                     | 0.2                      | V    |
| Top                  | Operating temperature range                                              |                                              | -40                      |                     | +85                      | °C   |

1. Typical specifications are not guaranteed.

2. 4 mA is the maximum driving capability, i.e. the maximum DC current that can be sourced/sunk by the digital pad in order to guarantee the correct digital output voltage levels V<sub>OH</sub> and V<sub>OL</sub>.

### 4.3 Temperature sensor characteristics

@ Vdd = 1.8 V, T = 25 °C unless otherwise noted.

**Table 5. Temperature sensor characteristics**

| Symbol              | Parameter                                     | Test condition | Min. | Typ. <sup>(1)</sup> | Max. | Unit   |
|---------------------|-----------------------------------------------|----------------|------|---------------------|------|--------|
| TODR <sup>(2)</sup> | Temperature refresh rate                      |                |      | 52                  |      | Hz     |
| Toff                | Temperature offset <sup>(3)</sup>             |                | -15  |                     | +15  | °C     |
| TSen                | Temperature sensitivity                       |                |      | 256                 |      | LSB/°C |
| TST                 | Temperature stabilization time <sup>(4)</sup> |                |      |                     | 500  | µs     |
| T_ADC_res           | Temperature ADC resolution                    |                |      | 16                  |      | bit    |
| Top                 | Operating temperature range                   |                | -40  |                     | +85  | °C     |

1. Typical specifications are not guaranteed.
2. When the accelerometer is in Low-Power mode and the gyroscope part is turned off, the TODR value is equal to the accelerometer ODR.
3. The output of the temperature sensor is 0 LSB (typ.) at 25 °C.
4. Time from power ON bit to valid data based on characterization data.

## 4.4 Communication interface characteristics

### 4.4.1 SPI - serial peripheral interface

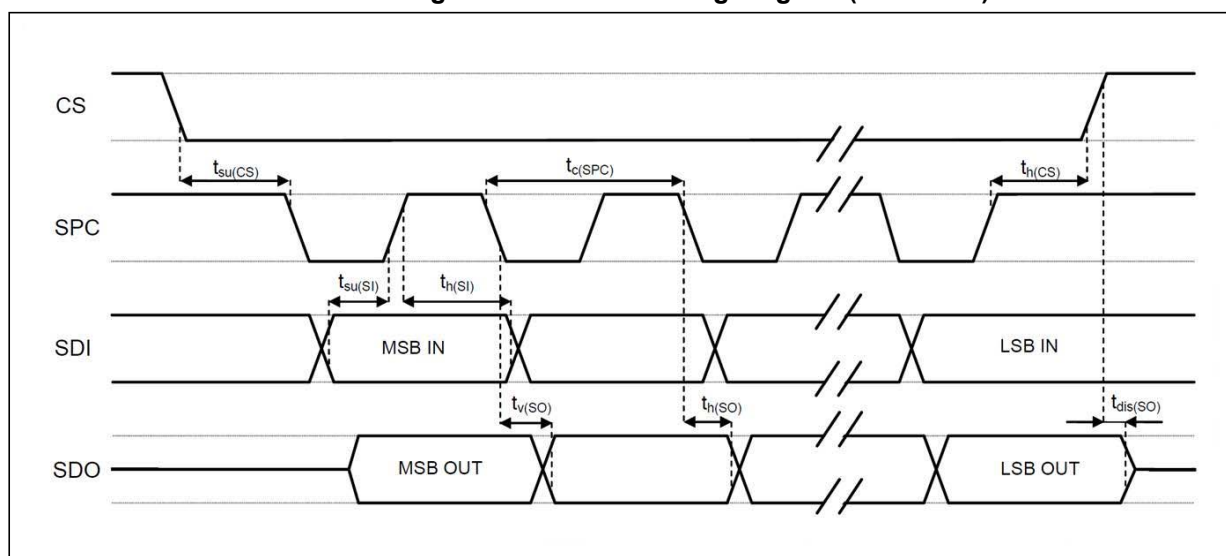
Subject to general operating conditions for Vdd and Top.

**Table 6. SPI slave timing values (in mode 3)**

| Symbol        | Parameter               | Value <sup>(1)</sup> |     | Unit |
|---------------|-------------------------|----------------------|-----|------|
|               |                         | Min                  | Max |      |
| $t_{c(SPC)}$  | SPI clock cycle         | 100                  |     | ns   |
| $f_{c(SPC)}$  | SPI clock frequency     |                      | 10  | MHz  |
| $t_{su(CS)}$  | CS setup time           | 5                    |     | ns   |
| $t_{h(CS)}$   | CS hold time            | 20                   |     |      |
| $t_{su(SI)}$  | SDI input setup time    | 5                    |     |      |
| $t_{h(SI)}$   | SDI input hold time     | 15                   |     |      |
| $t_{v(SO)}$   | SDO valid output time   |                      | 50  |      |
| $t_{h(SO)}$   | SDO output hold time    | 5                    |     |      |
| $t_{dis(SO)}$ | SDO output disable time |                      | 50  |      |

1. Values are guaranteed at 10 MHz clock frequency for SPI with both 4 and 3 wires, based on characterization results, not tested in production

**Figure 3. SPI slave timing diagram (in mode 3)**

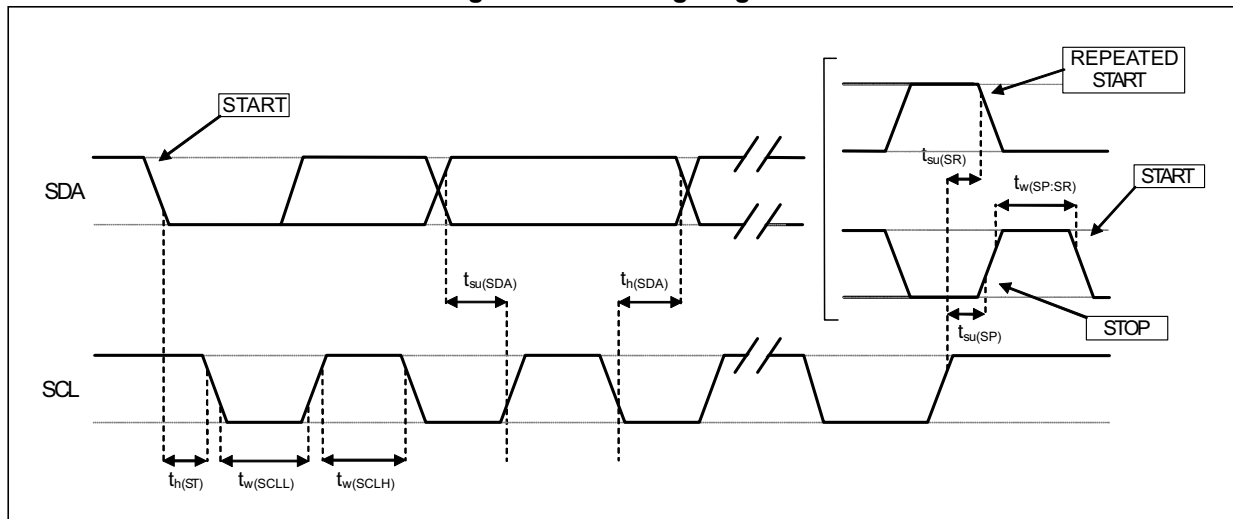


**Note:** Measurement points are done at  $0.2 \cdot V_{dd\_IO}$  and  $0.8 \cdot V_{dd\_IO}$ , for both input and output ports.

#### 4.4.2 I<sup>2</sup>C - inter-IC control interface

Subject to general operating conditions for Vdd and Top.

Figure 4. I<sup>2</sup>C timing diagram



Note: Measurement points are done at  $0.2 \cdot V_{dd\_IO}$  and  $0.8 \cdot V_{dd\_IO}$ , for both ports.

##### 4.4.2.1 I<sup>2</sup>C slave

Table 7. I<sup>2</sup>C slave timing values

| Symbol         | Parameter                                      | I <sup>2</sup> C standard mode <sup>(1)</sup> |      | I <sup>2</sup> C fast mode <sup>(1)</sup> |     | Unit    |
|----------------|------------------------------------------------|-----------------------------------------------|------|-------------------------------------------|-----|---------|
|                |                                                | Min                                           | Max  | Min                                       | Max |         |
| $f_{(SCL)}$    | SCL clock frequency                            | 0                                             | 100  | 0                                         | 400 | kHz     |
| $t_{w(SCLL)}$  | SCL clock low time                             | 4.7                                           |      | 1.3                                       |     | $\mu s$ |
| $t_{w(SCLH)}$  | SCL clock high time                            | 4.0                                           |      | 0.6                                       |     |         |
| $t_{su(SDA)}$  | SDA setup time                                 | 250                                           |      | 100                                       |     | ns      |
| $t_h(SDA)$     | SDA data hold time                             | 0                                             | 3.45 | 0                                         | 0.9 | $\mu s$ |
| $t_h(ST)$      | START condition hold time                      | 4                                             |      | 0.6                                       |     | $\mu s$ |
| $t_{su(SR)}$   | Repeated START condition setup time            | 4.7                                           |      | 0.6                                       |     |         |
| $t_{su(SP)}$   | STOP condition setup time                      | 4                                             |      | 0.6                                       |     |         |
| $t_{w(SP:SR)}$ | Bus free time between STOP and START condition | 4.7                                           |      | 1.3                                       |     |         |

1. Data based on standard I<sup>2</sup>C protocol requirement, not tested in production.

#### 4.4.2.2 I<sup>2</sup>C master

When in I<sup>2</sup>C Master Mode, an external sensor can be connected to LSM6DSL. LSM6DSL supports I<sup>2</sup>C Master - Fast Mode only.

**Table 8. I<sup>2</sup>C master timing values**

| Symbol                | Parameter                                      | I <sup>2</sup> C Master | I <sup>2</sup> C Fast Mode (min) | Unit |
|-----------------------|------------------------------------------------|-------------------------|----------------------------------|------|
| f <sub>(SCL)</sub>    | SCL clock frequency                            | 116.3                   | 0<br>(400 kHz max)               | kHz  |
| t <sub>w(SCLL)</sub>  | SCL clock low time                             | 5.86                    | 1.3                              | μs   |
| t <sub>w(SCLH)</sub>  | SCL clock high time                            | 2.74                    | 0.6                              | ns   |
|                       | Data valid time                                | 3.9                     | -                                | μs   |
|                       | SDA hold time                                  | ≥0                      | 0                                | ns   |
|                       | SDA setup time                                 | ≥100                    | 100                              | ns   |
| t <sub>su(SR)</sub>   | Repeated START condition setup time            | 1.56                    | 0.6                              | μs   |
| t <sub>su(HD)</sub>   | Repeated START condition hold time             | 1.56                    | 0.6                              | μs   |
| t <sub>su(SP)</sub>   | STOP condition setup time                      | 2.73                    | 0.6                              | μs   |
| t <sub>w(SP:SR)</sub> | Bus free time between STOP and START condition | 21                      | 1.3                              | μs   |

## 4.5 Absolute maximum ratings

Stresses above those listed as “Absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

**Table 9. Absolute maximum ratings**

| Symbol           | Ratings                                                                           | Maximum value       | Unit     |
|------------------|-----------------------------------------------------------------------------------|---------------------|----------|
| Vdd              | Supply voltage                                                                    | -0.3 to 4.8         | V        |
| T <sub>STG</sub> | Storage temperature range                                                         | -40 to +125         | °C       |
| Sg               | Acceleration <i>g</i> for 0.2 ms                                                  | 10,000              | <i>g</i> |
| ESD              | Electrostatic discharge protection (HBM)                                          | 2                   | kV       |
| Vin              | Input voltage on any control pin<br>(including CS, SCL/SPC, SDA/SDI/SDO, SDO/SA0) | -0.3 to Vdd_IO +0.3 | V        |

**Note:** *Supply voltage on any pin should never exceed 4.8 V.*



This device is sensitive to mechanical shock, improper handling can cause permanent damage to the part.



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

## 4.6 Terminology

### 4.6.1 Sensitivity

Linear acceleration sensitivity can be determined, for example, by applying 1 *g* acceleration to the device. Because the sensor can measure DC accelerations, this can be done easily by pointing the selected axis towards the ground, noting the output value, rotating the sensor 180 degrees (pointing towards the sky) and noting the output value again. By doing so,  $\pm 1$  *g* acceleration is applied to the sensor. Subtracting the larger output value from the smaller one, and dividing the result by 2, leads to the actual sensitivity of the sensor. This value changes very little over temperature and over time. The sensitivity tolerance describes the range of sensitivities of a large number of sensors (see [Table 3](#)).

An angular rate gyroscope is a device that produces a positive-going digital output for counterclockwise rotation around the axis considered. Sensitivity describes the gain of the sensor and can be determined by applying a defined angular velocity to it. This value changes very little over temperature and time (see [Table 3](#)).

### 4.6.2 Zero-g and zero-rate level

Linear acceleration zero-*g* level offset (TyOff) describes the deviation of an actual output signal from the ideal output signal if no acceleration is present. A sensor in a steady state on a horizontal surface will measure 0 *g* on both the X-axis and Y-axis, whereas the Z-axis will measure 1 *g*. Ideally, the output is in the middle of the dynamic range of the sensor (content of OUT registers 00h, data expressed as 2's complement number). A deviation from the ideal value in this case is called zero-*g* offset.

Offset is to some extent a result of stress to MEMS sensor and therefore the offset can slightly change after mounting the sensor onto a printed circuit board or exposing it to extensive mechanical stress. Offset changes little over temperature, see "Linear acceleration zero-*g* level change vs. temperature" in [Table 3](#). The zero-*g* level tolerance (TyOff) describes the standard deviation of the range of zero-*g* levels of a group of sensors.

Zero-rate level describes the actual output signal if there is no angular rate present. The zero-rate level of precise MEMS sensors is, to some extent, a result of stress to the sensor and therefore the zero-rate level can slightly change after mounting the sensor onto a printed circuit board or after exposing it to extensive mechanical stress. This value changes very little over temperature and time (see [Table 3](#)).

## 5 Functionality

### 5.1 Operating modes

In the LSM6DSL, the accelerometer and the gyroscope can be turned on/off independently of each other and are allowed to have different ODRs and power modes.

The LSM6DSL has three operating modes available:

- only accelerometer active and gyroscope in power-down
- only gyroscope active and accelerometer in power-down
- both accelerometer and gyroscope sensors active with independent ODR

The accelerometer is activated from power-down by writing ODR\_XL[3:0] in [CTRL1\\_XL \(10h\)](#) while the gyroscope is activated from power-down by writing ODR\_G[3:0] in [CTRL2\\_G \(11h\)](#). For combo-mode the ODRs are totally independent.

### 5.2 Gyroscope power modes

In the LSM6DSL, the gyroscope can be configured in four different operating modes: power-down, low-power, normal mode and high-performance mode. The operating mode selected depends on the value of the G\_HM\_MODE bit in [CTRL7\\_G \(16h\)](#). If G\_HM\_MODE is set to '0', high-performance mode is valid for all ODRs (from 12.5 Hz up to 6.66 kHz).

To enable the low-power and normal mode, the G\_HM\_MODE bit has to be set to '1'. Low-power mode is available for lower ODRs (12.5, 26, 52 Hz) while normal mode is available for ODRs equal to 104 and 208 Hz.

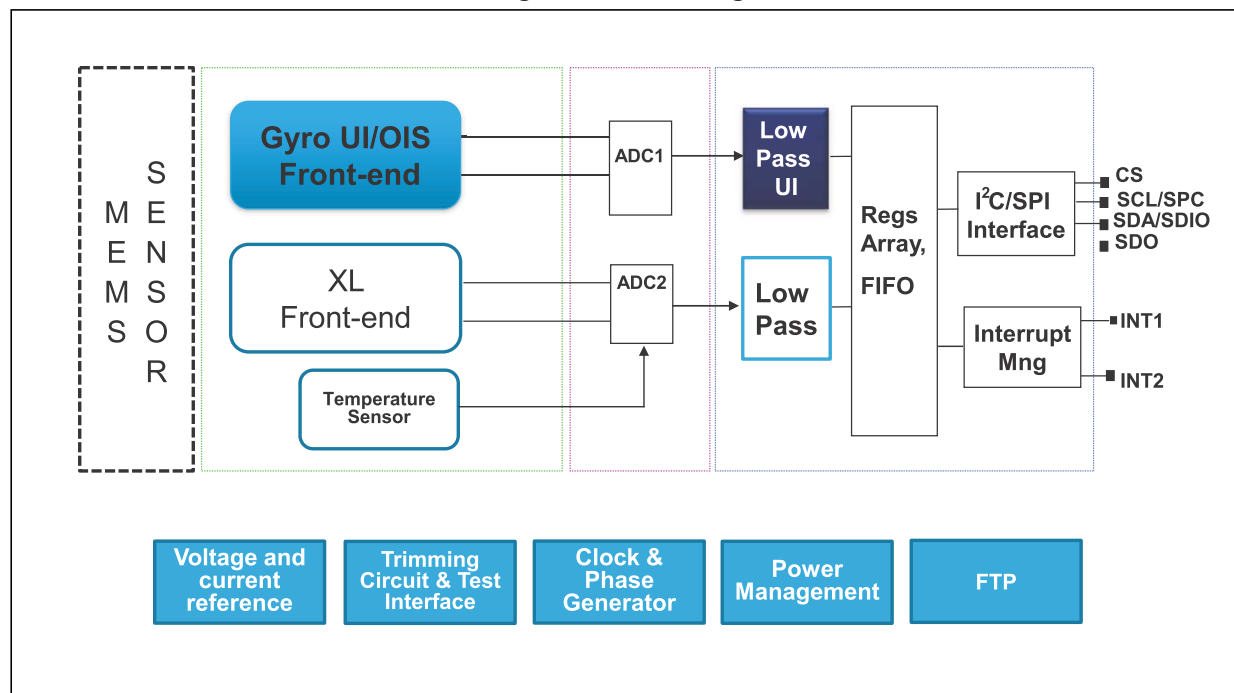
### 5.3 Accelerometer power modes

In the LSM6DSL, the accelerometer can be configured in four different operating modes: power-down, low-power, normal mode and high-performance mode. The operating mode selected depends on the value of the XL\_HM\_MODE bit in [CTRL6\\_C \(15h\)](#). If XL\_HM\_MODE is set to '0', high-performance mode is valid for all ODRs (from 12.5 Hz up to 6.66 kHz).

To enable the low-power and normal mode, the XL\_HM\_MODE bit has to be set to '1'. Low-power mode is available for lower ODRs (1.6, 12.5, 26, 52 Hz) while normal mode is available for ODRs equal to 104 and 208 Hz.

## 5.4 Block diagram of filters

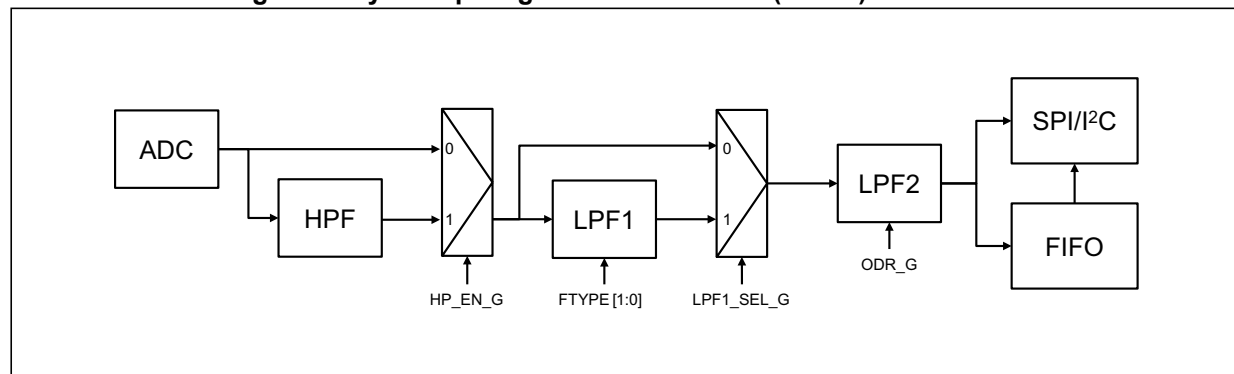
Figure 5. Block diagram of filters



### 5.4.1 Block diagrams of the gyroscope filters

The gyroscope filtering configuration for both Mode 1 (for User Interface (UI) and Electronic Image Stabilization (EIS) functionality) and Mode 2 is shown in [Figure 6](#).

Figure 6. Gyroscope digital chain - Mode 1 (UI/EIS) and Mode 2



The gyroscope ODR is selectable from 12.5 Hz up to 6.66 kHz. A low-pass filter (LPF1) is available, for more details about the filter characteristics see [Table 68: Gyroscope LPF1 bandwidth selection](#).

Data can be acquired from the output registers and FIFO over the I²C/SPI interface.

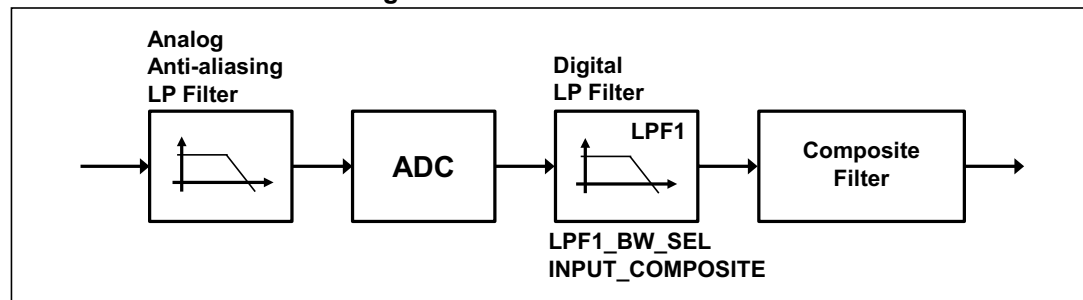
### 5.4.2 Block diagrams of the accelerometer filters

In the LSM6DSL, the filtering chain for the accelerometer part is composed of the following:

- Analog filter (anti-aliasing)
- Digital filter (LPF1)
- Composite filter

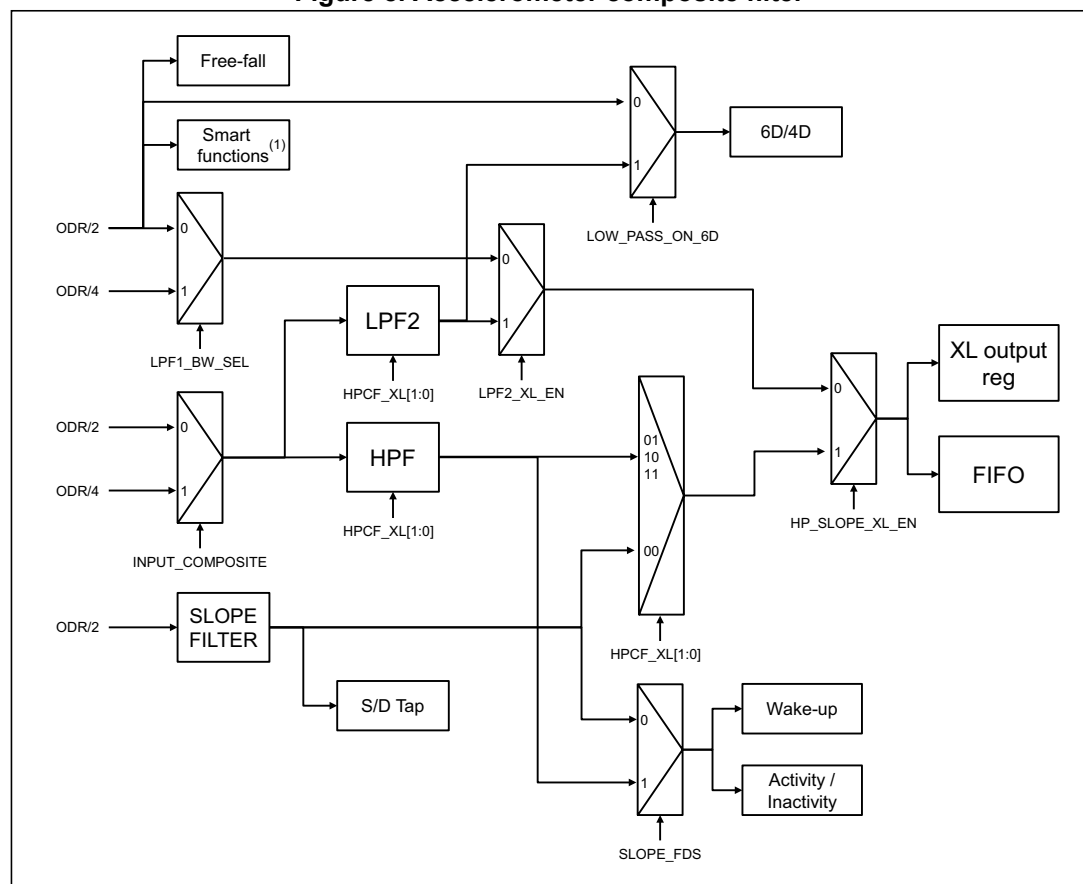
Details of the block diagram appear in the following figure.

**Figure 7. Accelerometer chain**



The configuration of the digital filter can be set using the LPF1\_BW\_SEL bit in [CTRL1\\_XL \(10h\)](#) and the INPUT\_COMPOSITE bit in [CTRL8\\_XL \(17h\)](#).

**Figure 8. Accelerometer composite filter**



1. Pedometer, step detector and step counter, significant motion and tilt functions.

## 5.5 FIFO

The presence of a FIFO allows consistent power saving for the system since the host processor does not need continuously poll data from the sensor, but it can wake up only when needed and burst the significant data out from the FIFO.

The LSM6DSL embeds 4 kbytes data FIFO to store the following data:

- gyroscope
- accelerometer
- external sensors
- step counter and timestamp
- temperature

Writing data in the FIFO can be configured to be triggered by the:

- accelerometer/gyroscope data-ready signal; in which case the ODR must be lower than or equal to both the accelerometer and gyroscope ODRs;
- sensor hub data-ready signal;
- step detection signal.

In addition, each data can be stored at a decimated data rate compared to FIFO ODR and it is configurable by the user, setting the *FIFO\_CTRL3 (08h)* and *FIFO\_CTRL4 (09h)* registers. The available decimation factors are 2, 3, 4, 8, 16, 32.

The programmable FIFO threshold can be set in *FIFO\_CTRL1 (06h)* and *FIFO\_CTRL2 (07h)* using the FTH [10:0] bits.

To monitor the FIFO status, dedicated registers *FIFO\_STATUS1 (3Ah)*, *FIFO\_STATUS2 (3Bh)*, *FIFO\_STATUS3 (3Ch)*, *FIFO\_STATUS4 (3Dh)* can be read to detect FIFO overrun events, FIFO full status, FIFO empty status, FIFO threshold status and the number of unread samples stored in the FIFO. To generate dedicated interrupts on the INT1 and INT2 pads of these status events, the configuration can be set in *INT1\_CTRL (0Dh)* and *INT2\_CTRL (0Eh)*.

The FIFO buffer can be configured according to five different modes:

- Bypass mode
- FIFO mode
- Continuous mode
- Continuous-to-FIFO mode
- Bypass-to-continuous mode

Each mode is selected by the FIFO\_MODE\_[2:0] bits in the *FIFO\_CTRL5 (0Ah)* register. To guarantee the correct acquisition of data during the switching into and out of FIFO mode, the first sample acquired must be discarded.

### 5.5.1 Bypass mode

In Bypass mode (*FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0] = 000), the FIFO is not operational and it remains empty.

Bypass mode is also used to reset the FIFO when in FIFO mode.

### 5.5.2 FIFO mode

In FIFO mode (*FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0] = 001) data from the output channels are stored in the FIFO until it is full.

To reset FIFO content, Bypass mode should be selected by writing *FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0]) to '000'. After this reset command, it is possible to restart FIFO mode by writing *FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0]) to '001'.

FIFO buffer memorizes up to 4096 samples of 16 bits each but the depth of the FIFO can be resized by setting the FTH [10:0] bits in *FIFO\_CTRL1 (06h)* and *FIFO\_CTRL2 (07h)*. If the STOP\_ON\_FTH bit in *FIFO\_CTRL4 (09h)* is set to '1', FIFO depth is limited up to FTH [10:0] bits in *FIFO\_CTRL1 (06h)* and *FIFO\_CTRL2 (07h)*.

### 5.5.3 Continuous mode

Continuous mode (*FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0] = 110) provides a continuous FIFO update: as new data arrives, the older data is discarded.

A FIFO threshold flag *FIFO\_STATUS2 (3Bh)*(FTH) is asserted when the number of unread samples in FIFO is greater than or equal to *FIFO\_CTRL1 (06h)* and *FIFO\_CTRL2 (07h)*(FTH [10:0]).

It is possible to route *FIFO\_STATUS2 (3Bh)*(FTH) to the INT1 pin by writing in register *INT1\_CTRL (0Dh)*(INT1\_FTH) = '1' or to the INT2 pin by writing in register *INT2\_CTRL (0Eh)*(INT2\_FTH) = '1'.

A full-flag interrupt can be enabled, *INT1\_CTRL (0Dh)*(INT\_FULL\_FLAG) = '1', in order to indicate FIFO saturation and eventually read its content all at once.

If an overrun occurs, at least one of the oldest samples in FIFO has been overwritten and the OVER\_RUN flag in *FIFO\_STATUS2 (3Bh)* is asserted.

In order to empty the FIFO before it is full, it is also possible to pull from FIFO the number of unread samples available in *FIFO\_STATUS1 (3Ah)* and *FIFO\_STATUS2 (3Bh)* (DIFF\_FIFO [10:0]).

### 5.5.4 Continuous-to-FIFO mode

In Continuous-to-FIFO mode *FIFO\_CTRL5 (0Ah)*(FIFO\_MODE\_[2:0] = 011), FIFO behavior changes according to the trigger event detected in one of the following interrupt registers *FUNC\_SRC1 (53h)*, *TAP\_SRC (1Ch)*, *WAKE\_UP\_SRC (1Bh)* and *D6D\_SRC (1Dh)*.

When the selected trigger bit is equal to '1', FIFO operates in FIFO mode.

When the selected trigger bit is equal to '0', FIFO operates in Continuous mode.

### 5.5.5 Bypass-to-Continuous mode

In Bypass-to-Continuous mode (*FIFO\_CTRL5 (0Ah)* (FIFO\_MODE\_[2:0] = '100'), data measurement storage inside FIFO operates in Continuous mode when selected triggers in one of the following interrupt registers *FUNC\_SRC1 (53h)*, *TAP\_SRC (1Ch)*, *WAKE\_UP\_SRC (1Bh)* and *D6D\_SRC (1Dh)* are equal to '1', otherwise FIFO content is reset (Bypass mode).

### 5.5.6 FIFO reading procedure

The data stored in FIFO are accessible from dedicated registers (*FIFO\_DATA\_OUT\_L (3Eh)* and *FIFO\_DATA\_OUT\_H (3Fh)*) and each FIFO sample is composed of 16 bits.

All FIFO status registers (*FIFO\_STATUS1 (3Ah)*, *FIFO\_STATUS2 (3Bh)*, *FIFO\_STATUS3 (3Ch)*, *FIFO\_STATUS4 (3Dh)*) can be read at the start of a reading operation, minimizing the intervention of the application processor.

Saving data in the FIFO buffer is organized in four FIFO data sets consisting of 6 bytes each:

The 1<sup>st</sup> FIFO data set is reserved for gyroscope data;

The 2<sup>nd</sup> FIFO data set is reserved for accelerometer data;

The 3<sup>rd</sup> FIFO data set is reserved for the external sensor data stored in the registers from *SENSORHUB1\_REG (2Eh)* to *SENSORHUB6\_REG (33h)*;

The 4<sup>th</sup> FIFO data set can be alternately associated to the external sensor data stored in the registers from *SENSORHUB7\_REG (34h)* to *SENSORHUB12\_REG (39h)*, to the step counter and timestamp info, or to the temperature sensor data.

## 6 Digital interfaces

### 6.1 I<sup>2</sup>C/SPI interface

The registers embedded inside the LSM6DSL may be accessed through both the I<sup>2</sup>C and SPI serial interfaces. The latter may be SW configured to operate either in 3-wire or 4-wire interface mode. The device is compatible with SPI modes 0 and 3.

The serial interfaces are mapped onto the same pins. To select/exploit the I<sup>2</sup>C interface, the CS line must be tied high (i.e connected to Vdd\_IO).

**Table 10. Serial interface pin description**

| Pin name    | Pin description                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CS          | SPI enable<br>I <sup>2</sup> C/SPI mode selection (1: SPI idle mode / I <sup>2</sup> C communication enabled;<br>0: SPI communication mode / I <sup>2</sup> C disabled) |
| SCL/SPC     | I <sup>2</sup> C Serial Clock (SCL)<br>SPI Serial Port Clock (SPC)                                                                                                      |
| SDA/SDI/SDO | I <sup>2</sup> C Serial Data (SDA)<br>SPI Serial Data Input (SDI)<br>3-wire Interface Serial Data Output (SDO)                                                          |
| SDO/SA0     | SPI Serial Data Output (SDO)<br>I <sup>2</sup> C less significant bit of the device address                                                                             |

### 6.2 Master I<sup>2</sup>C

If the LSM6DSL is configured in Mode2, a master I<sup>2</sup>C line is available. The master serial interface is mapped in the following dedicated pins.

**Table 11. Master I<sup>2</sup>C pin details**

| Pin name | Pin description                                         |
|----------|---------------------------------------------------------|
| MSCL     | I <sup>2</sup> C serial clock master                    |
| MSDA     | I <sup>2</sup> C serial data master                     |
| MDRDY    | I <sup>2</sup> C master external synchronization signal |

## 6.3 I<sup>2</sup>C serial interface

The LSM6DSL I<sup>2</sup>C is a bus slave. The I<sup>2</sup>C is employed to write the data to the registers, whose content can also be read back.

The relevant I<sup>2</sup>C terminology is provided in the table below.

**Table 12. I<sup>2</sup>C terminology**

| Term        | Description                                                                              |
|-------------|------------------------------------------------------------------------------------------|
| Transmitter | The device which sends data to the bus                                                   |
| Receiver    | The device which receives data from the bus                                              |
| Master      | The device which initiates a transfer, generates clock signals and terminates a transfer |
| Slave       | The device addressed by the master                                                       |

There are two signals associated with the I<sup>2</sup>C bus: the serial clock line (SCL) and the Serial Data line (SDA). The latter is a bidirectional line used for sending and receiving the data to/from the interface. Both the lines must be connected to Vdd\_IO through external pull-up resistors. When the bus is free, both the lines are high.

The I<sup>2</sup>C interface is implemented with fast mode (400 kHz) I<sup>2</sup>C standards as well as with the standard mode.

In order to disable the I<sup>2</sup>C block, (I2C\_disable) = 1 must be written in [CTRL4\\_C \(13h\)](#).

### 6.3.1 I<sup>2</sup>C operation

The transaction on the bus is started through a START (ST) signal. A START condition is defined as a HIGH to LOW transition on the data line while the SCL line is held HIGH. After this has been transmitted by the master, the bus is considered busy. The next byte of data transmitted after the start condition contains the address of the slave in the first 7 bits and the eighth bit tells whether the master is receiving data from the slave or transmitting data to the slave. When an address is sent, each device in the system compares the first seven bits after a start condition with its address. If they match, the device considers itself addressed by the master.

The Slave Address (SAD) associated to the LSM6DSL is 110101xb. The SDO/SA0 pin can be used to modify the less significant bit of the device address. If the SDO/SA0 pin is connected to the supply voltage, LSb is '1' (address 1101011b); else if the SDO/SA0 pin is connected to ground, the LSb value is '0' (address 1101010b). This solution permits to connect and address two different inertial modules to the same I<sup>2</sup>C bus.

Data transfer with acknowledge is mandatory. The transmitter must release the SDA line during the acknowledge pulse. The receiver must then pull the data line LOW so that it remains stable low during the HIGH period of the acknowledge clock pulse. A receiver which has been addressed is obliged to generate an acknowledge after each byte of data received.

The I<sup>2</sup>C embedded inside the LSM6DSL behaves like a slave device and the following protocol must be adhered to. After the start condition (ST) a slave address is sent, once a slave acknowledge (SAK) has been returned, an 8-bit sub-address (SUB) is transmitted. The increment of the address is configured by the [CTRL3\\_C \(12h\)](#) (IF\_INC).

The slave address is completed with a Read/Write bit. If the bit is '1' (Read), a repeated START (SR) condition must be issued after the two sub-address bytes; if the bit is '0' (Write) the master will transmit to the slave with direction unchanged. [Table 12](#) explains how the SAD+Read/Write bit pattern is composed, listing all the possible configurations.

**Table 13. SAD+Read/Write patterns**

| Command | SAD[6:1] | SAD[0] = SA0 | R/W | SAD+R/W        |
|---------|----------|--------------|-----|----------------|
| Read    | 110101   | 0            | 1   | 11010101 (D5h) |
| Write   | 110101   | 0            | 0   | 11010100 (D4h) |
| Read    | 110101   | 1            | 1   | 11010111 (D7h) |
| Write   | 110101   | 1            | 0   | 11010110 (D6h) |

**Table 14. Transfer when master is writing one byte to slave**

|        |    |         |     |     |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |    |

**Table 15. Transfer when master is writing multiple bytes to slave**

|        |    |         |     |     |     |      |     |      |     |    |
|--------|----|---------|-----|-----|-----|------|-----|------|-----|----|
| Master | ST | SAD + W |     | SUB |     | DATA |     | DATA |     | SP |
| Slave  |    |         | SAK |     | SAK |      | SAK |      | SAK |    |

**Table 16. Transfer when master is receiving (reading) one byte of data from slave**

|        |    |         |     |     |     |    |         |     |      |      |    |
|--------|----|---------|-----|-----|-----|----|---------|-----|------|------|----|
| Master | ST | SAD + W |     | SUB |     | SR | SAD + R |     |      | NMAK | SP |
| Slave  |    |         | SAK |     | SAK |    |         | SAK | DATA |      |    |

**Table 17. Transfer when master is receiving (reading) multiple bytes of data from slave**

|        |    |       |     |     |     |    |       |     |      |     |      |     |      |      |    |
|--------|----|-------|-----|-----|-----|----|-------|-----|------|-----|------|-----|------|------|----|
| Master | ST | SAD+W |     | SUB |     | SR | SAD+R |     |      | MAK |      | MAK |      | NMAK | SP |
| Slave  |    |       | SAK |     | SAK |    |       | SAK | DATA |     | DATA |     | DATA |      |    |

Data are transmitted in byte format (DATA). Each data transfer contains 8 bits. The number of bytes transferred per transfer is unlimited. Data is transferred with the Most Significant bit (MSb) first. If a receiver can't receive another complete byte of data until it has performed some other function, it can hold the clock line, SCL LOW to force the transmitter into a wait state. Data transfer only continues when the receiver is ready for another byte and releases the data line. If a slave receiver doesn't acknowledge the slave address (i.e. it is not able to receive because it is performing some real-time function) the data line must be left HIGH by the slave. The master can then abort the transfer. A LOW to HIGH transition on the SDA line while the SCL line is HIGH is defined as a STOP condition. Each data transfer must be terminated by the generation of a STOP (SP) condition.

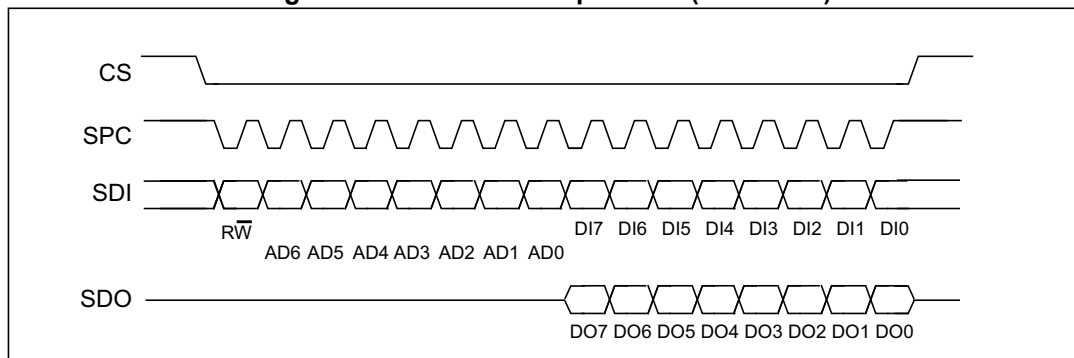
In the presented communication format MAK is Master acknowledge and NMAK is No Master Acknowledge.

## 6.4 SPI bus interface

The LSM6DSL SPI is a bus slave. The SPI allows writing and reading the registers of the device.

The serial interface communicates to the application using 4 wires: **CS**, **SPC**, **SDI** and **SDO**.

**Figure 9. Read and write protocol (in mode 3)**



**CS** is the serial port enable and it is controlled by the SPI master. It goes low at the start of the transmission and goes back high at the end. **SPC** is the serial port clock and it is controlled by the SPI master. It is stopped high when **CS** is high (no transmission). **SDI** and **SDO** are, respectively, the serial port data input and output. Those lines are driven at the falling edge of **SPC** and should be captured at the rising edge of **SPC**.

Both the read register and write register commands are completed in 16 clock pulses or in multiples of 8 in case of multiple read/write bytes. Bit duration is the time between two falling edges of **SPC**. The first bit (bit 0) starts at the first falling edge of **SPC** after the falling edge of **CS** while the last bit (bit 15, bit 23, ...) starts at the last falling edge of **SPC** just before the rising edge of **CS**.

**bit 0:**  $\overline{RW}$  bit. When 0, the data  $DI(7:0)$  is written into the device. When 1, the data  $DO(7:0)$  from the device is read. In latter case, the chip will drive **SDO** at the start of bit 8.

**bit 1-7:** address  $AD(6:0)$ . This is the address field of the indexed register.

**bit 8-15:** data  $DI(7:0)$  (write mode). This is the data that is written into the device (MSb first).

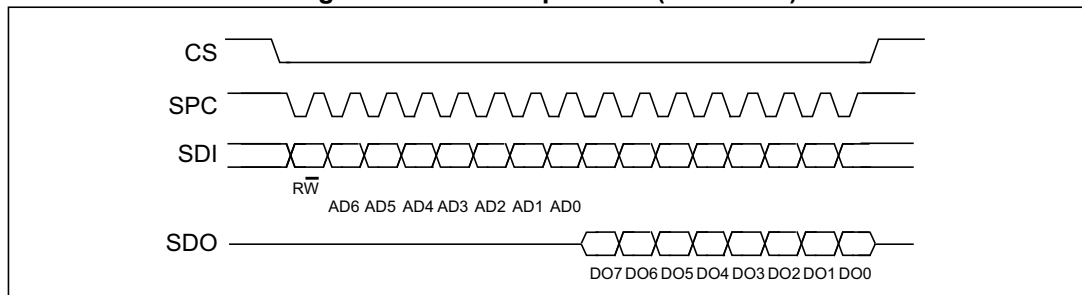
**bit 8-15:** data  $DO(7:0)$  (read mode). This is the data that is read from the device (MSb first).

In multiple read/write commands further blocks of 8 clock periods will be added. When the [CTRL3\\_C \(12h\)](#) (IF\_INC) bit is '0', the address used to read/write data remains the same for every block. When the [CTRL3\\_C \(12h\)](#) (IF\_INC) bit is '1', the address used to read/write data is increased at every block.

The function and the behavior of **SDI** and **SDO** remain unchanged.

### 6.4.1 SPI read

**Figure 10. SPI read protocol (in mode 3)**



The SPI Read command is performed with 16 clock pulses. A multiple byte read command is performed by adding blocks of 8 clock pulses to the previous one.

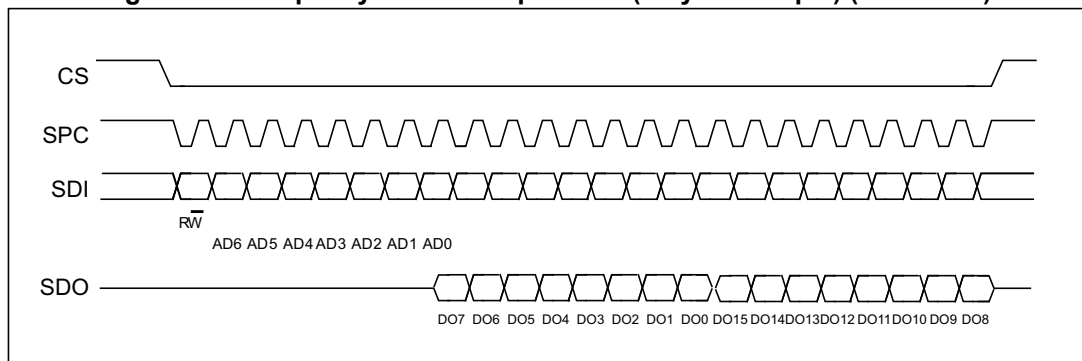
**bit 0:** READ bit. The value is 1.

**bit 1-7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DO(7:0) (read mode). This is the data that will be read from the device (MSb first).

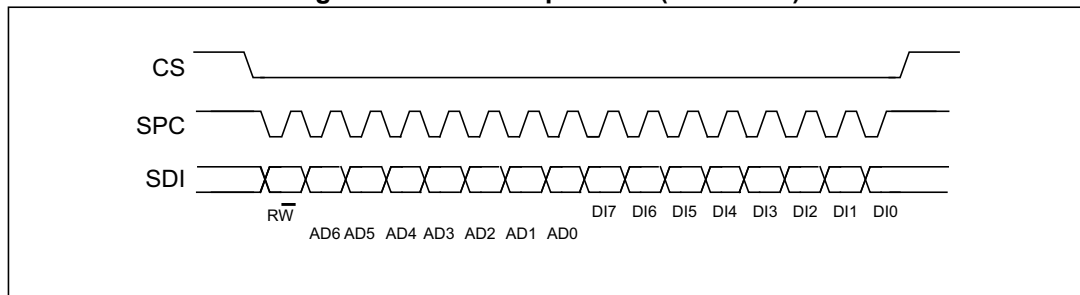
**bit 16-...:** data DO(...-8). Further data in multiple byte reads.

**Figure 11. Multiple byte SPI read protocol (2-byte example) (in mode 3)**



## 6.4.2 SPI write

Figure 12. SPI write protocol (in mode 3)



The SPI Write command is performed with 16 clock pulses. A multiple byte write command is performed by adding blocks of 8 clock pulses to the previous one.

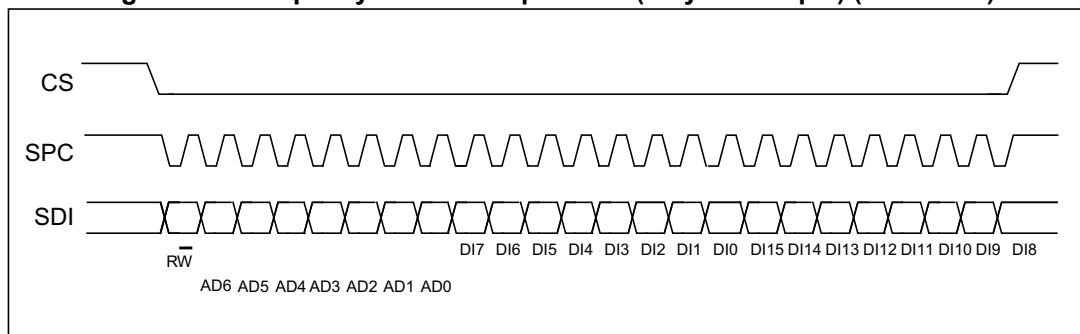
**bit 0:** WRITE bit. The value is 0.

**bit 1 -7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DI(7:0) (write mode). This is the data that is written inside the device (MSb first).

**bit 16-...** : data DI(...-8). Further data in multiple byte writes.

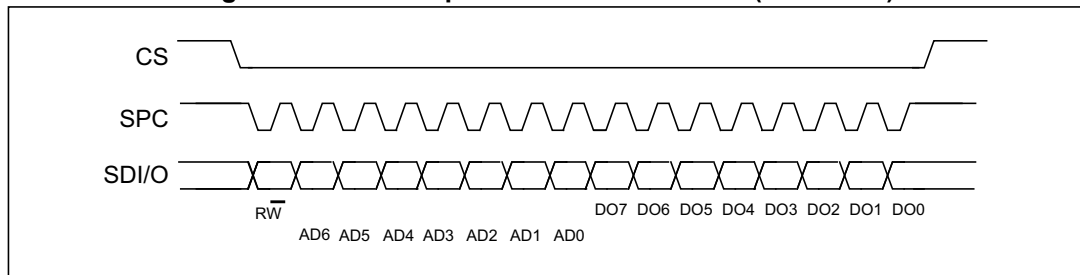
Figure 13. Multiple byte SPI write protocol (2-byte example) (in mode 3)



### 6.4.3 SPI read in 3-wire mode

A 3-wire mode is entered by setting the *CTRL3\_C (12h)* (SIM) bit equal to '1' (SPI serial interface mode selection).

**Figure 14. SPI read protocol in 3-wire mode (in mode 3)**



The SPI read command is performed with 16 clock pulses:

**bit 0:** READ bit. The value is 1.

**bit 1-7:** address AD(6:0). This is the address field of the indexed register.

**bit 8-15:** data DO(7:0) (read mode). This is the data that is read from the device (MSb first).

A multiple read command is also available in 3-wire mode.

## 7.1 LSM6DSL electrical connections in Mode 1

**Pin Configuration:**

- Pin 1:** SDO/SA0
- Pin 2:** SDx
- Pin 3:** SCx
- Pin 4:** INT1
- Pin 5:** VDDIO
- Pin 6:** GND
- Pin 7:** GND
- Pin 8:** VDD
- Pin 9:** NC<sup>(1)</sup>
- Pin 10:** NC<sup>(1)</sup>
- Pin 11:** INT2
- Pin 12:** CS
- Pin 13:** SCL
- Pin 14:** SDA

**I2C Configuration:**

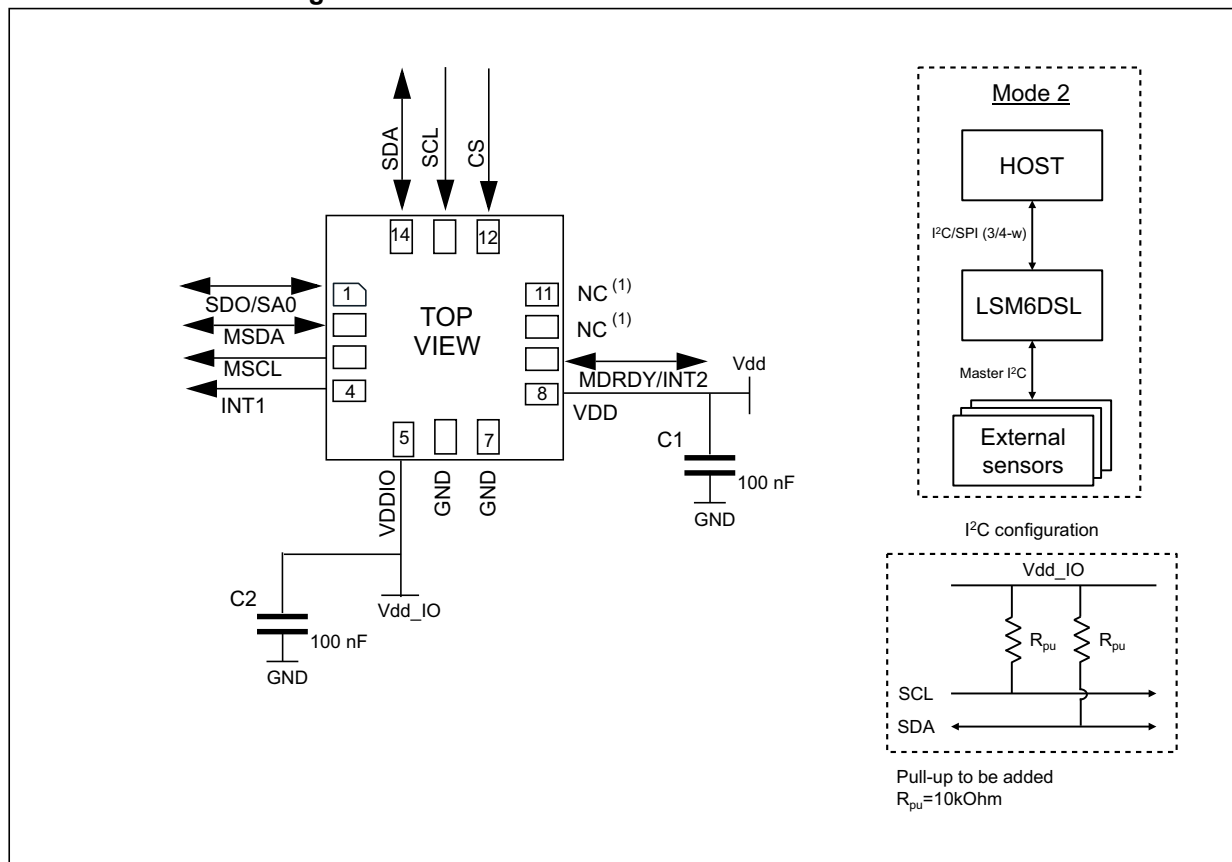
The I2C configuration shows the SCL and SDA lines connected to the Vdd\_IO supply. Pull-up resistors  $R_{pu}$  are connected to the Vdd\_IO supply. The pull-up to be added is  $R_{pu}=10k\Omega$ .

- The device core is supplied through the Vdd line. Power supply decoupling capacitors (C1, C2 = 100 nF ceramic) should be placed as near as possible to the supply pin of the device (common design practice).

The functions, the threshold and the timing of the two interrupt pins for each sensor can be completely programmed by the user through the SPI/I<sup>2</sup>C interface.

## 7.2 LSM6DSL electrical connections in Mode 2

Figure 16. LSM6DSL electrical connections in Mode 2



1. Leave pin electrically unconnected and soldered to PCB.

The device core is supplied through the Vdd line. Power supply decoupling capacitors (C1, C2 = 100 nF ceramic) should be placed as near as possible to the supply pin of the device (common design practice).

The functionality of the device and the measured acceleration/angular rate data is selectable and accessible through the SPI/I<sup>2</sup>C interface.

The functions, the threshold and the timing of the two interrupt pins for each sensor can be completely programmed by the user through the SPI/I<sup>2</sup>C interface.

**Table 18. Internal pin status**

| pin# | Name   | Mode 1 function                                                                                                                                                   | Mode 2 function                                                                                                                                                   | Pin status Mode 1                                                                                | Pin status Mode 2                                                                                |
|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1    | SDO    | SPI 4-wire interface serial data output (SDO)                                                                                                                     | SPI 4-wire interface serial data output (SDO)                                                                                                                     | Default: Input without pull-up.<br>Pull-up is enabled if bit SIM = 1<br>(SPI 3-wire) in reg 12h. | Default: Input without pull-up.<br>Pull-up is enabled if bit SIM = 1<br>(SPI 3-wire) in reg 12h. |
|      | SA0    | I <sup>2</sup> C least significant bit of the device address (SA0)                                                                                                | I <sup>2</sup> C least significant bit of the device address (SA0)                                                                                                |                                                                                                  |                                                                                                  |
| 2    | SDx    | Connect to VDDIO or GND                                                                                                                                           | I <sup>2</sup> C serial data master (MSDA)                                                                                                                        | Default: input without pull-up.<br>Pull-up is enabled if bit<br>PULL_UP_EN = 1 in reg 1Ah.       | Default: input without pull-up.<br>Pull-up is enabled if bit<br>PULL_UP_EN = 1 in reg 1Ah.       |
| 3    | SCx    | Connect to VDDIO or GND                                                                                                                                           | I <sup>2</sup> C serial clock master (MSCL)                                                                                                                       | Default: input without pull-up.<br>Pull-up is enabled if bit<br>PULL_UP_EN = 1 in reg 1Ah.       | Default: input without pull-up.<br>Pull-up is enabled if bit<br>PULL_UP_EN = 1 in reg 1Ah.       |
| 4    | INT1   | Programmable interrupt 1                                                                                                                                          | Programmable interrupt 1                                                                                                                                          | Default: Output forced to ground                                                                 | Default: Output forced to ground                                                                 |
| 5    | Vdd_IO | Power supply for I/O pins                                                                                                                                         | Power supply for I/O pins                                                                                                                                         |                                                                                                  |                                                                                                  |
| 6    | GND    | 0 V supply                                                                                                                                                        | 0 V supply                                                                                                                                                        |                                                                                                  |                                                                                                  |
| 7    | GND    | 0 V supply                                                                                                                                                        | 0 V supply                                                                                                                                                        |                                                                                                  |                                                                                                  |
| 8    | Vdd    | Power supply                                                                                                                                                      | Power supply                                                                                                                                                      |                                                                                                  |                                                                                                  |
| 9    | INT2   | Programmable interrupt 2 (INT2)<br>/ Data enabled (DEN)                                                                                                           | Programmable interrupt 2 (INT2)<br>/ Data enabled (DEN) /<br>I <sup>2</sup> C master external<br>synchronization signal (MDRDY)                                   | Default: Output forced to ground                                                                 | Default: Output forced to ground                                                                 |
| 10   | NC     | Leave unconnected <sup>(1)</sup>                                                                                                                                  | Leave unconnected <sup>(1)</sup>                                                                                                                                  | Default: Input with pull-up.<br>(See note below to disable pull-up)                              | Default: Input with pull-up.<br>(See note below to disable pull-up)                              |
| 11   | NC     | Leave unconnected <sup>(1)</sup>                                                                                                                                  | Leave unconnected <sup>(1)</sup>                                                                                                                                  | Default: Input with pull-up.<br>(See note below to disable pull-up)                              | Default: Input with pull-up.<br>(See note below to disable pull-up)                              |
| 12   | CS     | I <sup>2</sup> C/SPI mode selection<br>(1:SPI idle mode /<br>I <sup>2</sup> C communication enabled;<br>0: SPI communication mode /<br>I <sup>2</sup> C disabled) | I <sup>2</sup> C/SPI mode selection<br>(1:SPI idle mode /<br>I <sup>2</sup> C communication enabled;<br>0: SPI communication mode /<br>I <sup>2</sup> C disabled) | Default: Input with pull-up.<br>Pull-up is disabled<br>if bit I2C_disable = 1 in reg 13h.        | Default: Input with pull-up.<br>Pull-up is disabled<br>if bit I2C_disable = 1 in reg 13h.        |

**Table 18. Internal pin status (continued)**

| pin# | Name | Mode 1 function                                                                                              | Mode 2 function                                                                                              | Pin status Mode 1     | Pin status Mode 2     |
|------|------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|
| 13   | SCL  | I <sup>2</sup> C serial clock (SCL) / SPI serial port clock (SPC)                                            | I <sup>2</sup> C serial clock (SCL) / SPI serial port clock (SPC)                                            | Input without pull-up | Input without pull-up |
| 14   | SDA  | I <sup>2</sup> C serial data (SDA) / SPI serial data input (SDI) / 3-wire interface serial data output (SDO) | I <sup>2</sup> C serial data (SDA) / SPI serial data input (SDI) / 3-wire interface serial data output (SDO) | Input without pull-up | Input without pull-up |

1. Leave pin electrically unconnected and soldered to PCB.

Internal pull-up value is from 30 k $\Omega$  to 50 k $\Omega$ , depending on VDDIO.

**Note:** *The procedure to disable the pull-up on pins 10-11 is as follows:*

1. *AP side: write 80h in register at address 00h*
2. *AP side: write 01h in register at address 05h (disable the pull-up on pins 10 and 11 of LSM6DSL)*
3. *AP side: write 00h in register at address 00h*

## 8 Register mapping

The table given below provides a list of the 8/16-bit registers embedded in the device and the corresponding addresses.

**Table 19. Registers address map**

| Name                   | Type | Register address |           | Default  | Comment                                       |
|------------------------|------|------------------|-----------|----------|-----------------------------------------------|
|                        |      | Hex              | Binary    |          |                                               |
| RESERVED               | -    | 00               | 00000000  | -        | Reserved                                      |
| FUNC_CFG_ACCESS        | r/w  | 01               | 00000001  | 00000000 | Embedded functions configuration register     |
| RESERVED               | -    | 02               | 00000010  | -        | Reserved                                      |
| RESERVED               | -    | 03               | 00000011  | -        | Reserved                                      |
| SENSOR_SYNC_TIME_FRAME | r/w  | 04               | 00000100  | 00000000 | Sensor sync configuration register            |
| SENSOR_SYNC_RES_RATIO  | r/w  | 05               | 00000101  | 00000000 |                                               |
| FIFO_CTRL1             | r/w  | 06               | 00000110  | 00000000 | FIFO configuration registers                  |
| FIFO_CTRL2             | r/w  | 07               | 00000111  | 00000000 |                                               |
| FIFO_CTRL3             | r/w  | 08               | 00001000  | 00000000 |                                               |
| FIFO_CTRL4             | r/w  | 09               | 00001001  | 00000000 |                                               |
| FIFO_CTRL5             | r/w  | 0A               | 00001010  | 00000000 |                                               |
| DRDY_PULSE_CFG_G       | r/w  | 0B               | 00001011  | 00000000 |                                               |
| RESERVED               | -    | 0C               | 00001100  | -        | Reserved                                      |
| INT1_CTRL              | r/w  | 0D               | 00001101  | 00000000 | INT1 pin control                              |
| INT2_CTRL              | r/w  | 0E               | 00001110  | 00000000 | INT2 pin control                              |
| WHO_AM_I               | r    | 0F               | 00001111  | 01101010 | Who I am ID                                   |
| CTRL1_XL               | r/w  | 10               | 00010000  | 00000000 | Accelerometer and gyroscope control registers |
| CTRL2_G                | r/w  | 11               | 00010001  | 00000000 |                                               |
| CTRL3_C                | r/w  | 12               | 00010010  | 00000100 |                                               |
| CTRL4_C                | r/w  | 13               | 00010011  | 00000000 |                                               |
| CTRL5_C                | r/w  | 14               | 00010100  | 00000000 |                                               |
| CTRL6_C                | r/w  | 15               | 00010101  | 00000000 |                                               |
| CTRL7_G                | r/w  | 16               | 00010110  | 00000000 |                                               |
| CTRL8_XL               | r/w  | 17               | 0001 0111 | 00000000 |                                               |
| CTRL9_XL               | r/w  | 18               | 00011000  | 00000000 |                                               |
| CTRL10_C               | r/w  | 19               | 00011001  | 00000000 |                                               |

Table 19. Registers address map (continued)

| Name            | Type | Register address |          | Default  | Comment                                        |
|-----------------|------|------------------|----------|----------|------------------------------------------------|
|                 |      | Hex              | Binary   |          |                                                |
| MASTER_CONFIG   | r/w  | 1A               | 00011010 | 00000000 | I <sup>2</sup> C master configuration register |
| WAKE_UP_SRC     | r    | 1B               | 00011011 | output   | Interrupt registers                            |
| TAP_SRC         | r    | 1C               | 00011100 | output   |                                                |
| D6D_SRC         | r    | 1D               | 00011101 | output   |                                                |
| STATUS_REG      | r    | 1E               | 00011110 | output   | Status data register for user interface        |
| RESERVED        | -    | 1F               | 00011111 | -        |                                                |
| OUT_TEMP_L      | r    | 20               | 00100000 | output   | Temperature output data registers              |
| OUT_TEMP_H      | r    | 21               | 00100001 | output   |                                                |
| OUTX_L_G        | r    | 22               | 00100010 | output   | Gyroscope output registers for user interface  |
| OUTX_H_G        | r    | 23               | 00100011 | output   |                                                |
| OUTY_L_G        | r    | 24               | 00100100 | output   |                                                |
| OUTY_H_G        | r    | 25               | 00100101 | output   |                                                |
| OUTZ_L_G        | r    | 26               | 00100110 | output   |                                                |
| OUTZ_H_G        | r    | 27               | 00100111 | output   |                                                |
| OUTX_L_XL       | r    | 28               | 00101000 | output   | Accelerometer output registers                 |
| OUTX_H_XL       | r    | 29               | 00101001 | output   |                                                |
| OUTY_L_XL       | r    | 2A               | 00101010 | output   |                                                |
| OUTY_H_XL       | r    | 2B               | 00101011 | output   |                                                |
| OUTZ_L_XL       | r    | 2C               | 00101100 | output   |                                                |
| OUTZ_H_XL       | r    | 2D               | 00101101 | output   |                                                |
| SENSORHUB1_REG  | r    | 2E               | 00101110 | output   | Sensor hub output registers                    |
| SENSORHUB2_REG  | r    | 2F               | 00101111 | output   |                                                |
| SENSORHUB3_REG  | r    | 30               | 00110000 | output   |                                                |
| SENSORHUB4_REG  | r    | 31               | 00110001 | output   |                                                |
| SENSORHUB5_REG  | r    | 32               | 00110010 | output   |                                                |
| SENSORHUB6_REG  | r    | 33               | 00110011 | output   |                                                |
| SENSORHUB7_REG  | r    | 34               | 00110100 | output   |                                                |
| SENSORHUB8_REG  | r    | 35               | 00110101 | output   |                                                |
| SENSORHUB9_REG  | r    | 36               | 00110110 | output   |                                                |
| SENSORHUB10_REG | r    | 37               | 00110111 | output   |                                                |
| SENSORHUB11_REG | r    | 38               | 00111000 | output   |                                                |
| SENSORHUB12_REG | r    | 39               | 00111001 | output   |                                                |

Table 19. Registers address map (continued)

| Name             | Type | Register address |           | Default  | Comment                          |
|------------------|------|------------------|-----------|----------|----------------------------------|
|                  |      | Hex              | Binary    |          |                                  |
| FIFO_STATUS1     | r    | 3A               | 00111010  | output   | FIFO status registers            |
| FIFO_STATUS2     | r    | 3B               | 00111011  | output   |                                  |
| FIFO_STATUS3     | r    | 3C               | 00111100  | output   |                                  |
| FIFO_STATUS4     | r    | 3D               | 00111101  | output   |                                  |
| FIFO_DATA_OUT_L  | r    | 3E               | 00111110  | output   | FIFO data output registers       |
| FIFO_DATA_OUT_H  | r    | 3F               | 00111111  | output   |                                  |
| TIMESTAMP0_REG   | r    | 40               | 01000000  | output   | Timestamp output registers       |
| TIMESTAMP1_REG   | r    | 41               | 01000001  | output   |                                  |
| TIMESTAMP2_REG   | r/w  | 42               | 01000010  | output   |                                  |
| RESERVED         | -    | 43-48            |           | -        | Reserved                         |
| STEP_TIMESTAMP_L | r    | 49               | 0100 1001 | output   | Step counter timestamp registers |
| STEP_TIMESTAMP_H | r    | 4A               | 0100 1010 | output   |                                  |
| STEP_COUNTER_L   | r    | 4B               | 01001011  | output   | Step counter output registers    |
| STEP_COUNTER_H   | r    | 4C               | 01001100  | output   |                                  |
| SENSORHUB13_REG  | r    | 4D               | 01001101  | output   | Sensor hub output registers      |
| SENSORHUB14_REG  | r    | 4E               | 01001110  | output   |                                  |
| SENSORHUB15_REG  | r    | 4F               | 01001111  | output   |                                  |
| SENSORHUB16_REG  | r    | 50               | 01010000  | output   |                                  |
| SENSORHUB17_REG  | r    | 51               | 01010001  | output   |                                  |
| SENSORHUB18_REG  | r    | 52               | 01010010  | output   |                                  |
| FUNC_SRC1        | r    | 53               | 01010011  | output   | Interrupt registers              |
| FUNC_SRC2        | r    | 54               | 01010100  | output   |                                  |
| WRIST_TILT_IA    | r    | 55               | 01010101  | output   | Interrupt register               |
| RESERVED         | -    | 56-57            |           | -        | Reserved                         |
| TAP_CFG          | r/w  | 58               | 01011000  | 00000000 | Interrupt registers              |
| TAP_THS_6D       | r/w  | 59               | 01011001  | 00000000 |                                  |
| INT_DUR2         | r/w  | 5A               | 01011010  | 00000000 |                                  |
| WAKE_UP_THS      | r/w  | 5B               | 01011011  | 00000000 |                                  |
| WAKE_UP_DUR      | r/w  | 5C               | 01011100  | 00000000 |                                  |
| FREE_FALL        | r/w  | 5D               | 01011101  | 00000000 |                                  |
| MD1_CFG          | r/w  | 5E               | 01011110  | 00000000 |                                  |
| MD2_CFG          | r/w  | 5F               | 01011111  | 00000000 |                                  |
| MASTER_CMD_CODE  | r/w  | 60               | 01100000  | 00000000 |                                  |

Table 19. Registers address map (continued)

| Name                     | Type | Register address |           | Default  | Comment                                         |
|--------------------------|------|------------------|-----------|----------|-------------------------------------------------|
|                          |      | Hex              | Binary    |          |                                                 |
| SENS_SYNC_SPI_ERROR_CODE | r/w  | 61               | 0110 0001 | 00000000 |                                                 |
| RESERVED                 | -    | 62-65            |           | -        | Reserved                                        |
| OUT_MAG_RAW_X_L          | r    | 66               | 01100110  | output   | External magnetometer raw data output registers |
| OUT_MAG_RAW_X_H          | r    | 67               | 01100111  | output   |                                                 |
| OUT_MAG_RAW_Y_L          | r    | 68               | 01101000  | output   |                                                 |
| OUT_MAG_RAW_Y_H          | r    | 69               | 01101001  | output   |                                                 |
| OUT_MAG_RAW_Z_L          | r    | 6A               | 01101010  | output   |                                                 |
| OUT_MAG_RAW_Z_H          | r    | 6B               | 01101011  | output   |                                                 |
| RESERVED                 | -    | 6C-72            |           | -        | Reserved                                        |
| X_OFS_USR                | r/w  | 73               | 01110011  | 00000000 | Accelerometer user offset correction            |
| Y_OFS_USR                | r/w  | 74               | 01110100  | 00000000 |                                                 |
| Z_OFS_USR                | r/w  | 75               | 01110101  | 00000000 |                                                 |
| RESERVED                 | -    | 76-7F            |           | -        | Reserved                                        |

## 9 Register description

The device contains a set of registers which are used to control its behavior and to retrieve linear acceleration, angular rate and temperature data. The register addresses, made up of 7 bits, are used to identify them and to write the data through the serial interface.

### 9.1 FUNC\_CFG\_ACCESS (01h)

Enable embedded functions register (r/w).

**Table 20. FUNC\_CFG\_ACCESS register**

|             |                  |               |                  |                  |                  |                  |                  |
|-------------|------------------|---------------|------------------|------------------|------------------|------------------|------------------|
| FUNC_CFG_EN | 0 <sup>(1)</sup> | FUNC_CFG_EN_B | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|-------------|------------------|---------------|------------------|------------------|------------------|------------------|------------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 21. FUNC\_CFG\_ACCESS register description**

|               |                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| FUNC_CFG_EN   | Enable access to the embedded functions configuration registers bank A and B <sup>(1)</sup> .<br>Default value: 0. Refer to <a href="#">Table 22</a> . |
| FUNC_CFG_EN_B | Enable access to the embedded functions configuration register bank B <sup>(1)</sup> .<br>Default value: 0. Refer to <a href="#">Table 22</a> .        |

1. The embedded functions configuration registers details are available in [Section 10: Embedded functions register mapping](#), [Section 11: Embedded functions registers description - Bank A](#), and [Section 12: Embedded functions registers description - Bank B](#).

**Table 22. Configuration of embedded functions register banks**

| FUNC_CFG_EN | FUNC_CFG_EN_B | Status of embedded register banks |
|-------------|---------------|-----------------------------------|
| 0           | 0             | Bank A and B disabled (default)   |
| 0           | 1             | Forbidden                         |
| 1           | 0             | Bank A enabled                    |
| 1           | 1             | Bank B enabled                    |

### 9.2 SENSOR\_SYNC\_TIME\_FRAME (04h)

Sensor synchronization time frame register (r/w).

**Table 23. SENSOR\_SYNC\_TIME\_FRAME register**

|                  |                  |                  |                  |       |       |       |       |
|------------------|------------------|------------------|------------------|-------|-------|-------|-------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | TPH_3 | TPH_2 | TPH_1 | TPH_0 |
|------------------|------------------|------------------|------------------|-------|-------|-------|-------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 24. SENSOR\_SYNC\_TIME\_FRAME register description**

|            |                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| TPH_ [3:0] | Sensor synchronization time frame with the step of 500 ms and full range of 5 s.<br>Unsigned 8-bit.<br>Default value: 0000 0000 (sensor sync disabled) |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.3 SENSOR\_SYNC\_RES\_RATIO (05h)

Sensor synchronization resolution ratio (r/w)

**Table 25. SENSOR\_SYNC\_RES\_RATIO register**

|                  |                  |                  |                  |                  |                  |      |      |
|------------------|------------------|------------------|------------------|------------------|------------------|------|------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | RR_1 | RR_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------|------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 26. SENSOR\_SYNC\_RES\_RATIO register description**

|          |                                                            |
|----------|------------------------------------------------------------|
| RR_[1:0] | Resolution ratio of error code for sensor synchronization: |
|          | 00: SensorSync, Res_Ratio = 2-11                           |
|          | 01: SensorSync, Res_Ratio = 2-12                           |
|          | 10: SensorSync, Res_Ratio = 2-13                           |
|          | 11: SensorSync, Res_Ratio = 2-14                           |

### 9.4 FIFO\_CTRL1 (06h)

FIFO control register (r/w).

**Table 27. FIFO\_CTRL1 register**

|       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
| FTH_7 | FTH_6 | FTH_5 | FTH_4 | FTH_3 | FTH_2 | FTH_1 | FTH_0 |
|-------|-------|-------|-------|-------|-------|-------|-------|

**Table 28. FIFO\_CTRL1 register description**

|           |                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| FTH_[7:0] | FIFO threshold level setting <sup>(1)</sup> . Default value: 0000 0000.                                                             |
|           | Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level. |
|           | Minimum resolution for the FIFO is 1 LSB = 2 bytes (1 word) in FIFO                                                                 |

1. For a complete watermark threshold configuration, consider FTH\_[10:8] in [FIFO\\_CTRL2 \(07h\)](#).

## 9.5 FIFO\_CTRL2 (07h)

FIFO control register (r/w).

**Table 29. FIFO\_CTRL2 register**

|                    |                      |                  |                  |              |       |       |       |
|--------------------|----------------------|------------------|------------------|--------------|-------|-------|-------|
| TIMER_PEDO_FIF0_EN | TIMER_PEDO_FIF0_DRDY | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | FIFO_TEMP_EN | FTH10 | FTH_9 | FTH_8 |
|--------------------|----------------------|------------------|------------------|--------------|-------|-------|-------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 30. FIFO\_CTRL2 register description**

|                      |                                                                                                                                                                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TIMER_PEDO_FIF0_EN   | Enable pedometer step counter and timestamp as 4 <sup>th</sup> FIFO data set. Default: 0<br>(0: disable step counter and timestamp data as 4 <sup>th</sup> FIFO data set;<br>1: enable step counter and timestamp data as 4 <sup>th</sup> FIFO data set)                       |
| TIMER_PEDO_FIF0_DRDY | FIFO write mode <sup>(1)</sup> . Default: 0<br>(0: enable write in FIFO based on XL/Gyro data-ready;<br>1: enable write in FIFO at every step detected by step counter.)                                                                                                       |
| FIFO_TEMP_EN         | Enable the temperature data storage in FIFO. Default: 0.<br>(0: temperature not included in FIFO; 1: temperature included in FIFO)                                                                                                                                             |
| FTH_[10:8]           | FIFO threshold level setting <sup>(2)</sup> . Default value: 0000<br>Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level.<br>Minimum resolution for the FIFO is 1LSB = 2 bytes (1 word) in FIFO |

1. This bit is effective if the DATA\_VALID\_SEL\_FIF0 bit of the [MASTER\\_CONFIG \(1Ah\)](#) register is set to 0.  
2. For a complete watermark threshold configuration, consider FTH\_[7:0] in [FIFO\\_CTRL1 \(06h\)](#).

## 9.6 FIFO\_CTRL3 (08h)

FIFO control register (r/w).

**Table 31. FIFO\_CTRL3 register**

|                  |                  |                |                |                |              |              |              |
|------------------|------------------|----------------|----------------|----------------|--------------|--------------|--------------|
| 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | DEC_FIFO_GYRO2 | DEC_FIFO_GYRO1 | DEC_FIFO_GYRO0 | DEC_FIFO_XL2 | DEC_FIFO_XL1 | DEC_FIFO_XL0 |
|------------------|------------------|----------------|----------------|----------------|--------------|--------------|--------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 32. FIFO\_CTRL3 register description**

|                     |                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| DEC_FIFO_GYRO [2:0] | Gyro FIFO (first data set) decimation setting. Default: 000<br>For the configuration setting, refer to <a href="#">Table 33</a> .           |
| DEC_FIFO_XL [2:0]   | Accelerometer FIFO (second data set) decimation setting. Default: 000<br>For the configuration setting, refer to <a href="#">Table 34</a> . |

**Table 33. Gyro FIFO decimation setting**

| DEC_FIFO_GYRO [2:0] | Configuration             |
|---------------------|---------------------------|
| 000                 | Gyro sensor not in FIFO   |
| 001                 | No decimation             |
| 010                 | Decimation with factor 2  |
| 011                 | Decimation with factor 3  |
| 100                 | Decimation with factor 4  |
| 101                 | Decimation with factor 8  |
| 110                 | Decimation with factor 16 |
| 111                 | Decimation with factor 32 |

**Table 34. Accelerometer FIFO decimation setting**

| DEC_FIFO_XL [2:0] | Configuration                    |
|-------------------|----------------------------------|
| 000               | Accelerometer sensor not in FIFO |
| 001               | No decimation                    |
| 010               | Decimation with factor 2         |
| 011               | Decimation with factor 3         |
| 100               | Decimation with factor 4         |
| 101               | Decimation with factor 8         |
| 110               | Decimation with factor 16        |
| 111               | Decimation with factor 32        |

## 9.7 FIFO\_CTRL4 (09h)

FIFO control register (r/w).

**Table 35. FIFO\_CTRL4 register**

|             |                |               |               |               |               |               |               |
|-------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|
| STOP_ON_FTH | ONLY_HIGH_DATA | DEC_DS4_FIFO2 | DEC_DS4_FIFO1 | DEC_DS4_FIFO0 | DEC_DS3_FIFO2 | DEC_DS3_FIFO1 | DEC_DS3_FIFO0 |
|-------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 36. FIFO\_CTRL4 register description**

|                   |                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STOP_ON_FTH       | Enable FIFO threshold level use. Default value: 0.<br>(0: FIFO depth is not limited; 1: FIFO depth is limited to threshold level)                                              |
| ONLY_HIGH_DATA    | 8-bit data storage in FIFO. Default: 0<br>(0: disable MSByte only memorization in FIFO for XL and Gyro;<br>1: enable MSByte only memorization in FIFO for XL and Gyro in FIFO) |
| DEC_DS4_FIFO[2:0] | Fourth FIFO data set decimation setting. Default: 000<br>For the configuration setting, refer to <a href="#">Table 37</a> .                                                    |
| DEC_DS3_FIFO[2:0] | Third FIFO data set decimation setting. Default: 000<br>For the configuration setting, refer to <a href="#">Table 38</a> .                                                     |

**Table 37. Fourth FIFO data set decimation setting**

| DEC_DS4_FIFO[2:0] | Configuration                    |
|-------------------|----------------------------------|
| 000               | Fourth FIFO data set not in FIFO |
| 001               | No decimation                    |
| 010               | Decimation with factor 2         |
| 011               | Decimation with factor 3         |
| 100               | Decimation with factor 4         |
| 101               | Decimation with factor 8         |
| 110               | Decimation with factor 16        |
| 111               | Decimation with factor 32        |

**Table 38. Third FIFO data set decimation setting**

| DEC_DS3_FIFO[2:0] | Configuration                   |
|-------------------|---------------------------------|
| 000               | Third FIFO data set not in FIFO |
| 001               | No decimation                   |
| 010               | Decimation with factor 2        |
| 011               | Decimation with factor 3        |
| 100               | Decimation with factor 4        |
| 101               | Decimation with factor 8        |
| 110               | Decimation with factor 16       |
| 111               | Decimation with factor 32       |

## 9.8 FIFO\_CTRL5 (0Ah)

FIFO control register (r/w).

**Table 39. FIFO\_CTRL5 register**

|                  |            |            |            |            |             |             |             |
|------------------|------------|------------|------------|------------|-------------|-------------|-------------|
| 0 <sup>(1)</sup> | ODR_FIFO_3 | ODR_FIFO_2 | ODR_FIFO_1 | ODR_FIFO_0 | FIFO_MODE_2 | FIFO_MODE_1 | FIFO_MODE_0 |
|------------------|------------|------------|------------|------------|-------------|-------------|-------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 40. FIFO\_CTRL5 register description**

|                 |                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ODR_FIFO_[3:0]  | FIFO ODR selection, setting FIFO_MODE also. Default: 0000<br>For the configuration setting, refer to <a href="#">Table 41</a> .           |
| FIFO_MODE_[2:0] | FIFO mode selection bits, setting ODR_FIFO also. Default value: 000<br>For the configuration setting, refer to <a href="#">Table 42</a> . |

**Table 41. FIFO ODR selection**

| ODR_FIFO_[3:0] | Configuration <sup>(1)</sup> |
|----------------|------------------------------|
| 0000           | FIFO disabled                |
| 0001           | FIFO ODR is set to 12.5 Hz   |
| 0010           | FIFO ODR is set to 26 Hz     |
| 0011           | FIFO ODR is set to 52 Hz     |
| 0100           | FIFO ODR is set to 104 Hz    |
| 0101           | FIFO ODR is set to 208 Hz    |
| 0110           | FIFO ODR is set to 416 Hz    |
| 0111           | FIFO ODR is set to 833 Hz    |
| 1000           | FIFO ODR is set to 1.66 kHz  |
| 1001           | FIFO ODR is set to 3.33 kHz  |
| 1010           | FIFO ODR is set to 6.66 kHz  |

1. If the device is working at an ODR slower than the one selected, FIFO ODR is limited to that ODR value. Moreover, these bits are effective if both the DATA\_VALID\_SEL\_FIFO bit of [MASTER\\_CONFIG \(1Ah\)](#) and the TIMER\_PEDO\_FIFO\_DRDY bit of [FIFO\\_CTRL2 \(07h\)](#) are set to 0.

**Table 42. FIFO mode selection**

| FIFO_MODE_[2:0] | Configuration mode                                                             |
|-----------------|--------------------------------------------------------------------------------|
| 000             | Bypass mode. FIFO disabled.                                                    |
| 001             | FIFO mode. Stops collecting data when FIFO is full.                            |
| 010             | Reserved                                                                       |
| 011             | Continuous mode until trigger is deasserted, then FIFO mode.                   |
| 100             | Bypass mode until trigger is deasserted, then Continuous mode.                 |
| 101             | Reserved                                                                       |
| 110             | Continuous mode. If the FIFO is full, the new sample overwrites the older one. |
| 111             | Reserved                                                                       |

## 9.9 DRDY\_PULSE\_CFG\_G (0Bh)

DataReady configuration register (r/w).

**Table 43. DRDY\_PULSE\_CFG\_G register**

|             |                  |                  |                  |                  |                  |                  |                 |
|-------------|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|
| DRDY_PULSED | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | INT2_WRIST_TILT |
|-------------|------------------|------------------|------------------|------------------|------------------|------------------|-----------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 44. DRDY\_PULSE\_CFG\_G register description**

|                 |                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRDY_PULSED     | Enable pulsed DataReady mode. Default value: 0<br>(0: DataReady latched mode. Returns to 0 only after output data have been read;<br>1: DataReady pulsed mode. The DataReady pulses are 75 µs long.) |
| INT2_WRIST_TILT | Wrist tilt interrupt on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                      |

## 9.10 INT1\_CTRL (0Dh)

INT1 pad control register (r/w).

Each bit in this register enables a signal to be carried through INT1. The pad's output will supply the OR combination of the selected signals.

**Table 45. INT1\_CTRL register**

|                    |               |                |               |          |           |             |              |
|--------------------|---------------|----------------|---------------|----------|-----------|-------------|--------------|
| INT1_STEP_DETECTOR | INT1_SIGN_MOT | INT1_FULL_FLAG | INT1_FIFO_OVR | INT1_FTH | INT1_BOOT | INT1_DRDY_G | INT1_DRDY_XL |
|--------------------|---------------|----------------|---------------|----------|-----------|-------------|--------------|

**Table 46. INT1\_CTRL register description**

|                    |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| INT1_STEP_DETECTOR | Pedometer step recognition interrupt enable on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled) |
| INT1_SIGN_MOT      | Significant motion interrupt enable on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)         |
| INT1_FULL_FLAG     | FIFO full flag interrupt enable on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)             |
| INT1_FIFO_OVR      | FIFO overrun interrupt on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                      |
| INT1_FTH           | FIFO threshold interrupt on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                    |
| INT1_BOOT          | Boot status available on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                       |
| INT1_DRDY_G        | Gyroscope Data Ready on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                        |
| INT1_DRDY_XL       | Accelerometer Data Ready on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                    |

## 9.11 INT2\_CTRL (0Eh)

INT2 pad control register (r/w).

Each bit in this register enables a signal to be carried through INT2. The pad's output will supply the OR combination of the selected signals.

**Table 47. INT2\_CTRL register**

|                 |                    |                |               |          |                |             |              |
|-----------------|--------------------|----------------|---------------|----------|----------------|-------------|--------------|
| INT2_STEP_DELTA | INT2_STEP_COUNT_OV | INT2_FULL_FLAG | INT2_FIFO_OVR | INT2_FTH | INT2_DRDY_TEMP | INT2_DRDY_G | INT2_DRDY_XL |
|-----------------|--------------------|----------------|---------------|----------|----------------|-------------|--------------|

**Table 48. INT2\_CTRL register description**

|                    |                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| INT2_STEP_DELTA    | Pedometer step recognition interrupt on delta time <sup>(1)</sup> enable on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled) |
| INT2_STEP_COUNT_OV | Step counter overflow interrupt enable on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                   |
| INT2_FULL_FLAG     | FIFO full flag interrupt enable on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                          |
| INT2_FIFO_OVR      | FIFO overrun interrupt on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                   |
| INT2_FTH           | FIFO threshold interrupt on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                 |
| INT2_DRDY_TEMP     | Temperature Data Ready on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                   |
| INT2_DRDY_G        | Gyroscope Data Ready on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                     |
| INT2_DRDY_XL       | Accelerometer Data Ready on INT2 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                 |

1. Delta time value is defined in register [STEP\\_COUNT\\_DELTA \(15h\)](#).

## 9.12 WHO\_AM\_I (0Fh)

Who\_AM\_I register (r). This register is a read-only register. Its value is fixed at 6Ah.

**Table 49. WHO\_AM\_I register**

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
|---|---|---|---|---|---|---|---|

## 9.13 CTRL1\_XL (10h)

Linear acceleration sensor control register 1 (r/w).

**Table 50. CTRL1\_XL register**

|         |         |         |         |        |        |             |        |
|---------|---------|---------|---------|--------|--------|-------------|--------|
| ODR_XL3 | ODR_XL2 | ODR_XL1 | ODR_XL0 | FS_XL1 | FS_XL0 | LPF1_BW_SEL | BW0_XL |
|---------|---------|---------|---------|--------|--------|-------------|--------|

**Table 51. CTRL1\_XL register description**

|              |                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ODR_XL [3:0] | Output data rate and power mode selection. Default value: 0000 (see <a href="#">Table 52</a> ).                                      |
| FS_XL [1:0]  | Accelerometer full-scale selection. Default value: 00.<br>(00: $\pm 2$ g; 01: $\pm 16$ g; 10: $\pm 4$ g; 11: $\pm 8$ g)              |
| LPF1_BW_SEL  | Accelerometer digital LPF (LPF1) bandwidth selection. For bandwidth selection refer to <a href="#">CTRL8_XL (17h)</a> .              |
| BW0_XL       | Accelerometer analog chain bandwidth selection (only for accelerometer ODR $\geq 1.67$ kHz).<br>(0: BW @ 1.5 kHz;<br>1: BW @ 400 Hz) |

**Table 52. Accelerometer ODR register setting**

| ODR_XL3 | ODR_XL2 | ODR_XL1 | ODR_XL0 | ODR selection [Hz] when XL_HM_MODE = 1 | ODR selection [Hz] when XL_HM_MODE = 0 |
|---------|---------|---------|---------|----------------------------------------|----------------------------------------|
| 0       | 0       | 0       | 0       | Power-down                             | Power-down                             |
| 1       | 0       | 1       | 1       | 1.6 Hz (low power only)                | 12.5 Hz (high performance)             |
| 0       | 0       | 0       | 1       | 12.5 Hz (low power)                    | 12.5 Hz (high performance)             |
| 0       | 0       | 1       | 0       | 26 Hz (low power)                      | 26 Hz (high performance)               |
| 0       | 0       | 1       | 1       | 52 Hz (low power)                      | 52 Hz (high performance)               |
| 0       | 1       | 0       | 0       | 104 Hz (normal mode)                   | 104 Hz (high performance)              |
| 0       | 1       | 0       | 1       | 208 Hz (normal mode)                   | 208 Hz (high performance)              |
| 0       | 1       | 1       | 0       | 416 Hz (high performance)              | 416 Hz (high performance)              |
| 0       | 1       | 1       | 1       | 833 Hz (high performance)              | 833 Hz (high performance)              |
| 1       | 0       | 0       | 0       | 1.66 kHz (high performance)            | 1.66 kHz (high performance)            |
| 1       | 0       | 0       | 1       | 3.33 kHz (high performance)            | 3.33 kHz (high performance)            |
| 1       | 0       | 1       | 0       | 6.66 kHz (high performance)            | 6.66 kHz (high performance)            |
| 1       | 1       | x       | x       | Not allowed                            | Not allowed                            |

## 9.14 CTRL2\_G (11h)

Angular rate sensor control register 2 (r/w).

**Table 53. CTRL2\_G register**

|        |        |        |        |       |       |        |                  |
|--------|--------|--------|--------|-------|-------|--------|------------------|
| ODR_G3 | ODR_G2 | ODR_G1 | ODR_G0 | FS_G1 | FS_G0 | FS_125 | 0 <sup>(1)</sup> |
|--------|--------|--------|--------|-------|-------|--------|------------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 54. CTRL2\_G register description**

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| ODR_G [3:0] | Gyroscope output data rate selection. Default value: 0000<br>(Refer to <a href="#">Table 55</a> )           |
| FS_G [1:0]  | Gyroscope full-scale selection. Default value: 00<br>(00: 250 dps; 01: 500 dps; 10: 1000 dps; 11: 2000 dps) |
| FS_125      | Gyroscope full-scale at 125 dps. Default value: 0<br>(0: disabled; 1: enabled)                              |

**Table 55. Gyroscope ODR configuration setting**

| ODR_G3 | ODR_G2 | ODR_G1 | ODR_G0 | ODR [Hz] when<br>G_HM_MODE = 1 | ODR [Hz] when<br>G_HM_MODE = 0 |
|--------|--------|--------|--------|--------------------------------|--------------------------------|
| 0      | 0      | 0      | 0      | Power down                     | Power down                     |
| 0      | 0      | 0      | 1      | 12.5 Hz (low power)            | 12.5 Hz (high performance)     |
| 0      | 0      | 1      | 0      | 26 Hz (low power)              | 26 Hz (high performance)       |
| 0      | 0      | 1      | 1      | 52 Hz (low power)              | 52 Hz (high performance)       |
| 0      | 1      | 0      | 0      | 104 Hz (normal mode)           | 104 Hz (high performance)      |
| 0      | 1      | 0      | 1      | 208 Hz (normal mode)           | 208 Hz (high performance)      |
| 0      | 1      | 1      | 0      | 416 Hz (high performance)      | 416 Hz (high performance)      |
| 0      | 1      | 1      | 1      | 833 Hz (high performance)      | 833 Hz (high performance)      |
| 1      | 0      | 0      | 0      | 1.66 kHz (high performance)    | 1.66 kHz (high performance)    |
| 1      | 0      | 0      | 1      | 3.33 kHz (high performance)    | 3.33 kHz (high performance)    |
| 1      | 0      | 1      | 0      | 6.66 kHz (high performance)    | 6.66 kHz (high performance)    |
| 1      | 0      | 1      | 1      | Not available                  | Not available                  |

## 9.15 CTRL3\_C (12h)

Control register 3 (r/w).

**Table 56. CTRL3\_C register**

|      |     |           |       |     |        |     |          |
|------|-----|-----------|-------|-----|--------|-----|----------|
| BOOT | BDU | H_LACTIVE | PP_OD | SIM | IF_INC | BLE | SW_RESET |
|------|-----|-----------|-------|-----|--------|-----|----------|

**Table 57. CTRL3\_C register description**

|           |                                                                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOOT      | Reboot memory content. Default value: 0<br>(0: normal mode; 1: reboot memory content)                                                                                     |
| BDU       | Block Data Update. Default value: 0<br>(0: continuous update; 1: output registers not updated until MSB and LSB have been read)                                           |
| H_LACTIVE | Interrupt activation level. Default value: 0<br>(0: interrupt output pads active high; 1: interrupt output pads active low)                                               |
| PP_OD     | Push-pull/open-drain selection on INT1 and INT2 pads. Default value: 0<br>(0: push-pull mode; 1: open-drain mode)                                                         |
| SIM       | SPI Serial Interface Mode selection. Default value: 0<br>(0: 4-wire interface; 1: 3-wire interface).                                                                      |
| IF_INC    | Register address automatically incremented during a multiple byte access with a serial interface (I <sup>2</sup> C or SPI). Default value: 1<br>(0: disabled; 1: enabled) |
| BLE       | Big/Little Endian Data selection. Default value 0<br>(0: data LSB @ lower address; 1: data MSB @ lower address)                                                           |
| SW_RESET  | Software reset. Default value: 0<br>(0: normal mode; 1: reset device)<br>This bit is automatically cleared.                                                               |

## 9.16 CTRL4\_C (13h)

Control register 4 (r/w).

**Table 58. CTRL4\_C register**

|           |       |              |               |           |             |            |                  |
|-----------|-------|--------------|---------------|-----------|-------------|------------|------------------|
| DEN_XL_EN | SLEEP | INT2_on_INT1 | DEN_DRDY_INT1 | DRDY_MASK | I2C_disable | LPF1_SEL_G | 0 <sup>(1)</sup> |
|-----------|-------|--------------|---------------|-----------|-------------|------------|------------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 59. CTRL4\_C register description**

|               |                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEN_XL_EN     | Extend DEN functionality to accelerometer sensor. Default value: 0<br>(0: disabled; 1: enabled)                                                                                        |
| SLEEP         | Gyroscope sleep mode enable. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                             |
| DEN_DRDY_INT1 | DEN DRDY signal on INT1 pad. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                             |
| INT2_on_INT1  | All interrupt signals available on INT1 pad enable. Default value: 0<br>(0: interrupt signals divided between INT1 and INT2 pads;<br>1: all interrupt signals in logic or on INT1 pad) |
| DRDY_MASK     | Configuration 1 data available enable bit. Default value: 0<br>(0: DA timer disabled; 1: DA timer enabled)                                                                             |
| I2C_disable   | Disable I <sup>2</sup> C interface. Default value: 0<br>(0: both I <sup>2</sup> C and SPI enabled; 1: I <sup>2</sup> C disabled, SPI only enabled)                                     |
| LPF1_SEL_G    | Enable gyroscope digital LPF1. The bandwidth can be selected through FTYPE[1:0] in <a href="#">CTRL6_C (15h)</a> .<br>(0: disabled; 1: enabled)                                        |

## 9.17 CTRL5\_C (14h)

Control register 5 (r/w).

**Table 60. CTRL5\_C register**

|           |           |           |        |       |       |        |        |
|-----------|-----------|-----------|--------|-------|-------|--------|--------|
| ROUNDING2 | ROUNDING1 | ROUNDING0 | DEN_LH | ST1_G | ST0_G | ST1_XL | ST0_XL |
|-----------|-----------|-----------|--------|-------|-------|--------|--------|

**Table 61. CTRL5\_C register description**

|               |                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROUNDING[2:0] | Circular burst-mode (rounding) read from the output registers.<br>Default value: 000<br>(000: no rounding; Others: refer to <a href="#">Table 62</a> ) |
| DEN_LH        | DEN active level configuration. Default value: 0<br>(0: active low; 1: active high)                                                                    |
| ST_G [1:0]    | Angular rate sensor self-test enable. Default value: 00<br>(00: Self-test disabled; Other: refer to <a href="#">Table 63</a> )                         |
| ST_XL [1:0]   | Linear acceleration sensor self-test enable. Default value: 00<br>(00: Self-test disabled; Other: refer to <a href="#">Table 64</a> )                  |

Table 62. Output registers rounding pattern

| ROUNDING[2:0] | Rounding pattern                                                                                                                                                                                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 000           | No rounding                                                                                                                                                                                                              |
| 001           | Accelerometer only                                                                                                                                                                                                       |
| 010           | Gyroscope only                                                                                                                                                                                                           |
| 011           | Gyroscope + accelerometer                                                                                                                                                                                                |
| 100           | Registers from <a href="#">SENSORHUB1_REG (2Eh)</a> to <a href="#">SENSORHUB6_REG (33h)</a> only                                                                                                                         |
| 101           | Accelerometer + registers from <a href="#">SENSORHUB1_REG (2Eh)</a> to <a href="#">SENSORHUB6_REG (33h)</a>                                                                                                              |
| 110           | Gyroscope + accelerometer + registers from <a href="#">SENSORHUB1_REG (2Eh)</a> to <a href="#">SENSORHUB6_REG (33h)</a> and registers from <a href="#">SENSORHUB7_REG (34h)</a> to <a href="#">SENSORHUB12_REG (39h)</a> |
| 111           | Gyroscope + accelerometer + registers from <a href="#">SENSORHUB1_REG (2Eh)</a> to <a href="#">SENSORHUB6_REG (33h)</a>                                                                                                  |

Table 63. Angular rate sensor self-test mode selection

| ST1_G | ST0_G | Self-test mode          |
|-------|-------|-------------------------|
| 0     | 0     | Normal mode             |
| 0     | 1     | Positive sign self-test |
| 1     | 0     | Not allowed             |
| 1     | 1     | Negative sign self-test |

Table 64. Linear acceleration sensor self-test mode selection

| ST1_XL | ST0_XL | Self-test mode          |
|--------|--------|-------------------------|
| 0      | 0      | Normal mode             |
| 0      | 1      | Positive sign self-test |
| 1      | 0      | Negative sign self-test |
| 1      | 1      | Not allowed             |

## 9.18 CTRL6\_C (15h)

Angular rate sensor control register 6 (r/w).

**Table 65. CTRL6\_C register**

|         |        |         |            |           |                  |         |         |
|---------|--------|---------|------------|-----------|------------------|---------|---------|
| TRIG_EN | LVL_EN | LVL2_EN | XL_HM_MODE | USR_OFF_W | 0 <sup>(1)</sup> | FTYPE_1 | FTYPE_0 |
|---------|--------|---------|------------|-----------|------------------|---------|---------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 66. CTRL6\_C register description**

|            |                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRIG_EN    | DEN data edge-sensitive trigger enable. Refer to <a href="#">Table 67</a> .                                                                                                                    |
| LVL_EN     | DEN data level-sensitive trigger enable. Refer to <a href="#">Table 67</a> .                                                                                                                   |
| LVL2_EN    | DEN level-sensitive latched enable. Refer to <a href="#">Table 67</a> .                                                                                                                        |
| XL_HM_MODE | High-performance operating mode disable for accelerometer. Default value: 0<br>(0: high-performance operating mode enabled;<br>1: high-performance operating mode disabled)                    |
| USR_OFF_W  | Weight of XL user offset bits of registers <a href="#">X_OFS_USR (73h)</a> , <a href="#">Y_OFS_USR (74h)</a> ,<br><a href="#">Z_OFS_USR (75h)</a><br>0 = $2^{-10}$ g/LSB<br>1 = $2^{-6}$ g/LSB |
| FTYPE[1:0] | Gyroscope's low-pass filter (LPF1) bandwidth selection<br><a href="#">Table 68</a> shows the selectable bandwidth values.                                                                      |

**Table 67. Trigger mode selection**

| TRIG_EN, LVL1_EN, LVL2_EN | Trigger mode                                 |
|---------------------------|----------------------------------------------|
| 100                       | Edge-sensitive trigger mode is selected      |
| 010                       | Level-sensitive trigger mode is selected     |
| 011                       | Level-sensitive latched mode is selected     |
| 110                       | Level-sensitive FIFO enable mode is selected |

**Table 68. Gyroscope LPF1 bandwidth selection**

| FTYPE[1:0] | Bandwidth    |               |               |               |
|------------|--------------|---------------|---------------|---------------|
|            | ODR = 800 Hz | ODR = 1.6 kHz | ODR = 3.3 kHz | ODR = 6.6 kHz |
| 00         | 245 Hz       | 315 Hz        | 343 Hz        | 351 Hz        |
| 01         | 195 Hz       | 224 Hz        | 234 Hz        | 237 Hz        |
| 10         | 155 Hz       | 168 Hz        | 172 Hz        | 173 Hz        |
| 11         | 293 Hz       | 505 Hz        | 925 Hz        | 937 Hz        |

## 9.19 CTRL7\_G (16h)

Angular rate sensor control register 7 (r/w).

**Table 69. CTRL7\_G register**

|           |         |        |        |                  |                 |                  |                  |
|-----------|---------|--------|--------|------------------|-----------------|------------------|------------------|
| G_HM_MODE | HP_EN_G | HPM1_G | HPM0_G | 0 <sup>(1)</sup> | ROUNDING_STATUS | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|-----------|---------|--------|--------|------------------|-----------------|------------------|------------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 70. CTRL7\_G register description**

|                 |                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G_HM_MODE       | High-performance operating mode disable for gyroscope(1). Default: 0<br>(0: high-performance operating mode enabled;<br>1: high-performance operating mode disabled)                           |
| HP_EN_G         | Gyroscope digital high-pass filter enable. The filter is enabled only if the gyro is in HP mode. Default value: 0<br>(0: HPF disabled; 1: HPF enabled)                                         |
| HPM_G[1:0]      | Gyroscope digital HP filter cutoff selection. Default: 00<br>(00 = 16 mHz<br>01 = 65 mHz<br>10 = 260 mHz<br>11 = 1.04 Hz)                                                                      |
| ROUNDING_STATUS | Source register rounding function on WAKE_UP_SRC (1Bh), TAP_SRC (1Ch), D6D_SRC (1Dh), STATUS_REG (1Eh), and FUNC_SRC (53h).<br>Default value: 0<br>(0: Rounding disabled; 1: Rounding enabled) |

## 9.20 CTRL8\_XL (17h)

Linear acceleration sensor control register 8 (r/w).

**Table 71. CTRL8\_XL register**

|            |          |          |             |                 |                |                  |                |
|------------|----------|----------|-------------|-----------------|----------------|------------------|----------------|
| LPF2_XL_EN | HPCF_XL1 | HPCF_XL0 | HP_REF_MODE | INPUT_COMPOSITE | HP_SLOPE_XL_EN | 0 <sup>(1)</sup> | LOW_PASS_ON_6D |
|------------|----------|----------|-------------|-----------------|----------------|------------------|----------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 72. CTRL8\_XL register description**

|                 |                                                                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LPF2_XL_EN      | Accelerometer low-pass filter LPF2 selection. Refer to <a href="#">Figure 8</a> .                                                                                      |
| HPCF_XL[1:0]    | Accelerometer LPF2 and high-pass filter configuration and cutoff setting. Refer to <a href="#">Table 73</a> .                                                          |
| HP_REF_MODE     | Enable HP filter reference mode. Default value: 0<br>(0: disabled; 1: enabled <sup>(1)</sup> )                                                                         |
| INPUT_COMPOSITE | Composite filter input selection. Default: 0<br>(0: ODR/2 low pass filtered sent to composite filter (default)<br>1: ODR/4 low pass filtered sent to composite filter) |
| HP_SLOPE_XL_EN  | Accelerometer slope filter / high-pass filter selection. Refer to <a href="#">Figure 8</a> .                                                                           |
| LOW_PASS_ON_6D  | LPF2 on 6D function selection. Refer to <a href="#">Figure 8</a> .                                                                                                     |

1. When enabled, the first output data has to be discarded.

Table 73. Accelerometer bandwidth selection

| HP_SLOPE_XL_EN                       | LPF2_XL_EN | LPF1_BW_SEL | HPCF_XL[1:0] | INPUT_COMPOSITE                  | Bandwidth |
|--------------------------------------|------------|-------------|--------------|----------------------------------|-----------|
| 0<br>(low-pass path) <sup>(1)</sup>  | 0          | 0           | -            | -                                | ODR/2     |
|                                      |            | 1           | -            | -                                | ODR/4     |
|                                      | 1          | -           | 00           | 1 (low noise)<br>0 (low latency) | ODR/50    |
|                                      |            |             | 01           |                                  | ODR/100   |
|                                      |            |             | 10           |                                  | ODR/9     |
|                                      |            |             | 11           |                                  | ODR/400   |
| 1<br>(high-pass path) <sup>(2)</sup> | -          | -           | 00           | 0                                | ODR/4     |
|                                      |            |             | 01           |                                  | ODR/100   |
|                                      |            |             | 10           |                                  | ODR/9     |
|                                      |            |             | 11           |                                  | ODR/400   |

1. The bandwidth column is related to LPF1 if LPF2\_XL\_EN = 0 or to LPF2 if LPF2\_XL\_EN = 1.

2. The bandwidth column is related to the slope filter if HPCF\_XL[1:0] = 00 or to the HP filter if HPCF\_XL[1:0] = 01/10/11.

## 9.21 CTRL9\_XL (18h)

Linear acceleration sensor control register 9 (r/w).

Table 74. CTRL9\_XL register

| DEN_X | DEN_Y | DEN_Z | DEN_XL_G | 0 <sup>(1)</sup> | SOFT_EN | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> |
|-------|-------|-------|----------|------------------|---------|------------------|------------------|
|-------|-------|-------|----------|------------------|---------|------------------|------------------|

1. This bit must be set to '0' for the correct operation of the device.

Table 75. CTRL9\_XL register description

|          |                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEN_X    | DEN value stored in LSB of X-axis. Default value: 1<br>(0: DEN not stored in X-axis LSB; 1: DEN stored in X-axis LSB)                                                                                  |
| DEN_Y    | DEN value stored in LSB of Y-axis. Default value: 1<br>(0: DEN not stored in Y-axis LSB; 1: DEN stored in Y-axis LSB)                                                                                  |
| DEN_Z    | DEN value stored in LSB of Z-axis. Default value: 1<br>(0: DEN not stored in Z-axis LSB; 1: DEN stored in Z-axis LSB)                                                                                  |
| DEN_XL_G | DEN stamping sensor selection. Default value: 0<br>(0: DEN pin info stamped in the gyroscope axis selected by bits [7:5];<br>1: DEN pin info stamped in the accelerometer axis selected by bits [7:5]) |
| SOFT_EN  | Enable soft-iron correction algorithm for magnetometer <sup>(1)</sup> . Default value: 0<br>(0: soft-iron correction algorithm disabled;<br>1: soft-iron correction algorithm enabled)                 |

1. This bit is effective if the IRON\_EN bit of [MASTER\\_CONFIG \(1Ah\)](#) and the FUNC\_EN bit of [CTRL10\\_C \(19h\)](#) are set to 1.

## 9.22 CTRL10\_C (19h)

Control register 10 (r/w).

**Table 76. CTRL10\_C register**

|               |                  |          |         |         |         |               |                |
|---------------|------------------|----------|---------|---------|---------|---------------|----------------|
| WRIST_TILT_EN | 0 <sup>(1)</sup> | TIMER_EN | PEDO_EN | TILT_EN | FUNC_EN | PEDO_RST_STEP | SIGN_MOTION_EN |
|---------------|------------------|----------|---------|---------|---------|---------------|----------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 77. CTRL10\_C register description**

|                |                                                                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WRIST_TILT_EN  | Enable wrist tilt algorithm <sup>(1)(2)</sup> . Default value: 0<br>(0: wrist tilt algorithm disabled;<br>1: wrist tilt algorithm enabled)                                                                                                                                              |
| TIMER_EN       | Enable timestamp count. The count is saved in <a href="#">TIMESTAMP0_REG (40h)</a> , <a href="#">TIMESTAMP1_REG (41h)</a> and <a href="#">TIMESTAMP2_REG (42h)</a> . Default: 0<br>(0: timestamp count disabled; 1: timestamp count enabled)                                            |
| PEDO_EN        | Enable pedometer algorithm <sup>(1)</sup> . Default value: 0<br>(0: pedometer algorithm disabled;<br>1: pedometer algorithm enabled)                                                                                                                                                    |
| TILT_EN        | Enable tilt calculation <sup>(1)</sup> .                                                                                                                                                                                                                                                |
| FUNC_EN        | Enable embedded functionalities (pedometer, tilt, significant motion detection, sensor hub and ironing). Default value: 0<br>(0: disable functionalities of embedded functions and accelerometer filters;<br>1: enable functionalities of embedded functions and accelerometer filters) |
| PEDO_RST_STEP  | Reset pedometer step counter. Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                                                             |
| SIGN_MOTION_EN | Enable significant motion detection function <sup>(1)</sup> . Default value: 0<br>(0: disabled; 1: enabled)                                                                                                                                                                             |

1. This is effective if the FUNC\_EN bit is set to '1'.
2. By default, the wrist tilt algorithm is applied to the positive X-axis.

## 9.23 MASTER\_CONFIG (1Ah)

Master configuration register (r/w).

**Table 78. MASTER\_CONFIG register**

|              |                     |                  |              |            |                   |         |           |
|--------------|---------------------|------------------|--------------|------------|-------------------|---------|-----------|
| DRDY_ON_INT1 | DATA_VALID_SEL_FIFO | 0 <sup>(1)</sup> | START_CONFIG | PULL_UP_EN | PASS_THROUGH_MODE | IRON_EN | MASTER_ON |
|--------------|---------------------|------------------|--------------|------------|-------------------|---------|-----------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 79. MASTER\_CONFIG register description**

|                     |                                                                                                                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRDY_ON_INT1        | Manage the Master DRDY signal on INT1 pad. Default: 0<br>(0: disable Master DRDY on INT1; 1: enable Master DRDY on INT1)                                                                                                                                |
| DATA_VALID_SEL_FIFO | Selection of FIFO data-valid signal. Default value: 0<br>(0: data-valid signal used to write data in FIFO is the XL/Gyro data-ready or step detection <sup>(1)</sup> ;<br>1: data-valid signal used to write data in FIFO is the sensor hub data-ready) |
| START_CONFIG        | Sensor Hub trigger signal selection. Default value: 0<br>(0: Sensor hub signal is the XL/Gyro data-ready;<br>1: Sensor hub signal external from INT2 pad.)                                                                                              |
| PULL_UP_EN          | Auxiliary I <sup>2</sup> C pull-up. Default value: 0<br>(0: internal pull-up on auxiliary I <sup>2</sup> C line disabled;<br>1: internal pull-up on auxiliary I <sup>2</sup> C line enabled)                                                            |
| PASS_THROUGH_MODE   | I <sup>2</sup> C interface pass-through. Default value: 0<br>(0: pass-through disabled; 1: pass-through enabled)                                                                                                                                        |
| IRON_EN             | Enable hard-iron correction algorithm for magnetometer <sup>(2)</sup> . Default value: 0<br>(0: hard-iron correction algorithm disabled;<br>1: hard-iron correction algorithm enabled)                                                                  |
| MASTER_ON           | Sensor hub I <sup>2</sup> C master enable <sup>(2)</sup> . Default: 0<br>(0: master I <sup>2</sup> C of sensor hub disabled; 1: master I <sup>2</sup> C of sensor hub enabled)                                                                          |

1. If the TIMER\_PEDO\_FIFO\_DRDY bit in [FIFO\\_CTRL2 \(07h\)](#) is set to 0, the trigger for writing data in FIFO is XL/Gyro data-ready, otherwise it's the step detection.

2. This is effective if the FUNC\_EN bit is set to '1'.

## 9.24 WAKE\_UP\_SRC (1Bh)

Wake up interrupt source register (r).

**Table 80. WAKE\_UP\_SRC register**

|   |   |       |                |       |      |      |      |
|---|---|-------|----------------|-------|------|------|------|
| 0 | 0 | FF_IA | SLEEP_STATE_IA | WU_IA | X_WU | Y_WU | Z_WU |
|---|---|-------|----------------|-------|------|------|------|

**Table 81. WAKE\_UP\_SRC register description**

|                |                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| FF_IA          | Free-fall event detection status. Default: 0<br>(0: free-fall event not detected; 1: free-fall event detected)                            |
| SLEEP_STATE_IA | Sleep event status. Default value: 0<br>(0: sleep event not detected; 1: sleep event detected)                                            |
| WU_IA          | Wakeup event detection status. Default value: 0<br>(0: wakeup event not detected; 1: wakeup event detected.)                              |
| X_WU           | Wakeup event detection status on X-axis. Default value: 0<br>(0: wakeup event on X-axis not detected; 1: wakeup event on X-axis detected) |
| Y_WU           | Wakeup event detection status on Y-axis. Default value: 0<br>(0: wakeup event on Y-axis not detected; 1: wakeup event on Y-axis detected) |
| Z_WU           | Wakeup event detection status on Z-axis. Default value: 0<br>(0: wakeup event on Z-axis not detected; 1: wakeup event on Z-axis detected) |

## 9.25 TAP\_SRC (1Ch)

Tap source register (r).

**Table 82. TAP\_SRC register**

|   |        |            |            |          |       |       |       |
|---|--------|------------|------------|----------|-------|-------|-------|
| 0 | TAP_IA | SINGLE_TAP | DOUBLE_TAP | TAP_SIGN | X_TAP | Y_TAP | Z_TAP |
|---|--------|------------|------------|----------|-------|-------|-------|

**Table 83. TAP\_SRC register description**

|            |                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAP_IA     | Tap event detection status. Default: 0<br>(0: tap event not detected; 1: tap event detected)                                                                                  |
| SINGLE_TAP | Single-tap event status. Default value: 0<br>(0: single tap event not detected; 1: single tap event detected)                                                                 |
| DOUBLE_TAP | Double-tap event detection status. Default value: 0<br>(0: double-tap event not detected; 1: double-tap event detected.)                                                      |
| TAP_SIGN   | Sign of acceleration detected by tap event. Default: 0<br>(0: positive sign of acceleration detected by tap event;<br>1: negative sign of acceleration detected by tap event) |
| X_TAP      | Tap event detection status on X-axis. Default value: 0<br>(0: tap event on X-axis not detected; 1: tap event on X-axis detected)                                              |
| Y_TAP      | Tap event detection status on Y-axis. Default value: 0<br>(0: tap event on Y-axis not detected; 1: tap event on Y-axis detected)                                              |
| Z_TAP      | Tap event detection status on Z-axis. Default value: 0<br>(0: tap event on Z-axis not detected; 1: tap event on Z-axis detected)                                              |

## 9.26 D6D\_SRC (1Dh)

Portrait, landscape, face-up and face-down source register (r)

**Table 84. D6D\_SRC register**

|          |        |    |    |    |    |    |    |
|----------|--------|----|----|----|----|----|----|
| DEN_DRDY | D6D_IA | ZH | ZL | YH | YL | XH | XL |
|----------|--------|----|----|----|----|----|----|

**Table 85. D6D\_SRC register description**

|          |                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEN_DRDY | DEN data-ready signal. It is set high when data output is related to the data coming from a DEN active condition. <sup>(1)</sup>                              |
| D6D_IA   | Interrupt active for change position portrait, landscape, face-up, face-down. Default value: 0 (0: change position not detected; 1: change position detected) |
| ZH       | Z-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over threshold) detected)                                              |
| ZL       | Z-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)                                             |
| YH       | Y-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over-threshold) detected)                                              |
| YL       | Y-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)                                             |
| XH       | X-axis high event (over threshold). Default value: 0 (0: event not detected; 1: event (over threshold) detected)                                              |
| XL       | X-axis low event (under threshold). Default value: 0 (0: event not detected; 1: event (under threshold) detected)                                             |

1. The DEN data-ready signal can be latched or pulsed depending on the value of the dataready\_pulsed bit of the [DRDY\\_PULSE\\_CFG\\_G \(0Bh\)](#) register.

## 9.27 STATUS\_REG (1Eh)

The STATUS\_REG register is read by the SPI/I<sup>2</sup>C interface (r).

**Table 86. STATUS\_REG register**

|   |   |   |   |   |     |     |      |
|---|---|---|---|---|-----|-----|------|
| 0 | 0 | 0 | 0 | 0 | TDA | GDA | XLDA |
|---|---|---|---|---|-----|-----|------|

**Table 87. STATUS\_REG register description**

|      |                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TDA  | Temperature new data available. Default: 0 (0: no set of data is available at temperature sensor output; 1: a new set of data is available at temperature sensor output) |
| GDA  | Gyroscope new data available. Default value: 0 (0: no set of data available at gyroscope output; 1: a new set of data is available at gyroscope output)                  |
| XLDA | Accelerometer new data available. Default value: 0 (0: no set of data available at accelerometer output; 1: a new set of data is available at accelerometer output)      |

## 9.28 OUT\_TEMP\_L (20h), OUT\_TEMP\_H (21h)

Temperature data output register (r). L and H registers together express a 16-bit word in two's complement.

**Table 88. OUT\_TEMP\_L register**

|       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
| Temp7 | Temp6 | Temp5 | Temp4 | Temp3 | Temp2 | Temp1 | Temp0 |
|-------|-------|-------|-------|-------|-------|-------|-------|

**Table 89. OUT\_TEMP\_H register**

|        |        |        |        |        |        |       |       |
|--------|--------|--------|--------|--------|--------|-------|-------|
| Temp15 | Temp14 | Temp13 | Temp12 | Temp11 | Temp10 | Temp9 | Temp8 |
|--------|--------|--------|--------|--------|--------|-------|-------|

**Table 90. OUT\_TEMP register description**

|            |                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------|
| Temp[15:0] | Temperature sensor output data<br>The value is expressed as two's complement sign extended on the MSB. |
|------------|--------------------------------------------------------------------------------------------------------|

## 9.29 OUTX\_L\_G (22h)

Angular rate sensor pitch axis (X) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 91. OUTX\_L\_G register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 92. OUTX\_L\_G register description**

|        |                                                                                                                                                                                            |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[7:0] | Pitch axis (X) angular rate value (LSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain pitch axis output |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.30 OUTX\_H\_G (23h)

Angular rate sensor pitch axis (X) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 93. OUTX\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 94. OUTX\_H\_G register description**

|         |                                                                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[15:8] | Pitch axis (X) angular rate value (MSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain pitch axis output |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.31 OUTY\_L\_G (24h)

Angular rate sensor roll axis (Y) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 95. OUTY\_L\_G register**

| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|----|

**Table 96. OUTY\_L\_G register description**

|        |                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[7:0] | Roll axis (Y) angular rate value (LSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain roll axis output |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.32 OUTY\_H\_G (25h)

Angular rate sensor roll axis (Y) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 97. OUTY\_H\_G register**

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|
|-----|-----|-----|-----|-----|-----|----|----|

**Table 98. OUTY\_H\_G register description**

|         |                                                                                                                                                                                          |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[15:8] | Roll axis (Y) angular rate value (MSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain roll axis output |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.33 OUTZ\_L\_G (26h)

Angular rate sensor yaw axis (Z) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 99. OUTZ\_L\_G register**

| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|----|

**Table 100. OUTZ\_L\_G register description**

|        |                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[7:0] | Yaw axis (Z) angular rate value (LSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain yaw axis output |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.34 OUTZ\_H\_G (27h)

Angular rate sensor Yaw axis (Z) angular rate output register (r). The value is expressed as a 16-bit word in two's complement.

Data are according to the full-scale and ODR settings ([CTRL2\\_G \(11h\)](#)) of the gyro user interface.

**Table 101. OUTZ\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 102. OUTZ\_H\_G register description**

|         |                                                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D[15:8] | Yaw axis (Z) angular rate value (MSbyte)<br>D[15:0] expressed in two's complement and its value depends on the interface used:<br>SPI1/I <sup>2</sup> C: Gyro UI chain yaw axis output |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.35 OUTX\_L\_XL (28h)

Linear acceleration sensor X-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 103. OUTX\_L\_XL register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 104. OUTX\_L\_XL register description**

|        |                                           |
|--------|-------------------------------------------|
| D[7:0] | X-axis linear acceleration value (LSbyte) |
|--------|-------------------------------------------|

### 9.36 OUTX\_H\_XL (29h)

Linear acceleration sensor X-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 105. OUTX\_H\_XL register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 106. OUTX\_H\_XL register description**

|         |                                           |
|---------|-------------------------------------------|
| D[15:8] | X-axis linear acceleration value (MSbyte) |
|---------|-------------------------------------------|

### 9.37 OUTY\_L\_XL (2Ah)

Linear acceleration sensor Y-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 107. OUTY\_L\_XL register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 108. OUTY\_L\_XL register description**

|        |                                           |
|--------|-------------------------------------------|
| D[7:0] | Y-axis linear acceleration value (LSbyte) |
|--------|-------------------------------------------|

### 9.38 OUTY\_H\_XL (2Bh)

Linear acceleration sensor Y-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 109. OUTY\_H\_G register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 110. OUTY\_H\_G register description**

|         |                                           |
|---------|-------------------------------------------|
| D[15:8] | Y-axis linear acceleration value (MSbyte) |
|---------|-------------------------------------------|

### 9.39 OUTZ\_L\_XL (2Ch)

Linear acceleration sensor Z-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 111. OUTZ\_L\_XL register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 112. OUTZ\_L\_XL register description**

|        |                                           |
|--------|-------------------------------------------|
| D[7:0] | Z-axis linear acceleration value (LSbyte) |
|--------|-------------------------------------------|

### 9.40 OUTZ\_H\_XL (2Dh)

Linear acceleration sensor Z-axis output register (r). The value is expressed as a 16-bit word in two's complement.

**Table 113. OUTZ\_H\_XL register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 114. OUTZ\_H\_XL register description**

|         |                                           |
|---------|-------------------------------------------|
| D[15:8] | Z-axis linear acceleration value (MSbyte) |
|---------|-------------------------------------------|

### 9.41 SENSORHUB1\_REG (2Eh)

First byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 115. SENSORHUB1\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub1_7 | SHub1_6 | SHub1_5 | SHub1_4 | SHub1_3 | SHub1_2 | SHub1_1 | SHub1_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 116. SENSORHUB1\_REG register description**

|             |                                           |
|-------------|-------------------------------------------|
| SHub1_[7:0] | First byte associated to external sensors |
|-------------|-------------------------------------------|

## 9.42 SENSORHUB2\_REG (2Fh)

Second byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operations configurations (for external sensors from x = 0 to x = 3).

**Table 117. SENSORHUB2\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub2_7 | SHub2_6 | SHub2_5 | SHub2_4 | SHub2_3 | SHub2_2 | SHub2_1 | SHub2_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 118. SENSORHUB2\_REG register description**

|             |                                            |
|-------------|--------------------------------------------|
| SHub2_[7:0] | Second byte associated to external sensors |
|-------------|--------------------------------------------|

## 9.43 SENSORHUB3\_REG (30h)

Third byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operations configurations (for external sensors from x = 0 to x = 3).

**Table 119. SENSORHUB3\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub3_7 | SHub3_6 | SHub3_5 | SHub3_4 | SHub3_3 | SHub3_2 | SHub3_1 | SHub3_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 120. SENSORHUB3\_REG register description**

|             |                                           |
|-------------|-------------------------------------------|
| SHub3_[7:0] | Third byte associated to external sensors |
|-------------|-------------------------------------------|

## 9.44 SENSORHUB4\_REG (31h)

Fourth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 121. SENSORHUB4\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub4_7 | SHub4_6 | SHub4_5 | SHub4_4 | SHub4_3 | SHub4_2 | SHub4_1 | SHub4_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 122. SENSORHUB4\_REG register description**

|             |                                            |
|-------------|--------------------------------------------|
| SHub4_[7:0] | Fourth byte associated to external sensors |
|-------------|--------------------------------------------|

## 9.45 SENSORHUB5\_REG (32h)

Fifth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 123. SENSORHUB5\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub5_7 | SHub5_6 | SHub5_5 | SHub5_4 | SHub5_3 | SHub5_2 | SHub5_1 | SHub5_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 124. SENSORHUB5\_REG register description**

|             |                                           |
|-------------|-------------------------------------------|
| SHub5_[7:0] | Fifth byte associated to external sensors |
|-------------|-------------------------------------------|

## 9.46 SENSORHUB6\_REG (33h)

Sixth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 125. SENSORHUB6\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub6_7 | SHub6_6 | SHub6_5 | SHub6_4 | SHub6_3 | SHub6_2 | SHub6_1 | SHub6_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 126. SENSORHUB6\_REG register description**

|             |                                           |
|-------------|-------------------------------------------|
| SHub6_[7:0] | Sixth byte associated to external sensors |
|-------------|-------------------------------------------|

## 9.47 SENSORHUB7\_REG (34h)

Seventh byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 127. SENSORHUB7\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub7_7 | SHub7_6 | SHub7_5 | SHub7_4 | SHub7_3 | SHub7_2 | SHub7_1 | SHub7_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 128. SENSORHUB7\_REG register description**

|             |                                             |
|-------------|---------------------------------------------|
| SHub7_[7:0] | Seventh byte associated to external sensors |
|-------------|---------------------------------------------|

## 9.48 SENSORHUB8\_REG(35h)

Eighth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 129. SENSORHUB8\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub8_7 | SHub8_6 | SHub8_5 | SHub8_4 | SHub8_3 | SHub8_2 | SHub8_1 | SHub8_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 130. SENSORHUB8\_REG register description**

|             |                                            |
|-------------|--------------------------------------------|
| SHub8_[7:0] | Eighth byte associated to external sensors |
|-------------|--------------------------------------------|

## 9.49 SENSORHUB9\_REG (36h)

Ninth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 131. SENSORHUB9\_REG register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| SHub9_7 | SHub9_6 | SHub9_5 | SHub9_4 | SHub9_3 | SHub9_2 | SHub9_1 | SHub9_0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 132. SENSORHUB9\_REG register description**

|             |                                           |
|-------------|-------------------------------------------|
| SHub9_[7:0] | Ninth byte associated to external sensors |
|-------------|-------------------------------------------|

## 9.50 SENSORHUB10\_REG (37h)

Tenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 133. SENSORHUB10\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub10_7 | SHub10_6 | SHub10_5 | SHub10_4 | SHub10_3 | SHub10_2 | SHub10_1 | SHub10_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 134. SENSORHUB10\_REG register description**

|              |                                           |
|--------------|-------------------------------------------|
| SHub10_[7:0] | Tenth byte associated to external sensors |
|--------------|-------------------------------------------|

## 9.51 SENSORHUB11\_REG (38h)

Eleventh byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 135. SENSORHUB11\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub11_7 | SHub11_6 | SHub11_5 | SHub11_4 | SHub11_3 | SHub11_2 | SHub11_1 | SHub11_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 136. SENSORHUB11\_REG register description**

|              |                                              |
|--------------|----------------------------------------------|
| SHub11_[7:0] | Eleventh byte associated to external sensors |
|--------------|----------------------------------------------|

## 9.52 SENSORHUB12\_REG (39h)

Twelfth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 137. SENSORHUB12\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub12_7 | SHub12_6 | SHub12_5 | SHub12_4 | SHub12_3 | SHub12_2 | SHub12_1 | SHub12_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 138. SENSORHUB12\_REG register description**

|             |                                             |
|-------------|---------------------------------------------|
| SHub12[7:0] | Twelfth byte associated to external sensors |
|-------------|---------------------------------------------|

## 9.53 FIFO\_STATUS1 (3Ah)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 139. FIFO\_STATUS1 register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| DIFF_FIFO_7 | DIFF_FIFO_6 | DIFF_FIFO_5 | DIFF_FIFO_4 | DIFF_FIFO_3 | DIFF_FIFO_2 | DIFF_FIFO_1 | DIFF_FIFO_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 140. FIFO\_STATUS1 register description**

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| DIFF_FIFO_[7:0] | Number of unread words (16-bit axes) stored in FIFO <sup>(1)</sup> . |
|-----------------|----------------------------------------------------------------------|

1. For a complete number of unread samples, consider DIFF\_FIFO [10:8] in [FIFO\\_STATUS2 \(3Bh\)](#).

## 9.54 FIFO\_STATUS2 (3Bh)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 141. FIFO\_STATUS2 register**

|        |          |                   |             |   |               |              |              |
|--------|----------|-------------------|-------------|---|---------------|--------------|--------------|
| WaterM | OVER_RUN | FIFO_ FULL_ SMART | FIFO_ EMPTY | 0 | DIFF_ FIFO_10 | DIFF_ FIFO_9 | DIFF_ FIFO_8 |
|--------|----------|-------------------|-------------|---|---------------|--------------|--------------|

**Table 142. FIFO\_STATUS2 register description**

|                  |                                                                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WaterM           | FIFO watermark status. The watermark is set through bits FTH_[7:0] in <a href="#">FIFO_CTRL1 (06h)</a> . Default value: 0<br>(0: FIFO filling is lower than watermark level <sup>(1)</sup> ;<br>1: FIFO filling is equal to or higher than the watermark level) |
| OVER_RUN         | FIFO overrun status. Default value: 0<br>(0: FIFO is not completely filled; 1: FIFO is completely filled)                                                                                                                                                       |
| FIFO_FULL_ SMART | Smart FIFO full status. Default value: 0<br>(0: FIFO is not full; 1: FIFO will be full at the next ODR)                                                                                                                                                         |
| FIFO_EMPTY       | FIFO empty bit. Default value: 0<br>(0: FIFO contains data; 1: FIFO is empty)                                                                                                                                                                                   |
| DIFF_FIFO_[10:8] | Number of unread words (16-bit axes) stored in FIFO <sup>(2)</sup> .                                                                                                                                                                                            |

1. FIFO watermark level is set in FTH\_[10:0] in [FIFO\\_CTRL1 \(06h\)](#) and [FIFO\\_CTRL2 \(07h\)](#).

2. For a complete number of unread samples, consider DIFF\_FIFO [7:0] in [FIFO\\_STATUS1 \(3Ah\)](#).

## 9.55 FIFO\_STATUS3 (3Ch)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 143. FIFO\_STATUS3 register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| FIFO_ PATTERN _7 | FIFO_ PATTERN _6 | FIFO_ PATTERN _5 | FIFO_ PATTERN _4 | FIFO_ PATTERN _3 | FIFO_ PATTERN _2 | FIFO_ PATTERN _1 | FIFO_ PATTERN _0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 144. FIFO\_STATUS3 register description**

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| FIFO_ PATTERN_[7:0] | Word of recursive pattern read at the next reading. |
|---------------------|-----------------------------------------------------|

## 9.56 FIFO\_STATUS4 (3Dh)

FIFO status control register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 145. FIFO\_STATUS4 register**

|   |   |   |   |   |   |                |                |
|---|---|---|---|---|---|----------------|----------------|
| 0 | 0 | 0 | 0 | 0 | 0 | FIFO_PATTERN_9 | FIFO_PATTERN_8 |
|---|---|---|---|---|---|----------------|----------------|

**Table 146. FIFO\_STATUS4 register description**

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| FIFO_PATTERN_[9:8] | Word of recursive pattern read at the next reading. |
|--------------------|-----------------------------------------------------|

## 9.57 FIFO\_DATA\_OUT\_L (3Eh)

FIFO data output register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 147. FIFO\_DATA\_OUT\_L register**

|                        |                        |                        |                        |                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| DATA_OUT_7<br>FIFO_L_7 | DATA_OUT_6<br>FIFO_L_6 | DATA_OUT_5<br>FIFO_L_5 | DATA_OUT_4<br>FIFO_L_4 | DATA_OUT_3<br>FIFO_L_3 | DATA_OUT_2<br>FIFO_L_2 | DATA_OUT_1<br>FIFO_L_1 | DATA_OUT_0<br>FIFO_L_0 |
|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|

**Table 148. FIFO\_DATA\_OUT\_L register description**

|                       |                               |
|-----------------------|-------------------------------|
| DATA_OUT_FIFO_L_[7:0] | FIFO data output (first byte) |
|-----------------------|-------------------------------|

## 9.58 FIFO\_DATA\_OUT\_H (3Fh)

FIFO data output register (r). For a proper reading of the register, it is recommended to set the BDU bit in [CTRL3\\_C \(12h\)](#) to 1.

**Table 149. FIFO\_DATA\_OUT\_H register**

|                        |                        |                        |                        |                        |                        |                        |                        |
|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| DATA_OUT_7<br>FIFO_H_7 | DATA_OUT_6<br>FIFO_H_6 | DATA_OUT_5<br>FIFO_H_5 | DATA_OUT_4<br>FIFO_H_4 | DATA_OUT_3<br>FIFO_H_3 | DATA_OUT_2<br>FIFO_H_2 | DATA_OUT_1<br>FIFO_H_1 | DATA_OUT_0<br>FIFO_H_0 |
|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|

**Table 150. FIFO\_DATA\_OUT\_H register description**

|                       |                                |
|-----------------------|--------------------------------|
| DATA_OUT_FIFO_H_[7:0] | FIFO data output (second byte) |
|-----------------------|--------------------------------|

## 9.59 TIMESTAMP0\_REG (40h)

Timestamp first (least significant) byte data output register (r). The value is expressed as a 24-bit word and the bit resolution is defined by setting the value in [WAKE\\_UP\\_DUR \(5Ch\)](#).

**Table 151. TIMESTAMP0\_REG register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| TIMESTA<br>MP0_7 | TIMESTA<br>MP0_6 | TIMESTA<br>MP0_5 | TIMESTA<br>MP0_4 | TIMESTA<br>MP0_3 | TIMESTA<br>MP0_2 | TIMESTA<br>MP0_1 | TIMESTA<br>MP0_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 152. TIMESTAMP0\_REG register description**

|                  |                                  |
|------------------|----------------------------------|
| TIMESTAMP0_[7:0] | TIMESTAMP first byte data output |
|------------------|----------------------------------|

## 9.60 TIMESTAMP1\_REG (41h)

Timestamp second byte data output register (r). The value is expressed as a 24-bit word and the bit resolution is defined by setting value in [WAKE\\_UP\\_DUR \(5Ch\)](#).

**Table 153. TIMESTAMP1\_REG register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| TIMESTA<br>MP1_7 | TIMESTA<br>MP1_6 | TIMESTA<br>MP1_5 | TIMESTA<br>MP1_4 | TIMESTA<br>MP1_3 | TIMESTA<br>MP1_2 | TIMESTA<br>MP1_1 | TIMESTA<br>MP1_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 154. TIMESTAMP1\_REG register description**

|                  |                                   |
|------------------|-----------------------------------|
| TIMESTAMP1_[7:0] | TIMESTAMP second byte data output |
|------------------|-----------------------------------|

## 9.61 TIMESTAMP2\_REG (42h)

Timestamp third (most significant) byte data output register (r/w). The value is expressed as a 24-bit word and the bit resolution is defined by setting the value in [WAKE\\_UP\\_DUR \(5Ch\)](#). To reset the timer, the AAh value has to be stored in this register.

**Table 155. TIMESTAMP2\_REG register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| TIMESTA<br>MP2_7 | TIMESTA<br>MP2_6 | TIMESTA<br>MP2_5 | TIMESTA<br>MP2_4 | TIMESTA<br>MP2_3 | TIMESTA<br>MP2_2 | TIMESTA<br>MP2_1 | TIMESTA<br>MP2_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 156. TIMESTAMP2\_REG register description**

|                  |                                  |
|------------------|----------------------------------|
| TIMESTAMP2_[7:0] | TIMESTAMP third byte data output |
|------------------|----------------------------------|

## 9.62 STEP\_TIMESTAMP\_L (49h)

Step counter timestamp information register (r). When a step is detected, the value of TIMESTAMP\_REG1 register is copied in STEP\_TIMESTAMP\_L.

**Table 157. STEP\_TIMESTAMP\_L register**

|                    |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| STEP_TIMESTAMP_L_7 | STEP_TIMESTAMP_L_6 | STEP_TIMESTAMP_L_5 | STEP_TIMESTAMP_L_4 | STEP_TIMESTAMP_L_3 | STEP_TIMESTAMP_L_2 | STEP_TIMESTAMP_L_1 | STEP_TIMESTAMP_L_0 |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|

**Table 158. STEP\_TIMESTAMP\_L register description**

|                       |                                  |
|-----------------------|----------------------------------|
| STEP_TIMESTAMP_L[7:0] | Timestamp of last step detected. |
|-----------------------|----------------------------------|

## 9.63 STEP\_TIMESTAMP\_H (4Ah)

Step counter timestamp information register (r). When a step is detected, the value of TIMESTAMP\_REG2 register is copied in STEP\_TIMESTAMP\_H.

**Table 159. STEP\_TIMESTAMP\_H register**

|                    |                    |                    |                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| STEP_TIMESTAMP_H_7 | STEP_TIMESTAMP_H_6 | STEP_TIMESTAMP_H_5 | STEP_TIMESTAMP_H_4 | STEP_TIMESTAMP_H_3 | STEP_TIMESTAMP_H_2 | STEP_TIMESTAMP_H_1 | STEP_TIMESTAMP_H_0 |
|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|

**Table 160. STEP\_TIMESTAMP\_H register description**

|                       |                                  |
|-----------------------|----------------------------------|
| STEP_TIMESTAMP_H[7:0] | Timestamp of last step detected. |
|-----------------------|----------------------------------|

## 9.64 STEP\_COUNTER\_L (4Bh)

Step counter output register (r).

**Table 161. STEP\_COUNTER\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| STEP_COUNTER_L_7 | STEP_COUNTER_L_6 | STEP_COUNTER_L_5 | STEP_COUNTER_L_4 | STEP_COUNTER_L_3 | STEP_COUNTER_L_2 | STEP_COUNTER_L_1 | STEP_COUNTER_L_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 162. STEP\_COUNTER\_L register description**

|                     |                              |
|---------------------|------------------------------|
| STEP_COUNTER_L[7:0] | Step counter output (LSbyte) |
|---------------------|------------------------------|

## 9.65 STEP\_COUNTER\_H (4Ch)

Step counter output register (r).

**Table 163. STEP\_COUNTER\_H register**

|                          |                          |                          |                          |                          |                          |                          |                          |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| STEP_CO<br>UNTER_H<br>_7 | STEP_CO<br>UNTER_H<br>_6 | STEP_CO<br>UNTER_H<br>_5 | STEP_CO<br>UNTER_H<br>_4 | STEP_CO<br>UNTER_H<br>_3 | STEP_CO<br>UNTER_H<br>_2 | STEP_CO<br>UNTER_H<br>_1 | STEP_CO<br>UNTER_H<br>_0 |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|

**Table 164. STEP\_COUNTER\_H register description**

|                     |                              |
|---------------------|------------------------------|
| STEP_COUNTER_H[7:0] | Step counter output (MSbyte) |
|---------------------|------------------------------|

## 9.66 SENSORHUB13\_REG (4Dh)

Thirteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 165. SENSORHUB13\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub13_7 | SHub13_6 | SHub13_5 | SHub13_4 | SHub13_3 | SHub13_2 | SHub13_1 | SHub13_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 166. SENSORHUB13\_REG register description**

|             |                                                |
|-------------|------------------------------------------------|
| SHub13[7:0] | Thirteenth byte associated to external sensors |
|-------------|------------------------------------------------|

## 9.67 SENSORHUB14\_REG (4Eh)

Fourteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 167. SENSORHUB14\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub14_7 | SHub14_6 | SHub14_5 | SHub14_4 | SHub14_3 | SHub14_2 | SHub14_1 | SHub14_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 168. SENSORHUB14\_REG register description**

|             |                                                |
|-------------|------------------------------------------------|
| SHub14[7:0] | Fourteenth byte associated to external sensors |
|-------------|------------------------------------------------|

## 9.68 SENSORHUB15\_REG (4Fh)

Fifteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 169. SENSORHUB15\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub15_7 | SHub15_6 | SHub15_5 | SHub15_4 | SHub15_3 | SHub15_2 | SHub15_1 | SHub15_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 170. SENSORHUB15\_REG register description**

|             |                                               |
|-------------|-----------------------------------------------|
| SHub15[7:0] | Fifteenth byte associated to external sensors |
|-------------|-----------------------------------------------|

## 9.69 SENSORHUB16\_REG (50h)

Sixteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 171. SENSORHUB16\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub16_7 | SHub16_6 | SHub16_5 | SHub16_4 | SHub16_3 | SHub16_2 | SHub16_1 | SHub16_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 172. SENSORHUB16\_REG register description**

|              |                                               |
|--------------|-----------------------------------------------|
| SHub16_[7:0] | Sixteenth byte associated to external sensors |
|--------------|-----------------------------------------------|

## 9.70 SENSORHUB17\_REG (51h)

Seventeenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 173. SENSORHUB17\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub17_7 | SHub17_6 | SHub17_5 | SHub17_4 | SHub17_3 | SHub17_2 | SHub17_1 | SHub17_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 174. SENSORHUB17\_REG register description**

|              |                                                 |
|--------------|-------------------------------------------------|
| SHub17_[7:0] | Seventeenth byte associated to external sensors |
|--------------|-------------------------------------------------|

## 9.71 SENSORHUB18\_REG (52h)

Eighteenth byte associated to external sensors. The content of the register is consistent with the SLAVEx\_CONFIG number of read operation configurations (for external sensors from x = 0 to x = 3).

**Table 175. SENSORHUB18\_REG register**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| SHub18_7 | SHub18_6 | SHub18_5 | SHub18_4 | SHub18_3 | SHub18_2 | SHub18_1 | SHub18_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 176. SENSORHUB18\_REG register description**

|              |                                                |
|--------------|------------------------------------------------|
| SHub18_[7:0] | Eighteenth byte associated to external sensors |
|--------------|------------------------------------------------|

## 9.72 FUNC\_SRC1 (53h)

Significant motion, tilt, step detector, hard/soft-iron and sensor hub interrupt source register (r).

**Table 177. FUNC\_SRC1 register**

|                     |                |         |               |               |         |           |                   |
|---------------------|----------------|---------|---------------|---------------|---------|-----------|-------------------|
| STEP_COUNT_DELTA_IA | SIGN_MOTION_IA | TILT_IA | STEP_DETECTED | STEP_OVERFLOW | HI_FAIL | SI_END_OP | SENSOR_HUB_END_OP |
|---------------------|----------------|---------|---------------|---------------|---------|-----------|-------------------|

**Table 178. FUNC\_SRC1 register description**

|                     |                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STEP_COUNT_DELTA_IA | Pedometer step recognition on delta time status. Default value: 0<br>(0: no step recognized during delta time;<br>1: at least one step recognized during delta time) |
| SIGN_MOTION_IA      | Significant motion event detection status. Default value: 0<br>(0: significant motion event not detected; 1: significant motion event detected)                      |
| TILT_IA             | Tilt event detection status. Default value: 0<br>(0: tilt event not detected; 1: tilt event detected)                                                                |
| STEP_DETECTED       | Step detector event detection status. Default value: 0<br>(0: step detector event not detected; 1: step detector event detected)                                     |
| STEP_OVERFLOW       | Step counter overflow status. Default value: 0<br>(0: step counter value < 2 <sup>16</sup> ; 1: step counter value reached 2 <sup>16</sup> )                         |
| HI_FAIL             | Fail in hard/soft-ironing algorithm.                                                                                                                                 |
| SI_END_OP           | Hard/soft-iron calculation status. Default value: 0<br>(0: Hard/soft-iron calculation not concluded; 1: Hard/soft-iron calculation concluded)                        |
| SENSORHUB_END_OP    | Sensor hub communication status. Default value: 0<br>(0: sensor hub communication not concluded;<br>1: sensor hub communication concluded)                           |

## 9.73 FUNC\_SRC2 (54h)

Wrist tilt interrupt source register (r).

**Table 179. FUNC\_SRC2 register**

|   |             |             |             |             |   |   |               |
|---|-------------|-------------|-------------|-------------|---|---|---------------|
| 0 | SLAVE3_NACK | SLAVE2_NACK | SLAVE1_NACK | SLAVE0_NACK | 0 | 0 | WRIST_TILT_IA |
|---|-------------|-------------|-------------|-------------|---|---|---------------|

**Table 180. FUNC\_SRC2 register description**

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| SLAVE3_NACK   | This bit is set to 1 if Not acknowledge occurs on slave 3 communication. Default value: 0                               |
| SLAVE2_NACK   | This bit is set to 1 if Not acknowledge occurs on slave 2 communication. Default value: 0                               |
| SLAVE1_NACK   | This bit is set to 1 if Not acknowledge occurs on slave 1 communication. Default value: 0                               |
| SLAVE0_NACK   | This bit is set to 1 if Not acknowledge occurs on slave 0 communication. Default value: 0                               |
| WRIST_TILT_IA | Wrist tilt event detection status. Default value: 0<br>(0: Wrist tilt event not detected; 1: Wrist tilt event detected) |

## 9.74 WRIST\_TILT\_IA (55h)

Wrist tilt interrupt source register (r).

**Table 181. WRIST\_TILT\_IA register**

|                            |                            |                            |                            |                            |                            |   |   |
|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|---|---|
| WRIST_<br>TILT_IA_<br>Xpos | WRIST_<br>TILT_IA_<br>Xneg | WRIST_<br>TILT_IA_<br>Ypos | WRIST_<br>TILT_IA_<br>Yneg | WRIST_<br>TILT_IA_<br>Zpos | WRIST_<br>TILT_IA_<br>Zneg | 0 | 0 |
|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|---|---|

**Table 182. WRIST\_TILT\_IA register description**

|                         |                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WRIST_<br>TILT_IA_ Xpos | Absolute Wrist Tilt event detection status on X-positive axis. Default value: 0<br>(0: Absolute Wrist Tilt event on X-positive axis not detected;<br>1: Absolute Wrist Tilt event on X-positive axis detected) |
| WRIST_<br>TILT_IA_ Xneg | Absolute Wrist Tilt event detection status on X-negative axis. Default value: 0<br>(0: Absolute Wrist Tilt event on X-negative axis not detected;<br>1: Absolute Wrist Tilt event on X-negative axis detected) |
| WRIST_<br>TILT_IA_ Ypos | Absolute Wrist Tilt event detection status on Y-positive axis. Default value: 0<br>(0: Absolute Wrist Tilt event on Y-positive axis not detected;<br>1: Absolute Wrist Tilt event on Y-positive axis detected) |
| WRIST_<br>TILT_IA_ Yneg | Absolute Wrist Tilt event detection status on Y-negative axis. Default value: 0<br>(0: Absolute Wrist Tilt event on Y-negative axis not detected;<br>1: Absolute Wrist Tilt event on Y-negative axis detected) |
| WRIST_<br>TILT_IA_ Zpos | Absolute Wrist Tilt event detection status on Z-positive axis. Default value: 0<br>(0: Absolute Wrist Tilt event on Z-positive axis not detected;<br>1: Absolute Wrist Tilt event on Z-positive axis detected) |
| WRIST_<br>TILT_IA_ Zneg | Absolute Wrist Tilt event detection status on Z-negative axis. Default value: 0<br>(0: Absolute Wrist Tilt event on Z-negative axis not detected;<br>1: Absolute Wrist Tilt event on Z-negative axis detected) |

## 9.75 TAP\_CFG (58h)

Enables interrupt and inactivity functions, configuration of filtering and tap recognition functions (r/w).

**Table 183. TAP\_CFG register**

| INTERRUPTS_ENABLE | INACT_EN1 | INACT_EN0 | SLOPE_FDS | TAP_X_EN | TAP_Y_EN | TAP_Z_EN | LIR |
|-------------------|-----------|-----------|-----------|----------|----------|----------|-----|
|-------------------|-----------|-----------|-----------|----------|----------|----------|-----|

**Table 184. TAP\_CFG register description**

|                   |                                                                                                                                                                                                                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INTERRUPTS_ENABLE | Enable basic interrupts (6D/4D, free-fall, wake-up, tap, inactivity). Default 0.<br>(0: interrupt disabled; 1: interrupt enabled)                                                                                                                                                                                  |
| INACT_EN[1:0]     | Enable inactivity function. Default value: 00<br>(00: disabled<br>01: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro does not change;<br>10: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro to sleep mode;<br>11: sets accelerometer ODR to 12.5 Hz (low-power mode), gyro to power-down mode) |
| SLOPE_FDS         | HPF or SLOPE filter selection on wake-up and Activity/Inactivity functions. Refer to <a href="#">Figure 8</a> . Default value: 0<br>(0: SLOPE filter applied; 1: HPF applied)                                                                                                                                      |
| TAP_X_EN          | Enable X direction in tap recognition. Default value: 0<br>(0: X direction disabled; 1: X direction enabled)                                                                                                                                                                                                       |
| TAP_Y_EN          | Enable Y direction in tap recognition. Default value: 0<br>(0: Y direction disabled; 1: Y direction enabled)                                                                                                                                                                                                       |
| TAP_Z_EN          | Enable Z direction in tap recognition. Default value: 0<br>(0: Z direction disabled; 1: Z direction enabled)                                                                                                                                                                                                       |
| LIR               | Latched Interrupt. Default value: 0<br>(0: interrupt request not latched; 1: interrupt request latched)                                                                                                                                                                                                            |

## 9.76 TAP\_THS\_6D (59h)

Portrait/landscape position and tap function threshold register (r/w).

**Table 185. TAP\_THS\_6D register**

|        |               |               |              |              |              |              |              |
|--------|---------------|---------------|--------------|--------------|--------------|--------------|--------------|
| D4D_EN | SIXD_THS<br>1 | SIXD_THS<br>0 | TAP_THS<br>4 | TAP_THS<br>3 | TAP_THS<br>2 | TAP_THS<br>1 | TAP_THS<br>0 |
|--------|---------------|---------------|--------------|--------------|--------------|--------------|--------------|

**Table 186. TAP\_THS\_6D register description**

|               |                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| D4D_EN        | 4D orientation detection enable. Z-axis position detection is disabled.<br>Default value: 0<br>(0: enabled; 1: disabled) |
| SIXD_THS[1:0] | Threshold for 4D/6D function. Default value: 00<br>For details, refer to <a href="#">Table 187</a> .                     |
| TAP_THS[4:0]  | Threshold for tap recognition. Default value: 00000<br>1 LSB corresponds to $FS_{XL}/2^5$                                |

**Table 187. Threshold for D4D/D6D function**

| SIXD_THS[1:0] | Threshold value |
|---------------|-----------------|
| 00            | 80 degrees      |
| 01            | 70 degrees      |
| 10            | 60 degrees      |
| 11            | 50 degrees      |

## 9.77 INT\_DUR2 (5Ah)

Tap recognition function setting register (r/w).

**Table 188. INT\_DUR2 register**

|      |      |      |      |        |        |        |        |
|------|------|------|------|--------|--------|--------|--------|
| DUR3 | DUR2 | DUR1 | DUR0 | QUIET1 | QUIET0 | SHOCK1 | SHOCK0 |
|------|------|------|------|--------|--------|--------|--------|

**Table 189. INT\_DUR2 register description**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DUR[3:0]   | Duration of maximum time gap for double tap recognition. Default: 0000<br>When double tap recognition is enabled, this register expresses the maximum time between two consecutive detected taps to determine a double tap event. The default value of these bits is 0000b which corresponds to $16 \times ODR_{XL}$ time. If the DUR[3:0] bits are set to a different value, 1LSB corresponds to $32 \times ODR_{XL}$ time. |
| QUIET[1:0] | Expected quiet time after a tap detection. Default value: 00<br>Quiet time is the time after the first detected tap in which there must not be any overthreshold event. The default value of these bits is 00b which corresponds to $2 \times ODR_{XL}$ time. If the QUIET[1:0] bits are set to a different value, 1LSB corresponds to $4 \times ODR_{XL}$ time.                                                             |
| SHOCK[1:0] | Maximum duration of overthreshold event. Default value: 00<br>Maximum duration is the maximum time of an overthreshold signal detection to be recognized as a tap event. The default value of these bits is 00b which corresponds to $4 \times ODR_{XL}$ time. If the SHOCK[1:0] bits are set to a different value, 1LSB corresponds to $8 \times ODR_{XL}$ time.                                                            |

## 9.78 WAKE\_UP\_THS (5Bh)

Single and double-tap function threshold register (r/w).

**Table 190. WAKE\_UP\_THS register**

|                   |   |         |         |         |         |         |         |
|-------------------|---|---------|---------|---------|---------|---------|---------|
| SINGLE_DOUBLE_TAP | 0 | WK_THS5 | WK_THS4 | WK_THS3 | WK_THS2 | WK_THS1 | WK_THS0 |
|-------------------|---|---------|---------|---------|---------|---------|---------|

**Table 191. WAKE\_UP\_THS register description**

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| SINGLE_DOUBLE_TAP | Single/double-tap event enable. Default: 0<br>(0: only single-tap event enabled;<br>1: both single and double-tap events enabled) |
| WK_THS[5:0]       | Threshold for wakeup. Default value: 000000<br>1 LSb corresponds to $FS_{XL}/2^6$                                                 |

## 9.79 WAKE\_UP\_DUR (5Ch)

Free-fall, wakeup, timestamp and sleep mode functions duration setting register (r/w).

**Table 192. WAKE\_UP\_DUR register**

|         |           |           |          |            |            |            |            |
|---------|-----------|-----------|----------|------------|------------|------------|------------|
| FF_DUR5 | WAKE_DUR1 | WAKE_DUR0 | TIMER_HR | SLEEP_DUR3 | SLEEP_DUR2 | SLEEP_DUR1 | SLEEP_DUR0 |
|---------|-----------|-----------|----------|------------|------------|------------|------------|

**Table 193. WAKE\_UP\_DUR register description**

|                |                                                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FF_DUR5        | Free fall duration event. Default: 0<br>For the complete configuration of the free-fall duration, refer to FF_DUR[4:0] in <a href="#">FREE_FALL (5Dh)</a> configuration.<br>1 LSB = 1 ODR_time |
| WAKE_DUR[1:0]  | Wake up duration event. Default: 00<br>1LSB = 1 ODR_time                                                                                                                                       |
| TIMER_HR       | Timestamp register resolution setting <sup>(1)</sup> . Default value: 0<br>(0: 1LSB = 6.4 ms; 1: 1LSB = 25 $\mu$ s)                                                                            |
| SLEEP_DUR[3:0] | Duration to go in sleep mode. Default value: 0000 (this corresponds to 16 ODR)<br>1 LSB = 512 ODR                                                                                              |

1. Configuration of this bit affects [TIMESTAMP0\\_REG \(40h\)](#), [TIMESTAMP1\\_REG \(41h\)](#), [TIMESTAMP2\\_REG \(42h\)](#), [STEP\\_TIMESTAMP\\_L \(49h\)](#), [STEP\\_TIMESTAMP\\_H \(4Ah\)](#), and [CTRL6\\_C \(15h\)](#) registers.

## 9.80 FREE\_FALL (5Dh)

Free-fall function duration setting register (r/w).

**Table 194. FREE\_FALL register**

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| FF_DUR4 | FF_DUR3 | FF_DUR2 | FF_DUR1 | FF_DUR0 | FF_THS2 | FF_THS1 | FF_THS0 |
|---------|---------|---------|---------|---------|---------|---------|---------|

**Table 195. FREE\_FALL register description**

|             |                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FF_DUR[4:0] | Free-fall duration event. Default: 0<br>For the complete configuration of the free fall duration, refer to FF_DUR5 in <a href="#">WAKE_UP_DUR (5Ch)</a> configuration |
| FF_THS[2:0] | Free fall threshold setting. Default: 000<br>For details refer to <a href="#">Table 196</a> .                                                                         |

**Table 196. Threshold for free-fall function**

| FF_THS[2:0] | Threshold value |
|-------------|-----------------|
| 000         | 156 mg          |
| 001         | 219 mg          |
| 010         | 250 mg          |
| 011         | 312 mg          |
| 100         | 344 mg          |
| 101         | 406 mg          |
| 110         | 469 mg          |
| 111         | 500 mg          |

## 9.81 MD1\_CFG (5Eh)

Functions routing on INT1 register (r/w).

**Table 197. MD1\_CFG register**

|                  |                 |         |         |                 |         |           |            |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|------------|
| INT1_INACT_STATE | INT1_SINGLE_TAP | INT1_WU | INT1_FF | INT1_DOUBLE_TAP | INT1_6D | INT1_TILT | INT1_TIMER |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|------------|

**Table 198. MD1\_CFG register description**

|                  |                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT1_INACT_STATE | Routing on INT1 of inactivity mode. Default: 0<br>(0: routing on INT1 of inactivity disabled; 1: routing on INT1 of inactivity enabled)                                                           |
| INT1_SINGLE_TAP  | Single-tap recognition routing on INT1. Default: 0<br>(0: routing of single-tap event on INT1 disabled;<br>1: routing of single-tap event on INT1 enabled)                                        |
| INT1_WU          | Routing of wakeup event on INT1. Default value: 0<br>(0: routing of wakeup event on INT1 disabled;<br>1: routing of wakeup event on INT1 enabled)                                                 |
| INT1_FF          | Routing of free-fall event on INT1. Default value: 0<br>(0: routing of free-fall event on INT1 disabled;<br>1: routing of free-fall event on INT1 enabled)                                        |
| INT1_DOUBLE_TAP  | Routing of tap event on INT1. Default value: 0<br>(0: routing of double-tap event on INT1 disabled;<br>1: routing of double-tap event on INT1 enabled)                                            |
| INT1_6D          | Routing of 6D event on INT1. Default value: 0<br>(0: routing of 6D event on INT1 disabled; 1: routing of 6D event on INT1 enabled)                                                                |
| INT1_TILT        | Routing of tilt event on INT1. Default value: 0<br>(0: routing of tilt event on INT1 disabled; 1: routing of tilt event on INT1 enabled)                                                          |
| INT1_TIMER       | Routing of end counter event of timer on INT1. Default value: 0<br>(0: routing of end counter event of timer on INT1 disabled;<br>1: routing of end counter event of timer event on INT1 enabled) |

## 9.82 MD2\_CFG (5Fh)

Functions routing on INT2 register (r/w).

**Table 199. MD2\_CFG register**

|                  |                 |         |         |                 |         |           |           |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|-----------|
| INT2_INACT_STATE | INT2_SINGLE_TAP | INT2_WU | INT2_FF | INT2_DOUBLE_TAP | INT2_6D | INT2_TILT | INT2_IRON |
|------------------|-----------------|---------|---------|-----------------|---------|-----------|-----------|

**Table 200. MD2\_CFG register description**

|                  |                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INT2_INACT_STATE | Routing on INT2 of inactivity mode. Default: 0<br>(0: routing on INT2 of inactivity disabled; 1: routing on INT2 of inactivity enabled)                                                                                            |
| INT2_SINGLE_TAP  | Single-tap recognition routing on INT2. Default: 0<br>(0: routing of single-tap event on INT2 disabled;<br>1: routing of single-tap event on INT2 enabled)                                                                         |
| INT2_WU          | Routing of wakeup event on INT2. Default value: 0<br>(0: routing of wakeup event on INT2 disabled;<br>1: routing of wake-up event on INT2 enabled)                                                                                 |
| INT2_FF          | Routing of free-fall event on INT2. Default value: 0<br>(0: routing of free-fall event on INT2 disabled;<br>1: routing of free-fall event on INT2 enabled)                                                                         |
| INT2_DOUBLE_TAP  | Routing of tap event on INT2. Default value: 0<br>(0: routing of double-tap event on INT2 disabled;<br>1: routing of double-tap event on INT2 enabled)                                                                             |
| INT2_6D          | Routing of 6D event on INT2. Default value: 0<br>(0: routing of 6D event on INT2 disabled; 1: routing of 6D event on INT2 enabled)                                                                                                 |
| INT2_TILT        | Routing of tilt event on INT2. Default value: 0<br>(0: routing of tilt event on INT2 disabled; 1: routing of tilt event on INT2 enabled)                                                                                           |
| INT2_IRON        | Routing of soft-iron/hard-iron algorithm end event on INT2. Default value: 0<br>(0: routing of soft-iron/hard-iron algorithm end event on INT2 disabled;<br>1: routing of soft-iron/hard-iron algorithm end event on INT2 enabled) |

### 9.83 MASTER\_CMD\_CODE (60h)

Table 201. MASTER\_CMD\_CODE register

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| MASTER_CMD_CODE7 | MASTER_CMD_CODE6 | MASTER_CMD_CODE5 | MASTER_CMD_CODE4 | MASTER_CMD_CODE3 | MASTER_CMD_CODE2 | MASTER_CMD_CODE1 | MASTER_CMD_CODE0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

Table 202. MASTER\_CMD\_CODE register description

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| MASTER_CMD_CODE[7:0] | Master command code used for stamping for sensor sync. Default: 0 |
|----------------------|-------------------------------------------------------------------|

### 9.84 SENS\_SYNC\_SPI\_ERROR\_CODE (61h)

Table 203. SENS\_SYNC\_SPI\_ERROR\_CODE register

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| ERROR_CODE7 | ERROR_CODE6 | ERROR_CODE5 | ERROR_CODE4 | ERROR_CODE3 | ERROR_CODE2 | ERROR_CODE1 | ERROR_CODE0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

Table 204. SENS\_SYNC\_SPI\_ERROR\_CODE register description

|                 |                                                         |
|-----------------|---------------------------------------------------------|
| ERROR_CODE[7:0] | Error code used for sensor synchronization. Default: 0) |
|-----------------|---------------------------------------------------------|

### 9.85 OUT\_MAG\_RAW\_X\_L (66h)

External magnetometer raw data (r).

Table 205. OUT\_MAG\_RAW\_X\_L register

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

Table 206. OUT\_MAG\_RAW\_X\_L register description

|        |                                             |
|--------|---------------------------------------------|
| D[7:0] | X-axis external magnetometer value (LSbyte) |
|--------|---------------------------------------------|

### 9.86 OUT\_MAG\_RAW\_X\_H (67h)

External magnetometer raw data (r).

Table 207. OUT\_MAG\_RAW\_X\_H register

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

Table 208. OUT\_MAG\_RAW\_X\_H register description

|         |                                             |
|---------|---------------------------------------------|
| D[15:8] | X-axis external magnetometer value (MSbyte) |
|---------|---------------------------------------------|

**9.87 OUT\_MAG\_RAW\_Y\_L (68h)**

External magnetometer raw data (r).

**Table 209. OUT\_MAG\_RAW\_Y\_L register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 210. OUT\_MAG\_RAW\_Y\_L register description**

|        |                                             |
|--------|---------------------------------------------|
| D[7:0] | Y-axis external magnetometer value (LSbyte) |
|--------|---------------------------------------------|

**9.88 OUT\_MAG\_RAW\_Y\_H (69h)**

External magnetometer raw data (r).

**Table 211. OUT\_MAG\_RAW\_Y\_H register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 212. OUT\_MAG\_RAW\_Y\_H register description**

|         |                                             |
|---------|---------------------------------------------|
| D[15:8] | Y-axis external magnetometer value (MSbyte) |
|---------|---------------------------------------------|

**9.89 OUT\_MAG\_RAW\_Z\_L (6Ah)**

External magnetometer raw data (r).

**Table 213. OUT\_MAG\_RAW\_Z\_L register**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|

**Table 214. OUT\_MAG\_RAW\_Z\_L register description**

|        |                                             |
|--------|---------------------------------------------|
| D[7:0] | Z-axis external magnetometer value (LSbyte) |
|--------|---------------------------------------------|

**9.90 OUT\_MAG\_RAW\_Z\_H (6Bh)**

External magnetometer raw data (r).

**Table 215. OUT\_MAG\_RAW\_Z\_H register**

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 |
|-----|-----|-----|-----|-----|-----|----|----|

**Table 216. OUT\_MAG\_RAW\_Z\_H register description**

|         |                                             |
|---------|---------------------------------------------|
| D[15:8] | Z-axis external magnetometer value (MSbyte) |
|---------|---------------------------------------------|

## 9.91 X\_OFS\_USR (73h)

Accelerometer X-axis user offset correction (r/w). The offset value set in the X\_OFS\_USR offset register is internally added to the acceleration value measured on the X-axis.

**Table 217. X\_OFS\_USR register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| X_OFS_USR_7 | X_OFS_USR_6 | X_OFS_USR_5 | X_OFS_USR_4 | X_OFS_USR_3 | X_OFS_USR_2 | X_OFS_USR_1 | X_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 218. X\_OFS\_USR register description**

|               |                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| X_OFS_USR_7:0 | Accelerometer X-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127]. |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.92 Y\_OFS\_USR (74h)

Accelerometer Y-axis user offset correction (r/w). The offset value set in the Y\_OFS\_USR offset register is internally added to the acceleration value measured on the Y-axis.

**Table 219. Y\_OFS\_USR register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Y_OFS_USR_7 | Y_OFS_USR_6 | Y_OFS_USR_5 | Y_OFS_USR_4 | Y_OFS_USR_3 | Y_OFS_USR_2 | Y_OFS_USR_1 | Y_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 220. Y\_OFS\_USR register description**

|               |                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y_OFS_USR_7:0 | Accelerometer Y-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127]. |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.93 Z\_OFS\_USR (75h)

Accelerometer Z-axis user offset correction (r/w). The offset value set in the Z\_OFS\_USR offset register is internally subtracted from the acceleration value measured on the Z-axis.

**Table 221. Z\_OFS\_USR register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Z_OFS_USR_7 | Z_OFS_USR_6 | Z_OFS_USR_5 | Z_OFS_USR_4 | Z_OFS_USR_3 | Z_OFS_USR_2 | Z_OFS_USR_1 | Z_OFS_USR_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 222. Z\_OFS\_USR register description**

|               |                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Z_OFS_USR_7:0 | Accelerometer Z-axis user offset correction expressed in two's complement, weight depends on CTRL6_C(4) bit. The value must be in the range [-127 127]. |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 10 Embedded functions register mapping

The tables given below provide a list of the first (A) and second (B) bank registers related to the embedded functions available in the device and the corresponding addresses.

The embedded functions registers of bank A are accessible when FUNC\_CFG\_EN is set to '1' in [FUNC\\_CFG\\_ACCESS \(01h\)](#).

The embedded functions registers of bank B are accessible when both FUNC\_CFG\_EN and FUNC\_CFG\_EN\_B set to '1' in [FUNC\\_CFG\\_ACCESS \(01h\)](#).

**Note:** All modifications of the content of the embedded functions registers have to be performed with the device in power-down mode.

**Table 223. Register address map - Bank A - embedded functions**

| Name                        | Type | Register address |           | Default  | Comment  |
|-----------------------------|------|------------------|-----------|----------|----------|
|                             |      | Hex              | Binary    |          |          |
| SLV0_ADD                    | r/w  | 02               | 00000010  | 00000000 |          |
| SLV0_SUBADD                 | r/w  | 03               | 00000011  | 00000000 |          |
| SLAVE0_CONFIG               | r/w  | 04               | 00000100  | 00000000 |          |
| SLV1_ADD                    | r/w  | 05               | 00000101  | 00000000 |          |
| SLV1_SUBADD                 | r/w  | 06               | 00000110  | 00000000 |          |
| SLAVE1_CONFIG               | r/w  | 07               | 00000111  | 00000000 |          |
| SLV2_ADD                    | r/w  | 08               | 00001000  | 00000000 |          |
| SLV2_SUBADD                 | r/w  | 09               | 00001001  | 00000000 |          |
| SLAVE2_CONFIG               | r/w  | 0A               | 00001010  | 00000000 |          |
| SLV3_ADD                    | r/w  | 0B               | 00001011  | 00000000 |          |
| SLV3_SUBADD                 | r/w  | 0C               | 00001100  | 00000000 |          |
| SLAVE3_CONFIG               | r/w  | 0D               | 00001101  | 00000000 |          |
| DATAWRITE_SRC_MODE_SUB_SLV0 | r/w  | 0E               | 00001110  | 00000000 |          |
| CONFIG_PEDO_THS_MIN         | r/w  | 0F               | 00001111  | 00010000 |          |
| RESERVED                    | -    | 10-12            |           |          | Reserved |
| SM_THS                      | r/w  | 13               | 00010011  | 00000110 |          |
| PEDO_DEB_REG                | r/w  | 14               | 00010100  | 01101110 |          |
| STEP_COUNT_DELTA            | r/w  | 15               | 0001 0101 | 00000000 |          |
| MAG_SI_XX                   | r/w  | 24               | 00100100  | 00001000 |          |
| MAG_SI_XY                   | r/w  | 25               | 00100101  | 00000000 |          |
| MAG_SI_XZ                   | r/w  | 26               | 00100110  | 00000000 |          |
| MAG_SI_YX                   | r/w  | 27               | 00100111  | 00000000 |          |
| MAG_SI_YY                   | r/w  | 28               | 00101000  | 00001000 |          |

**Table 223. Register address map - Bank A - embedded functions (continued)**

| Name       | Type | Register address |          | Default  | Comment |
|------------|------|------------------|----------|----------|---------|
|            |      | Hex              | Binary   |          |         |
| MAG_SI_YZ  | r/w  | 29               | 00101001 | 00000000 |         |
| MAG_SI_ZX  | r/w  | 2A               | 00101010 | 00000000 |         |
| MAG_SI_ZY  | r/w  | 2B               | 00101011 | 00000000 |         |
| MAG_SI_ZZ  | r/w  | 2C               | 00101100 | 00001000 |         |
| MAG_OFFX_L | r/w  | 2D               | 00101101 | 00000000 |         |
| MAG_OFFX_H | r/w  | 2E               | 00101110 | 00000000 |         |
| MAG_OFFY_L | r/w  | 2F               | 00101111 | 00000000 |         |
| MAG_OFFY_H | r/w  | 30               | 00110000 | 00000000 |         |
| MAG_OFFZ_L | r/w  | 31               | 00110001 | 00000000 |         |
| MAG_OFFZ_H | r/w  | 32               | 00110010 | 00000000 |         |

**Table 224. Register address map - Bank B - embedded functions**

| Name              | Type | Register address |          | Default  | Comment  |
|-------------------|------|------------------|----------|----------|----------|
|                   |      | Hex              | Binary   |          |          |
| A_WRIST_TILT_LAT  | r/w  | 50               | 01010000 | 00001111 |          |
| RESERVED          | -    | 51-53            |          |          | Reserved |
| A_WRIST_TILT_THS  | r/w  | 54               | 01010100 | 00100000 |          |
| RESERVED          | -    | 55-58            |          |          | Reserved |
| A_WRIST_TILT_Mask | r/w  | 59               | 01011001 | 11000000 |          |

Registers marked as *Reserved* must not be changed. Writing to those registers may cause permanent damage to the device.

The content of the registers that are loaded at boot should not be changed. They contain the factory calibration values. Their content is automatically restored when the device is powered up.

## 11 Embedded functions registers description - Bank A

*Note:* All modifications of the content of the embedded functions registers have to be performed with the device in power-down mode.

### 11.1 SLV0\_ADD (02h)

I<sup>2</sup>C slave address of the first external sensor (Sensor1) register (r/w).

**Table 225. SLV0\_ADD register**

|             |             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|
| Slave0_add6 | Slave0_add5 | Slave0_add4 | Slave0_add3 | Slave0_add2 | Slave0_add1 | Slave0_add0 | rw_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------|

**Table 226. SLV0\_ADD register description**

|                 |                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------|
| Slave0_add[6:0] | I <sup>2</sup> C slave address of Sensor1 that can be read by sensor hub.<br>Default value: 0000000 |
| rw_0            | Read/write operation on Sensor1. Default value: 0<br>(0: write operation; 1: read operation)        |

### 11.2 SLV0\_SUBADD (03h)

Address of register on the first external sensor (Sensor1) register (r/w).

**Table 227. SLV0\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave0_reg7 | Slave0_reg6 | Slave0_reg5 | Slave0_reg4 | Slave0_reg3 | Slave0_reg2 | Slave0_reg1 | Slave0_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 228. SLV0\_SUBADD register description**

|                 |                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave0_reg[7:0] | Address of register on Sensor1 that has to be read/write according to the rw_0 bit value in <a href="#">SLV0_ADD (02h)</a> . Default value: 00000000 |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.3 SLAVE0\_CONFIG (04h)

First external sensor (Sensor1) configuration and sensor hub settings register (r/w).

**Table 229. SLAVE0\_CONFIG register**

|              |              |              |              |          |               |               |               |
|--------------|--------------|--------------|--------------|----------|---------------|---------------|---------------|
| Slave0_rate1 | Slave0_rate0 | Aux_sens_on1 | Aux_sens_on0 | Src_mode | Slave0_numop2 | Slave0_numop1 | Slave0_numop0 |
|--------------|--------------|--------------|--------------|----------|---------------|---------------|---------------|

**Table 230. SLAVE0\_CONFIG register description**

|                   |                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave0_rate[1:0]  | Decimation of read operation on Sensor1 starting from the sensor hub trigger.<br>Default value: 00<br>(00: no decimation<br>01: update every 2 samples<br>10: update every 4 samples<br>11: update every 8 samples) |
| Aux_sens_on[1:0]  | Number of external sensors to be read by sensor hub. Default value: 00<br>(00: one sensor<br>01: two sensors<br>10: three sensors<br>11: four sensors)                                                              |
| Src_mode          | Source mode conditioned read <sup>(1)</sup> . Default value: 0<br>(0: source mode read disabled; 1: source mode read enabled)                                                                                       |
| Slave0_numop[2:0] | Number of read operations on Sensor1.                                                                                                                                                                               |

1. Read conditioned by the content of the register at address specified in the [DATAWRITE\\_SRC\\_MODE\\_SUB\\_SLV0 \(0Eh\)](#) register. If the content is non-zero, the operation continues with the reading of the address specified in the [SLV0\\_SUBADD \(03h\)](#) register, else the operation is interrupted.

## 11.4 SLV1\_ADD (05h)

I<sup>2</sup>C slave address of the second external sensor (Sensor2) register (r/w).

**Table 231. SLV1\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave1_add6 | Slave1_add5 | Slave1_add4 | Slave1_add3 | Slave1_add2 | Slave1_add1 | Slave1_add0 | r_1 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 232. SLV1\_ADD register description**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Slave1_add[6:0] | I <sup>2</sup> C slave address of Sensor2 that can be read by sensor hub.<br>Default value: 0000000           |
| r_1             | Read operation on Sensor2 enable. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

## 11.5 SLV1\_SUBADD (06h)

Address of register on the second external sensor (Sensor2) register (r/w).

**Table 233. SLV1\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave1_reg7 | Slave1_reg6 | Slave1_reg5 | Slave1_reg4 | Slave1_reg3 | Slave1_reg2 | Slave1_reg1 | Slave1_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 234. SLV1\_SUBADD register description**

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Slave1_reg[7:0] | Address of register on Sensor2 that has to be read according to the r_1 bit value in <a href="#">SLV1_ADD (05h)</a> . Default value: 00000000 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

## 11.6 SLAVE1\_CONFIG (07h)

Second external sensor (Sensor2) configuration register (r/w).

**Table 235. SLAVE1\_CONFIG register**

|              |              |            |                  |                  |               |               |               |
|--------------|--------------|------------|------------------|------------------|---------------|---------------|---------------|
| Slave1_rate1 | Slave1_rate0 | write_once | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | Slave1_numop2 | Slave1_numop1 | Slave1_numop0 |
|--------------|--------------|------------|------------------|------------------|---------------|---------------|---------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 236. SLAVE1\_CONFIG register description**

|                   |                                                                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave1_rate[1:0]  | Decimation of read operation on Sensor2 starting from the sensor hub trigger.<br>Default value: 00<br>(00: no decimation<br>01: update every 2 samples<br>10: update every 4 samples<br>11: update every 8 samples)   |
| write_once        | Slave 0 write operation is performed only at the first sensor hub cycle. <sup>(1)</sup><br>Default value: 0<br>0: write operation for each sensor hub cycle<br>1: write operation only for the first sensor hub cycle |
| Slave1_numop[2:0] | Number of read operations on Sensor2.                                                                                                                                                                                 |

1. This is effective if the Aux\_sens\_on[1:0] field in [SLAVE0\\_CONFIG \(04h\)](#) is set to a value other than 00.

## 11.7 SLV2\_ADD (08h)

I<sup>2</sup>C slave address of the third external sensor (Sensor3) register (r/w).

**Table 237. SLV2\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave2_add6 | Slave2_add5 | Slave2_add4 | Slave2_add3 | Slave2_add2 | Slave2_add1 | Slave2_add0 | r_2 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 238. SLV2\_ADD register description**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Slave2_add[6:0] | I <sup>2</sup> C slave address of Sensor3 that can be read by sensor hub.<br>Default value: 0000000           |
| r_2             | Read operation on Sensor3 enable. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

## 11.8 SLV2\_SUBADD (09h)

Address of register on the third external sensor (Sensor3) register (r/w).

**Table 239. SLV2\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave2_reg7 | Slave2_reg6 | Slave2_reg5 | Slave2_reg4 | Slave2_reg3 | Slave2_reg2 | Slave2_reg1 | Slave2_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 240. SLV2\_SUBADD register description**

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Slave2_reg[7:0] | Address of register on Sensor3 that has to be read according to the r_2 bit value in <a href="#">SLV2_ADD (08h)</a> . Default value: 00000000 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

## 11.9 SLAVE2\_CONFIG (0Ah)

Third external sensor (Sensor3) configuration register (r/w).

**Table 241. SLAVE2\_CONFIG register**

|              |              |                  |                  |                  |               |               |               |
|--------------|--------------|------------------|------------------|------------------|---------------|---------------|---------------|
| Slave2_rate1 | Slave2_rate0 | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | Slave2_numop2 | Slave2_numop1 | Slave2_numop0 |
|--------------|--------------|------------------|------------------|------------------|---------------|---------------|---------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 242. SLAVE2\_CONFIG register description**

|                   |                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave2_rate[1:0]  | Decimation of read operation on Sensor3 starting from the sensor hub trigger.<br>Default value: 00<br>(00: no decimation<br>01: update every 2 samples<br>10: update every 4 samples<br>11: update every 8 samples) |
| Slave2_numop[2:0] | Number of read operations on Sensor3.                                                                                                                                                                               |

## 11.10 SLV3\_ADD (0Bh)

I<sup>2</sup>C slave address of the fourth external sensor (Sensor4) register (r/w).

**Table 243. SLV3\_ADD register**

|             |             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Slave3_add6 | Slave3_add5 | Slave3_add4 | Slave3_add3 | Slave3_add2 | Slave3_add1 | Slave3_add0 | r_3 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-----|

**Table 244. SLV3\_ADD register description**

|                 |                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Slave3_add[6:0] | I <sup>2</sup> C slave address of Sensor4 that can be read by the sensor hub.<br>Default value: 0000000       |
| r_3             | Read operation on Sensor4 enable. Default value: 0<br>(0: read operation disabled; 1: read operation enabled) |

## 11.11 SLV3\_SUBADD (0Ch)

Address of register on the fourth external sensor (Sensor4) register (r/w).

**Table 245. SLV3\_SUBADD register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Slave3_reg7 | Slave3_reg6 | Slave3_reg5 | Slave3_reg4 | Slave3_reg3 | Slave3_reg2 | Slave3_reg1 | Slave3_reg0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 246. SLV3\_SUBADD register description**

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Slave3_reg[7:0] | Address of register on Sensor4 that has to be read according to the r_3 bit value in <a href="#">SLV3_ADD (0Bh)</a> . Default value: 00000000 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

## 11.12 SLAVE3\_CONFIG (0Dh)

Fourth external sensor (Sensor4) configuration register (r/w).

**Table 247. SLAVE3\_CONFIG register**

|              |              |                  |                  |                  |               |               |               |
|--------------|--------------|------------------|------------------|------------------|---------------|---------------|---------------|
| Slave3_rate1 | Slave3_rate0 | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | 0 <sup>(1)</sup> | Slave3_numop2 | Slave3_numop1 | Slave3_numop0 |
|--------------|--------------|------------------|------------------|------------------|---------------|---------------|---------------|

1. This bit must be set to '0' for the correct operation of the device.

**Table 248. SLAVE3\_CONFIG register description**

|                   |                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave3_rate[1:0]  | Decimation of read operation on Sensor4 starting from the sensor hub trigger.<br>Default value: 00<br>(00: no decimation<br>01: update every 2 samples<br>10: update every 4 samples<br>11: update every 8 samples) |
| Slave3_numop[2:0] | Number of read operations on Sensor4.                                                                                                                                                                               |

## 11.13 DATAWRITE\_SRC\_MODE\_SUB\_SLV0 (0Eh)

Data to be written into the slave device register (r/w).

**Table 249. DATAWRITE\_SRC\_MODE\_SUB\_SLV0 register**

|              |              |              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Slave_dataw7 | Slave_dataw6 | Slave_dataw5 | Slave_dataw4 | Slave_dataw3 | Slave_dataw2 | Slave_dataw1 | Slave_dataw0 |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|

**Table 250. DATAWRITE\_SRC\_MODE\_SUB\_SLV0 register description**

|                  |                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Slave_dataw[7:0] | Data to be written into the slave device according to the rw_0 bit in <a href="#">SLV0_ADD (02h)</a> register or address to be read in source mode.<br>Default value: 00000000 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 11.14 CONFIG\_PEDO\_THS\_MIN (0Fh)

**Table 251. CONFIG\_PEDO\_THS\_MIN register**

|         |   |   |           |           |           |           |           |
|---------|---|---|-----------|-----------|-----------|-----------|-----------|
| PEDO_FS | 0 | 0 | ths_min_4 | ths_min_3 | ths_min_2 | ths_min_1 | ths_min_0 |
|---------|---|---|-----------|-----------|-----------|-----------|-----------|

**Table 252. CONFIG\_PEDO\_THS\_MIN register description**

|               |                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------|
| PEDO_FS       | Pedometer data elaboration at 4 g.<br>(0: elaboration of 2 g data;<br>1: elaboration of 4 g data) |
| ths_min_[4:0] | Minimum threshold to detect a peak. Default is 10h.                                               |

## 11.15 SM\_THS (13h)

Significant motion configuration register (r/w).

**Table 253. SM\_THS register**

| SM_THS_7 | SM_THS_6 | SM_THS_5 | SM_THS_4 | SM_THS_3 | SM_THS_2 | SM_THS_1 | SM_THS_0 |
|----------|----------|----------|----------|----------|----------|----------|----------|
|----------|----------|----------|----------|----------|----------|----------|----------|

**Table 254. SM\_THS register description**

|             |                                                       |
|-------------|-------------------------------------------------------|
| SM_THS[7:0] | Significant motion threshold. Default value: 00000110 |
|-------------|-------------------------------------------------------|

## 11.16 PEDO\_DEB\_REG (14h)

**Table 255. PEDO\_DEB\_REG register default values**

| DEB_TIME4 | DEB_TIME3 | DEB_TIME2 | DEB_TIME1 | DEB_TIME0 | DEB_STEP2 | DEB_STEP1 | DEB_STEP0 |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 0         | 1         | 1         | 0         | 1         | 1         | 1         | 0         |

**Table 256. PEDO\_DEB\_REG register description**

|               |                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| DEB_TIME[4:0] | Debounce time. If the time between two consecutive steps is greater than DEB_TIME*80ms, the debouncer is reactivated. Default value: 01101 |
| DEB_STEP[2:0] | Debounce threshold. Minimum number of steps to increment step counter (debounce). Default value: 110                                       |

## 11.17 STEP\_COUNT\_DELTA (15h)

Time period register for step detection on delta time (r/w).

**Table 257. STEP\_COUNT\_DELTA register**

| SC_DELTA_7 | SC_DELTA_6 | SC_DELTA_5 | SC_DELTA_4 | SC_DELTA_3 | SC_DELTA_2 | SC_DELTA_1 | SC_DELTA_0 |
|------------|------------|------------|------------|------------|------------|------------|------------|
|------------|------------|------------|------------|------------|------------|------------|------------|

**Table 258. STEP\_COUNT\_DELTA register description**

|               |                                                    |
|---------------|----------------------------------------------------|
| SC_DELTA[7:0] | Time period value <sup>(1)</sup> (1LSB = 1.6384 s) |
|---------------|----------------------------------------------------|

1. This value is effective if the TIMER\_EN bit of [CTRL10\\_C \(19h\)](#) is set to 1 and the TIMER\_HR bit of [WAKE\\_UP\\_DUR \(5Ch\)](#) register is set to 0.

## 11.18 MAG\_SI\_XX (24h)

Soft-iron matrix correction register (r/w).

**Table 259. MAG\_SI\_XX register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_XX_7 | MAG_SI_XX_6 | MAG_SI_XX_5 | MAG_SI_XX_4 | MAG_SI_XX_3 | MAG_SI_XX_2 | MAG_SI_XX_1 | MAG_SI_XX_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 260. MAG\_SI\_XX register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_XX[7:0] | Soft-iron correction row1 col1 coefficient <sup>(1)</sup> . Default value: 00001000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.19 MAG\_SI\_XY (25h)

Soft-iron matrix correction register (r/w).

**Table 261. MAG\_SI\_XY register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_XY_7 | MAG_SI_XY_6 | MAG_SI_XY_5 | MAG_SI_XY_4 | MAG_SI_XY_3 | MAG_SI_XY_2 | MAG_SI_XY_1 | MAG_SI_XY_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 262. MAG\_SI\_XY register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_XY[7:0] | Soft-iron correction row1 col2 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.20 MAG\_SI\_XZ (26h)

Soft-iron matrix correction register (r/w).

**Table 263. MAG\_SI\_XZ register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_XZ_7 | MAG_SI_XZ_6 | MAG_SI_XZ_5 | MAG_SI_XZ_4 | MAG_SI_XZ_3 | MAG_SI_XZ_2 | MAG_SI_XZ_1 | MAG_SI_XZ_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 264. MAG\_SI\_XZ register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_XZ[7:0] | Soft-iron correction row1 col3 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.21 MAG\_SI\_YX (27h)

Soft-iron matrix correction register (r/w).

**Table 265. MAG\_SI\_YX register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_YX_7 | MAG_SI_YX_6 | MAG_SI_YX_5 | MAG_SI_YX_4 | MAG_SI_YX_3 | MAG_SI_YX_2 | MAG_SI_YX_1 | MAG_SI_YX_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 266. MAG\_SI\_YX register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_YX[7:0] | Soft-iron correction row2 col1 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.22 MAG\_SI\_YY (28h)

Soft-iron matrix correction register (r/w).

**Table 267. MAG\_SI\_YY register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_YY_7 | MAG_SI_YY_6 | MAG_SI_YY_5 | MAG_SI_YY_4 | MAG_SI_YY_3 | MAG_SI_YY_2 | MAG_SI_YY_1 | MAG_SI_YY_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 268. MAG\_SI\_YY register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_YY[7:0] | Soft-iron correction row2 col2 coefficient <sup>(1)</sup> . Default value: 00001000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.23 MAG\_SI\_YZ (29h)

Soft-iron matrix correction register (r/w).

**Table 269. MAG\_SI\_YZ register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_YZ_7 | MAG_SI_YZ_6 | MAG_SI_YZ_5 | MAG_SI_YZ_4 | MAG_SI_YZ_3 | MAG_SI_YZ_2 | MAG_SI_YZ_1 | MAG_SI_YZ_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 270. MAG\_SI\_YZ register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_YZ[7:0] | Soft-iron correction row2 col3 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.24 MAG\_SI\_ZX (2Ah)

Soft-iron matrix correction register (r/w).

**Table 271. MAG\_SI\_ZX register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_ZX_7 | MAG_SI_ZX_6 | MAG_SI_ZX_5 | MAG_SI_ZX_4 | MAG_SI_ZX_3 | MAG_SI_ZX_2 | MAG_SI_ZX_1 | MAG_SI_ZX_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 272. MAG\_SI\_ZX register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_ZX[7:0] | Soft-iron correction row3 col1 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.25 MAG\_SI\_ZY (2Bh)

Soft-iron matrix correction register (r/w).

**Table 273. MAG\_SI\_ZY register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_ZY_7 | MAG_SI_ZY_6 | MAG_SI_ZY_5 | MAG_SI_ZY_4 | MAG_SI_ZY_3 | MAG_SI_ZY_2 | MAG_SI_ZY_1 | MAG_SI_ZY_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 274. MAG\_SI\_ZY register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_ZY[7:0] | Soft-iron correction row3 col2 coefficient <sup>(1)</sup> . Default value: 00000000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.26 MAG\_SI\_ZZ (2Ch)

Soft-iron matrix correction register (r/w).

**Table 275. MAG\_SI\_ZZ register**

|             |             |             |             |             |             |             |             |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MAG_SI_ZZ_7 | MAG_SI_ZZ_6 | MAG_SI_ZZ_5 | MAG_SI_ZZ_4 | MAG_SI_ZZ_3 | MAG_SI_ZZ_2 | MAG_SI_ZZ_1 | MAG_SI_ZZ_0 |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|

**Table 276. MAG\_SI\_ZZ register description**

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| MAG_SI_ZZ[7:0] | Soft-iron correction row3 col3 coefficient <sup>(1)</sup> . Default value: 00001000 |
|----------------|-------------------------------------------------------------------------------------|

1. Value is expressed in sign-module format.

## 11.27 MAG\_OFFX\_L (2Dh)

Offset for X-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 277. MAG\_OFFX\_L register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| MAG_OFF_X_L_7 | MAG_OFF_X_L_6 | MAG_OFF_X_L_5 | MAG_OFF_X_L_4 | MAG_OFF_X_L_3 | MAG_OFF_X_L_2 | MAG_OFF_X_L_1 | MAG_OFF_X_L_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 278. MAG\_OFFX\_L register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFX_L[7:0] | Offset for X-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

## 11.28 MAG\_OFFX\_H (2Eh)

Offset for X-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 279. MAG\_OFFX\_H register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| MAG_OFF_X_H_7 | MAG_OFF_X_H_6 | MAG_OFF_X_H_5 | MAG_OFF_X_H_4 | MAG_OFF_X_H_3 | MAG_OFF_X_H_2 | MAG_OFF_X_H_1 | MAG_OFF_X_H_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 280. MAG\_OFFX\_H register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFX_H[7:0] | Offset for X-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

## 11.29 MAG\_OFFY\_L (2Fh)

Offset for Y-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 281. MAG\_OFFY\_L register**

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| MAG_OFF_Y_L_7 | MAG_OFF_Y_L_6 | MAG_OFF_Y_L_5 | MAG_OFF_Y_L_4 | MAG_OFF_Y_L_3 | MAG_OFF_Y_L_2 | MAG_OFF_Y_L_1 | MAG_OFF_Y_L_0 |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|

**Table 282. MAG\_OFFY\_L register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFY_L[7:0] | Offset for Y-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

### 11.30 MAG\_OFFY\_H (30h)

Offset for Y-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 283. MAG\_OFFY\_H register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| MAG_OFF<br>Y_H_7 | MAG_OFF<br>Y_H_6 | MAG_OFF<br>Y_H_5 | MAG_OFF<br>Y_H_4 | MAG_OFF<br>Y_H_3 | MAG_OFF<br>Y_H_2 | MAG_OFF<br>Y_H_1 | MAG_OFF<br>Y_H_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 284. MAG\_OFFY\_H register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFY_H[7:0] | Offset for Y-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

### 11.31 MAG\_OFFZ\_L (31h)

Offset for Z-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 285. MAG\_OFFZ\_L register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| MAG_OFF<br>Z_L_7 | MAG_OFF<br>Z_L_6 | MAG_OFF<br>Z_L_5 | MAG_OFF<br>Z_L_4 | MAG_OFF<br>Z_L_3 | MAG_OFF<br>Z_L_2 | MAG_OFF<br>Z_L_1 | MAG_OFF<br>Z_L_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 286. MAG\_OFFZ\_L register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFZ_L[7:0] | Offset for Z-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

### 11.32 MAG\_OFFZ\_H (32h)

Offset for Z-axis hard-iron compensation register (r/w). The value is expressed as a 16-bit word in two's complement.

**Table 287. MAG\_OFFZ\_H register**

|                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| MAG_OFF<br>Z_H_7 | MAG_OFF<br>Z_H_6 | MAG_OFF<br>Z_H_5 | MAG_OFF<br>Z_H_4 | MAG_OFF<br>Z_H_3 | MAG_OFF<br>Z_H_2 | MAG_OFF<br>Z_H_1 | MAG_OFF<br>Z_H_0 |
|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|

**Table 288. MAG\_OFFZ\_H register description**

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| MAG_OFFZ_H[7:0] | Offset for Z-axis hard-iron compensation. Default value: 00000000 |
|-----------------|-------------------------------------------------------------------|

## 12 Embedded functions registers description - Bank B

### 12.1 A\_WRIST\_TILT\_LAT (50h)

Absolute Wrist Tilt latency register (r/w).

**Table 289. A\_WRIST\_TILT\_LAT register**

|                   |                   |                   |                   |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| WRIST_TILT_TIMER7 | WRIST_TILT_TIMER6 | WRIST_TILT_TIMER5 | WRIST_TILT_TIMER4 | WRIST_TILT_TIMER3 | WRIST_TILT_TIMER2 | WRIST_TILT_TIMER1 | WRIST_TILT_TIMER0 |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|

**Table 290. A\_WRIST\_TILT\_LAT register description**

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| WRIST_TILT_TIMER[7:0] | Absolute wrist tilt latency parameters. 1 LSB = 40 ms.<br>Default value: 0Fh (600 ms) |
|-----------------------|---------------------------------------------------------------------------------------|

### 12.2 A\_WRIST\_TILT\_THS (54h)

Absolute Wrist Tilt threshold register (r/w).

**Table 291. A\_WRIST\_TILT\_THS register**

|                 |                 |                 |                 |                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| WRIST_TILT_THS7 | WRIST_TILT_THS6 | WRIST_TILT_THS5 | WRIST_TILT_THS4 | WRIST_TILT_THS3 | WRIST_TILT_THS2 | WRIST_TILT_THS1 | WRIST_TILT_THS0 |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|

**Table 292. A\_WRIST\_TILT\_THS register description**

|                     |                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| WRIST_TILT_THS[7:0] | Absolute wrist tilt threshold parameters. 1 LSB = 15.625 mg.<br>Default value: 20h (500 mg) |
|---------------------|---------------------------------------------------------------------------------------------|

### 12.3 A\_WRIST\_TILT\_Mask (59h)

Absolute Wrist Tilt mask register (r/w).

**Table 293. A\_WRIST\_TILT\_Mask register**

|                      |                      |                      |                      |                      |                      |   |   |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---|---|
| WRIST_TILT_MASK_Xpos | WRIST_TILT_MASK_Xneg | WRIST_TILT_MASK_Ypos | WRIST_TILT_MASK_Yneg | WRIST_TILT_MASK_Zpos | WRIST_TILT_MASK_Zneg | 0 | 0 |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---|---|

**Table 294. A\_WRIST\_TILT\_Mask register description**

|                      |                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------|
| WRIST_TILT_MASK_Xpos | Absolute wrist tilt positive X-axis enable. Default value: 1<br>(0: disable; 1: enable) |
| WRIST_TILT_MASK_Xneg | Absolute wrist tilt negative X-axis enable. Default value: 1<br>(0: disable; 1: enable) |
| WRIST_TILT_MASK_Ypos | Absolute wrist tilt positive Y-axis enable. Default value: 0<br>(0: disable; 1: enable) |
| WRIST_TILT_MASK_Yneg | Absolute wrist tilt negative Y-axis enable. Default value: 0<br>(0: disable; 1: enable) |
| WRIST_TILT_MASK_Zpos | Absolute wrist tilt positive Z-axis enable. Default value: 0<br>(0: disable; 1: enable) |
| WRIST_TILT_MASK_Zneg | Absolute wrist tilt negative Z-axis enable. Default value: 0<br>(0: disable; 1: enable) |

## 13 Soldering information

The LGA package is compliant with the ECOPACK®, RoHS and "Green" standard. It is qualified for soldering heat resistance according to JEDEC J-STD-020.

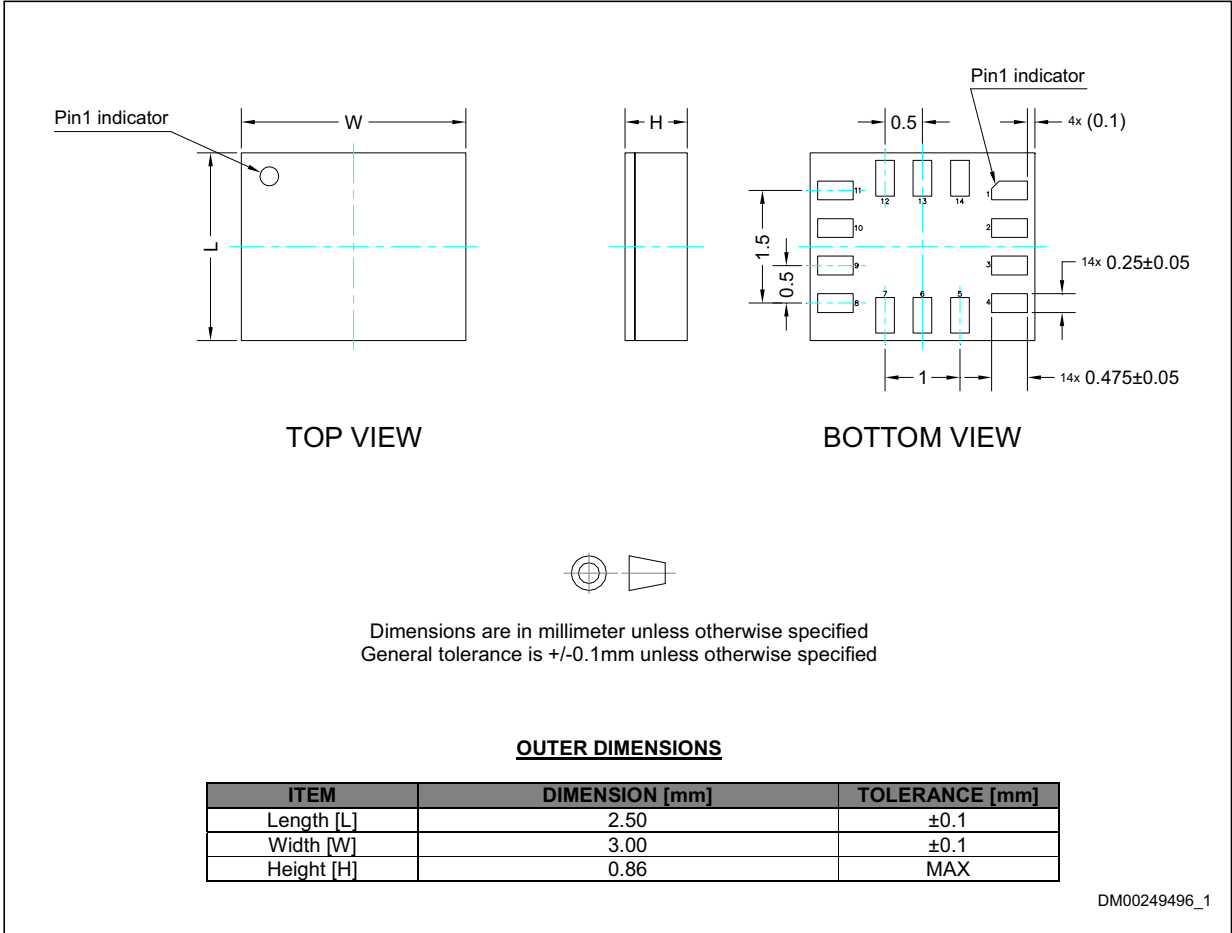
Land pattern and soldering recommendations are available at [www.st.com/mems](http://www.st.com/mems).

# 14 Package information

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: [www.st.com](http://www.st.com). ECOPACK is an ST trademark.

## 14.1 LGA-14 package information

Figure 17. LGA-14 2.5x3x0.86 mm package outline and mechanical data



14.2 LGA-14 packing information

Figure 18. Carrier tape information for LGA-14 package

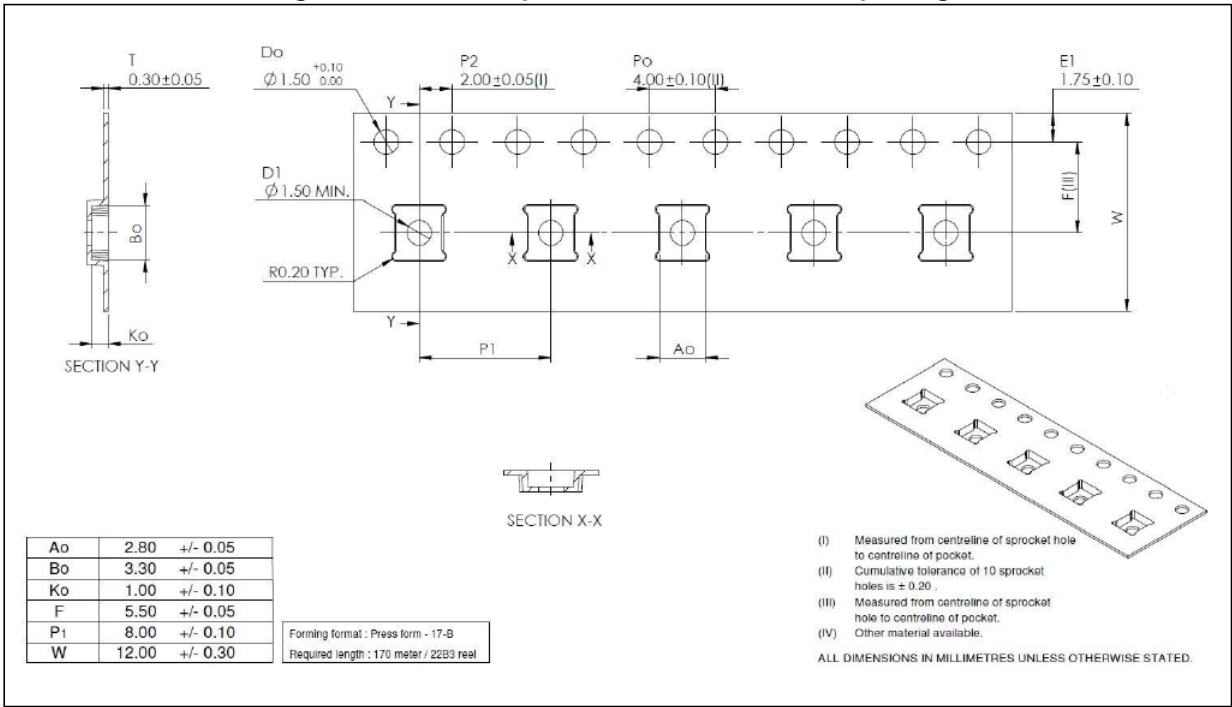
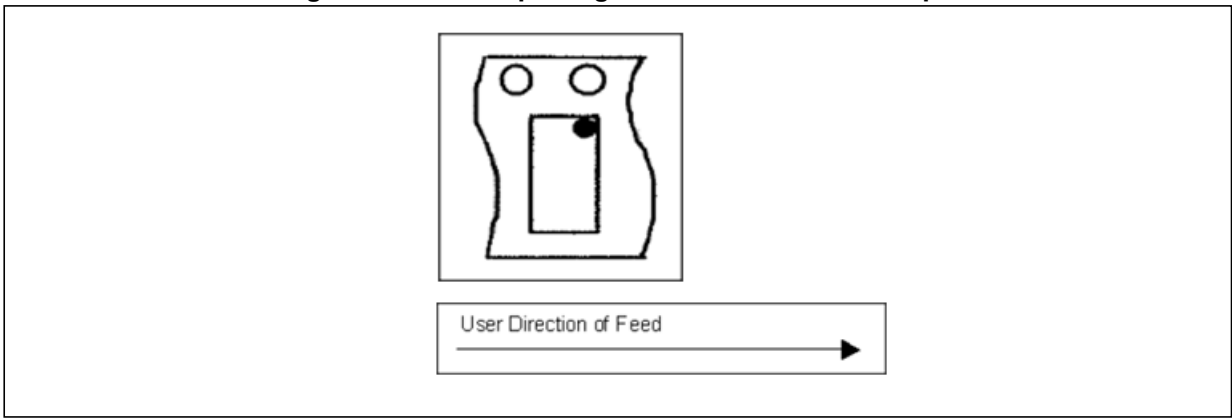
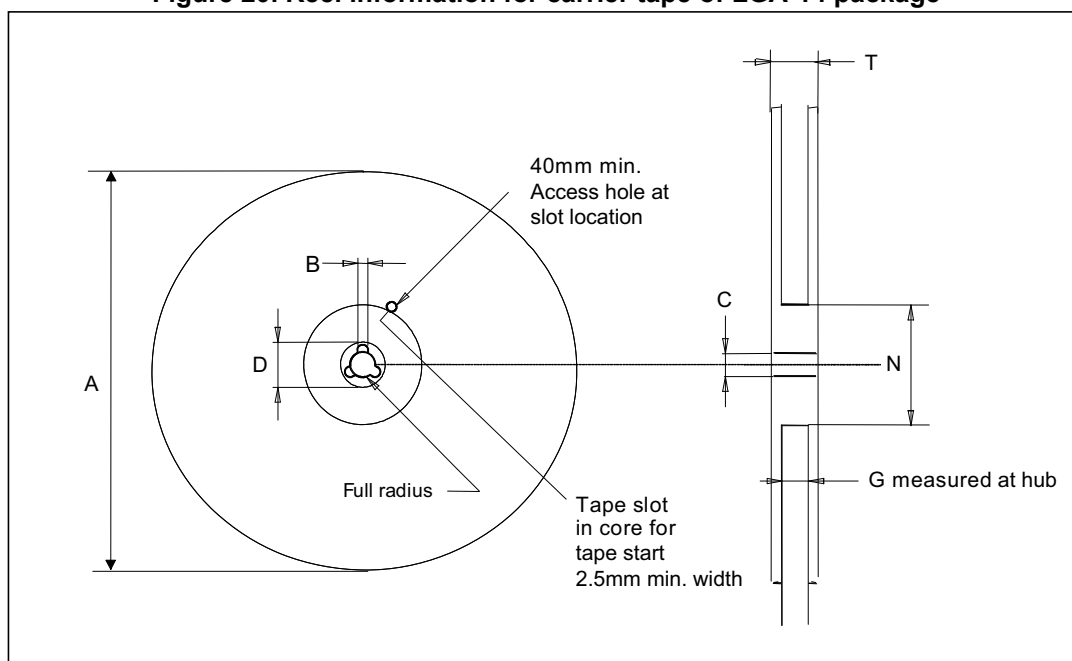


Figure 19. LGA-14 package orientation in carrier tape



**Figure 20. Reel information for carrier tape of LGA-14 package****Table 295. Reel dimensions for carrier tape of LGA-14 package**

| Reel dimensions (mm) |            |
|----------------------|------------|
| A (max)              | 330        |
| B (min)              | 1.5        |
| C                    | 13 ±0.25   |
| D (min)              | 20.2       |
| N (min)              | 60         |
| G                    | 12.4 +2/-0 |
| T (max)              | 18.4       |

## 15 Revision history

**Table 296. Document revision history**

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03-May-2017 | 6        | <p>Updated <a href="#">Section 4.4.2: I<sup>2</sup>C - inter-IC control interface</a> (added <a href="#">Table 8: I<sup>2</sup>C master timing values</a>)</p> <p>Updated <a href="#">Figure 11</a> and <a href="#">Figure 13</a></p> <p>Updated pin status mode 1 and 2 for pins 10 and 11 as well as adding procedure to disable pull-ups in <a href="#">Table 18: Internal pin status</a></p> <p>Updated bit 0 in <a href="#">CTRL1_XL (10h)</a></p> <p>Updated <a href="#">CTRL8_XL (17h)</a></p> <p>Updated description of SW_RESET bit in <a href="#">Table 57: CTRL3_C register description</a></p> <p>Updated description of <a href="#">X_OFS_USR (73h)</a>, <a href="#">Y_OFS_USR (74h)</a>, <a href="#">Z_OFS_USR (75h)</a></p> <p>Minor textual updates</p> |
| 29-Sep-2017 | 7        | <p>Updated <a href="#">Table 3: Mechanical characteristics</a></p> <p>Specified SPI mode 3 in <a href="#">Section 4.4.1: SPI - serial peripheral interface</a> and throughout <a href="#">Section 6: Digital interfaces</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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